The observed temperature power spectrum through fourth order in a radiation-only universe: analytical representation and evaluated limits

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Abstract

We compute the observed temperature anisotropy through third order, and its angular power spectrum through fourth order, in the primordial amplitude ϵ. The setting is a flat radiation-only universe in general relativity, with Planck test photons emitted on $T_{\rm phys} = T_\star$ and an observer on a fluid worldline. In the fluid's temperature frame the observed temperature is an exact line integral of the shear of a conformally rescaled metric. The observer's own temperature cancels exactly from the normalised map. A triangular scheme solves all ten Einstein equations at every order: the second-order kernels are in closed form, and the third-order kernels are explicit integrals. The linear transfer T_ℓ is given in closed form for every ℓ. The fourth-order spectrum is reduced to exact finite-dimensional integrals with every sky integral done. A compact cubic representation eliminates the independent third-order field solve and the second-order ray, using only acoustic and wave Green functions. On a monochromatic shell the leading-gradient quadrupole has the exact Gaussian factor $e^{8v}(1 + 5v/3)$, with $v = \epsilon^2\sigma^2$; its fourth-order correction is $29/3 = 8 + 5/3$, an observer dilation plus a curvature contribution. The corresponding octupole coefficient is $46 = 18 + 28$. The complete next-gradient quadrupole and its observer-dilation decomposition are also evaluated. The general finite-wavelength fourth-order spectrum is *not* evaluated in closed form. The results at a glance are listed in §1, and the open problems in §11. Analytical derivations, exact computer-algebra checks and supplementary numerical validation are distinguished; the detailed verification record is in the appendices.

1 Problem and summary of results

The task (`problem_statement.tex`) asks for $C_\ell = \epsilon^2 C_\ell^{(11)} + \epsilon^4[\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}] + O(\epsilon^6)$ for the observed map

$$\Theta(\hat{n}) = \frac{T_{\rm obs}(\hat{n})}{\frac{1}{4\pi}\int \mathrm{d}\Omega'\, T_{\rm obs}(\hat{n}')} - 1, \qquad T_{\rm obs} = T_\star \frac{E_o}{E_e}, \quad E = -u_\mu p^\mu.$$

The physical model is a perfect radiation fluid with $P = \rho/3$, $\rho = \alpha T_{\rm phys}^4$ and $\Lambda = 0$, with Gaussian adiabatic primordial perturbations and no independent higher-order modes. The calculation must be analytical and keep every term, with all normalisations fixed. A monochromatic specialisation is to be made only at the end.

Results at a glance

1. *A complete finite-time spectrum.* The linear transfer T_ℓ is in closed form for every ℓ (Prop. 16), and inside the horizon it reduces to the Sachs–Wolfe plus Doppler form (Prop. 30); the fourth-order answer is an exact integral representation with all sky integrals evaluated (§8). On a monochromatic shell, holding $\langle g^2\rangle = \sigma^2$ fixed and putting $v = \epsilon^2\sigma^2$, it is

$$C_\ell = 4\pi v T_\ell^2 + v^2[Q_\ell + 4\pi T_\ell \operatorname{Re} R_\ell] + O(v^3).$$

   The nonnegative quadratic term Q_ℓ fills linear-transfer nodes; the cubic term interferes with the linear response ((59), Prop. 23).

2. *Curvature multipoles and evaluated nonlinear corrections.* The leading quadrupole is the local trace-free Ricci curvature of the primordial seed; the octupole is its covariant gradient (§§9.1–9.2). On the shell,

$$\frac{C_{2,\mathrm{lead}}}{\epsilon^2 C^{(11)}_{2,\mathrm{lead}}} = e^{8v}(1+5v/3) = 1 + 29v/3 + O(v^2), \qquad \frac{C_{3,\mathrm{lead}}}{\epsilon^2 C^{(11)}_{3,\mathrm{lead}}} = 1 + 46v + O(v^2).$$

   The coefficients split as $29/3 = 8 + 5/3$ and $46 = 18 + 28$: observer dilation plus intrinsic curvature. The shell maximises both relative leading-gradient corrections at fixed variance ((75)). These resummations concern the leading-gradient maps, not the full finite-wavelength solution.

3. *The next-gradient quadrupole.* The complete $O(\epsilon^4 k_0^6)$ coefficient is evaluated with distinct emission, observation and physical initial times (§9.3). Three local tensors determine the map, and after observer dilation is subtracted only the spectral moment product $m_2 m_4$ remains ((73), (74)).

4. *Why the observable simplifies.* In the temperature frame the redshift is a single shear integral and the observer temperature cancels exactly (Lemmas 1, 2). A regular triangular solution satisfies all ten Einstein equations (Thm. 7). The map retains the ray, physical endpoints, clock, Fermi–Walker frame and random-monopole normalisation (§6).

5. *Exact structural reductions.* The spectrum has log-map and observer-dilation decompositions (§§8.2–8.3). The combined cubic response needs no independent third-order field solve, second ray or first field adjoint; the constraints remove hyperbolic Green kernels (§10). The surviving terms have at most three independent time integrations. On the shell the second-order metric is two amplitudes, and its tensor is the known radiation-era induced-wave kernel plus an elementary gauge correction (Prop. 10, (17)).

Scope. Exact definite integrals are an analytical answer under the problem's stated requirement. The general finite-wavelength fourth-order spectrum is not evaluated in closed form; its late-time limit and convergence for noncompact spectra remain open (§11). Existence is established at finite times for finite spectral measures of compact support in $0 < k < \infty$. Verification scopes and the requirements checklist are in Apps. A and E; numerical validation is separate (App. F) and is not used in the analytical derivation.

Two reading paths. For *results and interpretation*, go to the spectrum (§8), the curvature multipoles and next-gradient result (§9), then the reduced cubic response (§10). For the *derivation*, follow the experiment, exact frame, regular fields and complete map in §§2–6, with the detailed moment and adjoint formulas in Apps. B–C.

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2 Conventions and the experiment

Signature $(-,+,+,+)$, $c=1$. The background is $a=a_1 t$ and $\bar{T}=\mathcal{T}/(a_1 t)$, with $8\pi G\alpha\mathcal{T}^4=3a_1^2$.

Fourier and perturbative conventions.

- $g(X)=\int_k g_k e^{ik\cdot X}$, with $\int_k=\int \mathrm{d}^3k/(2\pi)^3$ and $\langle g_k g_{k'}\rangle=(2\pi)^3\delta^3(k+k')P(k)$; also $\mathcal{P}(k)=k^3P/2\pi^2$.
- Every field is $Y=\sum_r \epsilon^r Y^{(r)}/r!$ with $Y^{(r)}=\int_{k_1\ldots k_r} K_r(k_1,\ldots,k_r)\, g_{k_1}\cdots g_{k_r} e^{iK\cdot X}$ and $K=\sum k_a$. The symmetric kernels carry no extra $1/r!$.
- *Multilinear rule:* K_r is the coefficient of $\epsilon^r z_1\cdots z_r$ in the *full* series Y when $g=\sum_a z_a e^{ik_a\cdot X}$ is inserted and only monomials in distinct labels are kept. Equivalently, the coefficient of $z_1\cdots z_r$ in $Y^{(r)}$ is $r!\,K_r$, as the integral representation above requires. This fixes every combinatorial factor; for example $K_2(k,0)=-k\cdot\partial_k K_1$ in §7.
- Spherical harmonics $Y_{\ell m}$ are complex, with the Condon–Shortley phase.

Measurement labels.

- *Emission:* the surface $T_{\rm phys}=T_\star$. Emission is isotropic and Planckian in the fluid rest frame. The intersection is the branch continuously connected to the background one.
- *Observer:* the fluid line $X=0$. The initial event is $T_{\rm phys}=T_I>T_\star$ on that line, and observation follows after fixed proper time τ_o.
- *Tetrad:* Fermi–Walker transported, with initial orientation $\gamma(t_I,0)^{-1/2}$ (the symmetric square root, aligned with the marked labels).
- *Background times:* $\bar{t}_\star=\mathcal{T}/(a_1T_\star)$, $\bar{t}_I=\mathcal{T}/(a_1T_I)$ and $\bar{t}_o^2=\bar{t}_I^2+2\tau_o/a_1$. We write $x\equiv kt/\sqrt{3}=k/(\sqrt{3}\,aH)$.

3 Exact framework

Lemma 1 (Temperature frame)**.** *For an irrotational radiation fluid, $Tu_\mu=-\mathcal{T}\partial_\mu t$ defines the velocity potential t. In coordinates (t,X^i), with comoving labels X^i,*

$$g=N^2\big(-\mathrm{d}t^2+\gamma_{ij}\mathrm{d}X^i\mathrm{d}X^j\big),\qquad N=\mathcal{T}/T_{\rm phys},\qquad u=N^{-1}\partial_t$$

exactly. Euler's equation holds identically, and continuity is equivalent to $\partial_t\det\gamma=0$. Hence $\gamma=e^{2\zeta(X)}\exp H$ with $\operatorname{tr}H=0$, and the shear $\sigma_{ij}=\frac{1}{2}\partial_t\gamma_{ij}$ is traceless.

Lemma 2 (Exact redshift)**.** *Null geodesics of g and of $\tilde{g}=g/N^2$ coincide, and $E=\tilde{E}/N$. Hence*

$$T_{\rm obs}(\hat{n})=T_{\rm fluid}(o)\,e^{-I(\hat{n})},\qquad I=\int_{t_e(\hat{n})}^{t_o}\sigma_{ij}(t,X(t))\,e^i e^j\,\mathrm{d}t,$$

where $\mathrm{d}X^i/\mathrm{d}t=e^i$, $\gamma_{ij}e^ie^j=1$ and $\mathrm{d}e^i/\mathrm{d}t=-2S^i{}_je^j-\Gamma^i{}_{jk}e^je^k+e^i\sigma_{jk}e^je^k$, with $S=\gamma^{-1}\sigma$. The emission event is the root of $\ln t_e+\nu(t_e,X(t_e))=\ln\bar{t}_\star$, with $N=a_1te^\nu$.

$T_{\rm fluid}(o)$ is direction-independent. It therefore cancels from Θ *exactly*. The emission-surface temperature shift, Sachs–Wolfe, Doppler and integrated Sachs–Wolfe terms are all contained in the single integral I.

Proposition 3 (Redshift driver in orthonormal photon directions)**.** *Let $v = e^{\zeta} e^{H/2} e$ be the photon direction in the symmetric orthonormal spatial frame, so that $v^T v = 1$. Then exactly*

$$\sigma_{ij} e^i e^j = v^T \mathcal{B} v, \qquad \mathcal{B} = \tfrac{1}{2} e^{-H/2} (e^H)' e^{-H/2},$$

and for arbitrary non-commuting $H = \epsilon H_1 + \frac{\epsilon^2}{2} H_2 + \frac{\epsilon^3}{6} H_3$,

$$\boxed{\mathcal{B} = \tfrac{1}{2} H' + \tfrac{1}{48} \big[H, [H, H']\big] + O(\epsilon^4).}$$

The conformal factor $e^{2\zeta}$ drops out exactly, and no quadratic matrix product appears. On the monochromatic shell $H_1(t, X) = f(t) B(X)$, so the commutator vanishes and $\sigma_{ij} e^i e^j = \frac{1}{2} v^T H' v + O(\epsilon^4)$.

Proof. $\frac{\mathrm{d}}{\mathrm{d}t} e^H = \int_0^1 e^{sH} H' e^{(1-s)H} \mathrm{d}s$. Conjugating by $e^{-H/2}$ gives $\int_0^1 e^{(s-\frac{1}{2})\mathrm{ad}_H} H' \mathrm{d}s$. The odd moments of $s - \frac{1}{2}$ vanish and its second moment is $\frac{1}{12}$. Checked for generic non-commuting matrices in `solution/check_orthonormal_driver.py`. □

In these variables the $e^{2\zeta}$ and $(H_1^2)'$ shear terms are absorbed into the direction transformation. Those are the terms that cancel strongly against the direction terms in the exploratory recombination of App. F. The observer condition becomes simply $v_o = -\hat{n}$ in the symmetric frame. The ray equation for v still carries the spatial connection, and the emission surface and clock are unchanged, so propagation effects are not removed; they are reorganised.

The reorganisation applies to the whole map. Because $\sigma_{ij} e^i e^j = v^T \mathcal{B} v$ holds exactly at the displaced point, the integrand coefficients F_r of §6 are given by the same multilinear formulas with

$$\sigma_r \to \mathcal{B}_r, \qquad e_r \to v_r, \qquad \mathcal{B}_1 = \tfrac{1}{2} H_1', \quad \mathcal{B}_2 = \tfrac{1}{2} H_2', \quad \mathcal{B}_3 = \tfrac{1}{2} H_3' + \tfrac{1}{8} \big[H_1, [H_1, H_1']\big],$$

where v_r are the Taylor coefficients of $e^{\zeta} e^{H/2} e$ along the ray and $\bar{v} = \bar{e} = -\hat{n}$. In particular

$$\boxed{F_2 = \tfrac{1}{2} H_{2,nn}' + X_1 \cdot \partial H_{1,nn}' - 2 H_1'(n, v_1),}$$

with no local products of first-order fields. The first-order orthonormal direction satisfies $v_1(t_o) = 0$ and

$$v_1' = P_\perp \big(\tfrac{1}{2} H_1' n + \nabla \zeta_1\big) + \tfrac{1}{2} \big[\nabla H_{1,nn} - (n \cdot \partial)(H_1 n)\big], \qquad P_\perp = 1 - n n^T,$$

which is transverse, $n \cdot v_1' = 0$, so v stays a unit vector at this order. The positions still follow $X' = e = e^{-\zeta} e^{-H/2} v$, because they are comoving labels. Both statements are checked exactly for generic fields in `solution/check_orthonormal_map.py`.

Remark 4 (Longitudinal-gauge form). The same measurement can be written exactly in longitudinal (Poisson) variables, $\mathrm{d}s^2 = a^2[-e^{2\Phi} \mathrm{d}\eta^2 + h_{ij} (\mathrm{d}y^i + \beta^i \mathrm{d}\eta)(\mathrm{d}y^j + \beta^j \mathrm{d}\eta)]$ with $h = e^{-2\Psi} M$, keeping the induced shift, the tensor shear, the nonlinear Doppler factors $\mathcal{D} = \Gamma(1 - h_{ij} v^i e^j)$, and the displaced endpoints:

$$T_{\rm obs} = T_\star \frac{a_e}{a_o} \frac{\mathcal{D}_o}{\mathcal{D}_e} \exp\Big[\Phi_e - \Phi_o + \int_{\eta_e}^{\eta_o} \mathcal{S} \, \mathrm{d}\eta\Big], \qquad \mathcal{S} = (\partial_\eta - \beta \cdot \partial)(\Phi + \Psi) - \tfrac{1}{2} e^{-2\Psi} e^i e^j \big[\partial_\eta M - \mathcal{L}_\beta M\big]_{ij}.$$

It is related to the temperature-frame exponent exactly by $I = \big[\Phi - \ln(N_T/a) - \ln \mathcal{D}\big]_e^o - \int \mathcal{S}\, \mathrm{d}\eta$, with $N_T = \mathcal{T}/T_{\mathrm{phys}}$. To reach this gauge one needs both a time shift and a spatial transformation. A purely spatial relabelling leaves a shift vector and an unsuppressed $O(1)$ lapse oscillation, so it does not reach longitudinal gauge. This is the identity behind the late-time recombination proposal (App. F). The derivation is in `report/longitudinal_redshift.md`, and the check is `solution/check_longitudinal_redshift.py`. The second- and third-order longitudinal generators are not solved.

Proposition 5 (Unified field equations)**.** *With $N = a_1 t e^{\nu}$ (Lemma 2), Einstein's equations are equivalent to*

$$(N) \quad 3\nu'' - \Delta_\gamma \nu - 2|D\nu|^2 + \operatorname{tr} S^2 + \tfrac{3}{t^2}(e^{-2\nu} - 1) = 0, \tag{1}$$

$$(T) \quad \mathrm{TF}\left[S' + 2(t^{-1} + \nu')S + R^k{}_j[\gamma] - 2D^k D_j \nu + 2D^k \nu D_j \nu\right] = 0, \tag{2}$$

$$(M) \quad D_j S^j{}_i - 2\partial_i \nu' + \tfrac{2}{t}\partial_i \nu + 2S^j{}_i \partial_j \nu + 2\nu' \partial_i \nu = 0, \tag{3}$$

$$(\mathrm{Ham}) \quad R - 4\Delta\nu - 2|D\nu|^2 - \operatorname{tr} S^2 + 6(t^{-1} + \nu')^2 - 6e^{-2\nu}/t^2 = 0. \tag{4}$$

In the mixed form (T), the $\sigma\sigma$ terms of the spatial equation cancel, because $\gamma^{ki}\sigma'_{ij} = (S^k{}_j)' + 2S^k{}_i S^i{}_j$. This is proved for generic γ (six free functions, t-independent determinant) and arbitrary N, on independent symbolic 2-jets.

4 Primordial state and initial data

Definition 6 (Gaussian variable; absence of independent higher-order modes)**.** The state is the solution of Prop. 5 with the following properties:

(G) $\zeta = \epsilon g$ exactly, with g a zero-mean, homogeneous, isotropic Gaussian field;

(T) $\gamma \to e^{2\zeta}\delta_{ij}$ as $t \to 0$. The limit is conformally flat in the marked conformally Cartesian material chart, so there is no tensor seed;

(V) $\mathrm{d}(Tu) = 0$, so there is no vorticity seed;

(R) ν and H are even and analytic in t at fixed momenta, with $\nu(0) = 0$ and $H(0) = 0$, so there are no decaying modes.

"No independent higher-order modes" means $\zeta^{(r\geq 2)} \equiv 0$ together with (T), (V) and (R) at every order.

Because $\det\gamma$ is t-independent, $\zeta = \frac{1}{6}\ln\det\gamma$ at *every* time. As $t \to 0$, constant-t slices become uniform-density slices, so ζ is the nonlinear uniform-density (δN) curvature perturbation.

The residual conformal relabellings are fixed as follows: the dilation by $\langle g\rangle = 0$; special conformal transformations by statistical homogeneity; translations and rotations are statistical symmetries.

Exactly in ζ, the gradient-expansion data are

$$\nu = \tfrac{t^2}{18}e^{-2\zeta}\big(2\nabla^2\zeta + |\nabla\zeta|^2\big) + O(t^4), \quad H_{ij} = \tfrac{t^2}{3}e^{-2\zeta}\ \mathrm{TF}\big[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\big] + O(t^4). \tag{5}$$

The initial data at any t_i are the regular solution of §5 evaluated at t_i, with $\rho = 3a_1^2/(8\pi G N^4)$ and $u = N^{-1}\partial_t$. Higher-order data are nonzero. For example, $\gamma^{(2)}(0) = 4g^2\delta$, and the $\nu^{(2)}$ kernel is $\frac{t^2}{9}[2(k_1^2 + k_2^2) - k_1\cdot k_2] + O(t^4)$. The forced $K = 0$ mode ($5k^2t^2/9$ at $k_2 = -k_1$) is kept. The data evolve back into the same regular solution, so t_i drops out identically.

5 Field solution at every order

5.1 Triangular scheme

At total momentum K, the order-r kernels (ν_r, A_r), with A_r the order-r part of H, solve $\mathcal{L}_K[\nu_r, A_r] + c^{(r)} = 0$. The operators are the same at every order:

$$L_N = 3\nu'' + K^2\nu - \tfrac{6}{t^2}\nu, \qquad (L_M)_i = \tfrac{i}{2}K_j A'_{ji} - 2iK_i(\nu' - \nu/t),$$
$$L_T = \tfrac{1}{2}(A'' + \tfrac{2}{t}A') + \mathrm{TF}\, R_1[A] + 2\,\mathrm{TF}(K_iK_j)\nu.$$

Here $R_1[A]_{ij} = \frac{1}{2}(K^2A_{ij} - K_aK_iA_{aj} - K_aK_jA_{ai})$. The source $c^{(r)}$ is the multilinear coefficient with $(\nu_r, A_r) = 0$; it is explicit in the lower orders.

Write $c = K/\sqrt{3}$, $x = ct$, $u_1 = \sin x/x - \cos x$ and $u_2 = -\cos x/x - \sin x$. The coefficient c_r is the t^2 coefficient of (5) at order r. In particular $c_2 = [2(k_1^2 + k_2^2) - k_1 \cdot k_2]/9$ and $c_3 = \frac{2}{9}[-2\sum k_a^2 + \sum_{a<b} k_a \cdot k_b]$. Then

$$\nu_r = \frac{9c_r}{K^2}u_1(x) - \frac{1}{3c}\Big[u_2(x)\int_0^t u_1(cs)c_N^{(r)}\mathrm{d}s - u_1(x)\int_0^t u_2(cs)c_N^{(r)}\mathrm{d}s\Big], \tag{6}$$

$$h_S = \tfrac{3}{2K^2}\big[4K^2\nu_r + 12\nu_r'/t + 12\nu_r/t^2 + c_{\mathrm{Ham}}^{(r)}\big] \quad \text{(algebraic, from (Ham))}, \tag{7}$$

$$w = \tfrac{2i}{K^2}\int_0^t P_\perp c_M^{(r)}\mathrm{d}s, \qquad h = -2\int_0^t \tfrac{s}{t}\tfrac{\sin K(t-s)}{K}\,\mathrm{TT}[c_T^{(r)}](s)\,\mathrm{d}s, \tag{8}$$

$$A_r = (\hat{K}\hat{K} - \tfrac{1}{3}\delta)h_S + (K \otimes w + w \otimes K) + h. \tag{9}$$

At $K = 0$ no projector appears: $A_r = -2\int_0^t \frac{s}{t}(t-s)c_T^{(r)}(s)\mathrm{d}s$ and $\nu_r = c_r t^2 - \frac{1}{3}\int_0^t \frac{t^3 - s^3}{3ts}c_N^{(r)}\mathrm{d}s$. The lapse particular part is $O(t^4)$; the spatial solution generally starts at $O(t^2)$. The free regular lapse coefficient c_r is fixed by the leading, $O(t^0)$ Hamiltonian constraint, equivalently by the $O(t^2)$ lapse data in (5).

Theorem 7 (Consistency at every order)**.** *The triangular solution satisfies all ten field equations at every order.*

Proof. Let $\mathbb{E}_{\mu\nu} = G_{\mu\nu} - 8\pi G T_{\mu\nu}$. Matter conservation is exact (Lemma 1), so $\nabla_\mu \mathbb{E}^\mu{}_\nu = 0$. Inductively assume that all residuals vanish below order r. At order r this is the background Bianchi identity, with $\mathcal{H} = 1/t$. Set $f = \mathbb{E}^t{}_t$, $v_i = \mathbb{E}^t{}_i$, and $B^j{}_i = \mathbb{E}^j{}_i$, suppressing the order label. Equation (N), which is solved by construction, gives $f = \frac{1}{2}\mathbb{E}$, hence $\mathrm{tr}\, B = f$. The two Bianchi identities are therefore (at this order B is symmetric, since all lower residuals vanish)

$$f' + \frac{2}{t}f - iK_iv_i = 0, \qquad v_i' + \frac{4}{t}v_i + iK_jB^j{}_i = 0. \tag{10}$$

For $K \neq 0$, the scalar spatial solution imposes (Ham), so $f = 0$; the vector solution imposes $P_\perp v = 0$. The first identity in (10) now gives $K \cdot v = 0$, and thus the *longitudinal* momentum constraint follows rather than being assumed. Consequently $v = 0$, $\mathrm{tr}\, B = 0$, and the second identity gives $K_jB^j{}_i = 0$. The tensor solution imposes $\mathrm{TT}[B] = 0$. A symmetric tensor that is trace-free and transverse is itself TT, so $B = 0$.

For $K = 0$, the full trace-free spatial equation is solved, and (10) gives $v_i = b_i/t^4$ and $f = b/t^2$. The regular field series has $(M)_i = O(t)$, hence $v_i = O(t^{-1})$; this excludes $b_i \neq 0$. It does *not* exclude $b \neq 0$. Here the initial Hamiltonian constraint is essential: the coefficient c_r from (5) cancels the $O(t^0)$ term of (Ham). Since $\mathbb{E}^t{}_t = -(\mathrm{Ham})/(2N^2)$ and $N \sim a_1 t$, this gives $\lim_{t\to 0} t^2 f = 0$, hence $b = 0$. Finally $\mathrm{tr}\, B = f = 0$ and $\mathrm{TF}\, B = 0$ give $B = 0$. This completes the induction. □

Proposition 8 (Projector-free Cartesian solution)**.** *Define, on symmetric trace-free matrices,* $\mathfrak{B}_K A = \mathrm{TF}[K^2 A - K \otimes (AK) - (AK) \otimes K]$*. Its eigenvalues are* K^2 *on TT tensors,* 0 *on vector tensors* $K \otimes w + w \otimes K$ *(*$w \perp K$*), and* $-K^2/3$ *on* $\mathrm{TF}(K \otimes K)$*. The trace-free spatial equation at order* r *is*

$$A_r'' + \tfrac{2}{t} A_r' + \mathfrak{B}_K A_r = -4\,\mathrm{TF}(K \otimes K)\,\nu_r - 2c_T^{(r)},$$

and its unique regular solution with $A_r(0) = 0$ *is*

$$\boxed{\begin{aligned} A_r(t) &= \int_0^t \mathfrak{G}_K(t,s)\big[-4\,\mathrm{TF}(K \otimes K)\nu_r(s) - 2c_T^{(r)}(s)\big]\mathrm{d}s, \\ \mathfrak{G}_K(t,s) &= \frac{s}{t}\sum_{n\geq 0} \frac{(-1)^n (t-s)^{2n+1}}{(2n+1)!}\mathfrak{B}_K^n. \end{aligned}}$$

$\mathfrak{G}_K$ *is an entire function of the Cartesian components of* K*, with no projectors. Hence, by induction on* r *(the sources and* ν_r *are analytic in the momenta),* A_r *is analytic in* K *at fixed internal momenta, including at* $K = 0$*. There it reduces to the exact* $K = 0$ *kernel* $-2\int_0^t \frac{s}{t}(t-s)c_T^{(r)}\mathrm{d}s$*.*

Proof. The equation is $L_T + c_T = 0$ with $\mathrm{TF}\, R_1[A] = \frac{1}{2}\mathfrak{B}_K A$. In each eigen-sector with eigenvalue λ, the homogeneous solutions are $\sin(\sqrt{\lambda}t)/t$ and $\cos(\sqrt{\lambda}t)/t$. For $\lambda = 0$ they are 1 and $1/t$. None of these is both regular and zero at $t = 0$, so the regular solution with $A_r(0) = 0$ is unique. It is therefore the triangular solution, which satisfies the full trace-free equation (Thm. 7) and has $H(0) = 0$. `solution/check_cartesian_green.py` verifies the eigenvalues for generic K. It also verifies that the regular series of the Cartesian equation, whose coefficients are polynomial in the Cartesian K, equals the projector-based kernel coefficient by coefficient through $O(t^7)$. That holds for the pairs (k, q) and $(k, -q)$, for the zero-momentum pair $(q, -q)$, and for the third-order kernel at $(k, q, -q)$. So no inverse power of $|K|$ survives. □

The projector-based triangular scheme remains the efficient way to *evaluate* the kernels, especially at large Kt, where the scalar sector's sinh behaviour makes the Cartesian series cancel strongly. The Cartesian form is the clean statement of their *regularity*.

Remark 9 (Why initial constraint matching cannot be replaced by regularity). At linear order and $K = 0$, take $\nu = \beta t^2$, $H = 0$, $\zeta = 0$ with arbitrary constant β. Then (N), (M) and (T) hold, but (Ham) $= 36\beta$. The metric variables are even and analytic. This mode is excluded by (5), not by analyticity alone.

5.2 Compact field-source representation

The large component expressions in the ancillary files need not be the working form of the answer. Keep the matrix products and the scalar/vector/tensor projections factored. In particular, put $M = e^H$, $\chi = \zeta + 2\nu$, and let ∇_M and $\mathrm{Ric}[M]$ refer to the spatial metric M. The conformal transformation of spatial Ricci curvature and of the lapse Hessian gives the exact identity

$$\begin{aligned} \mathcal{F} &\equiv \mathrm{TF}\left[R^i{}_j[\gamma] - 2D^i D_j \nu + 2D^i \nu D_j \nu\right] \\ &= e^{-2\zeta}\,\mathrm{TF}\big\{M^{-1}\left[\mathrm{Ric}[M] - \nabla_M \mathrm{d}\chi + \mathrm{d}\chi \otimes \mathrm{d}\chi - 2\mathrm{d}\nu \otimes \mathrm{d}\nu\right]\big\}. \end{aligned} \tag{11}$$

Here $\nabla_M \mathrm{d}\chi$ is the covariant Hessian. Thus (T) is simply $S' + 2(t^{-1} + \nu')S + \mathcal{F} = 0$. Its matrix shear can be generated through cubic order as

$$S = \frac{1}{2}H' + \frac{1}{4}[H', H] + \frac{1}{12}[[H', H], H] + O(\epsilon^4). \tag{12}$$

These commutators must not be dropped: matrices at different orders or momenta need not commute. Together with (N), (M), (Ham) and the Green operators above, these identities give a compact source prescription at arbitrary momenta: retain lower-order fields, set the unknown top-order kernels to zero, and extract the multilinear coefficient. No momenta, endpoint terms or tensor sectors are discarded. The ancillary files contain the expanded coefficients of this factored representation. This is a structural simplification, not a claim that the remaining finite-time integrals collapse to a smaller set of evaluated scalar functions.

5.3 Explicit kernels

First order [closed form].

$$\nu_1 = -(\sin x/x - \cos x), \qquad H_1 = -\tfrac{6}{k^2}\big(\tfrac{2\sin x}{x} - \cos x - 1\big)\big(k_i k_j - \tfrac{1}{3}\delta_{ij}k^2\big), \qquad \zeta_1 = 1.$$

Second order [closed form, general momenta]. Every integral above is of the form $\int_0^t B(s)\mathrm{d}s$, where $B = \sum_\omega e^{i\omega s}R_\omega(s)$ and each R_ω is a Laurent polynomial. The frequencies ω are integer combinations of $p/\sqrt{3}$, $q/\sqrt{3}$, $K/\sqrt{3}$ and K (with $p = |k_1|$, $q = |k_2|$, $K = |k_1 + k_2|$); they vanish on resonance surfaces. We integrate term by term in *entire* functions:

$$\int_0^t s^m e^{i\omega s}\mathrm{d}s = t^{m+1}\mathfrak{e}_m(i\omega t), \qquad \mathfrak{e}_m(z) = \int_0^1 u^m e^{zu}\mathrm{d}u \qquad (m \geq 0), \tag{13}$$

$$\int_0^t s^{-M}\Big[e^{i\omega s} - \sum_{j<M}\tfrac{(i\omega s)^j}{j!}\Big]\mathrm{d}s = J_M(\omega, t), \quad J_1 = -\operatorname{Cin}(\omega t) + i\operatorname{Si}(\omega t), \tag{14}$$

$$J_M = \tfrac{t^{1-M}E_M}{1-M} + \tfrac{i\omega}{M-1}J_{M-1}, \qquad E_M = e^{i\omega t} - \sum_{j<M}\tfrac{(i\omega t)^j}{j!}. \tag{15}$$

Here $\operatorname{Cin}(x) = \int_0^x (1 - \cos u)u^{-1}\mathrm{d}u$ is entire and even, and Si is entire and odd. The subtracted Taylor pieces sum to the principal part of B at $s = 0$, which vanishes (this is asserted in the code). *No division by any frequency occurs*, so the kernels are manifestly finite on every resonance surface:

- the scalar collinear surfaces $\mu = \pm 1$;
- the tensor sound cone $\mu = -1/3$ (for $|k_1| = |k_2|$). On this shell the second-order vector response vanishes identically (Prop. 10), so the resonance belongs to the tensor sector alone.

The removable $\ln|\alpha|$ terms that Ci would introduce never appear. The complete expressions for ν_2, h_S, w and h are in `solution/results/second_order_closed_forms.txt`. Each antiderivative is verified by exact differentiation. Substituting the assembled ν_2 back into the lapse equation (N) gives an exact zero in closed form. The scalar part h_S satisfies (Ham) by construction.

The momentum and trace-free equations follow from the Green identities and Thm. 7. The scope of the independent closed-form and finite-series substitution checks is recorded in App. E.

Proposition 10 (Two-amplitude form on the shell)**.** *For any pair (k_1, k_2), reflection through the plane of the two momenta leaves a single TT polarisation. If moreover $|k_1| = |k_2|$ and $K = k_1 + k_2 \neq 0$, then $P_\perp c_M^{(2)} = 0$ identically in t, so the second-order vector response vanishes. With $u = \hat{K}$, $v = (k_1 - k_2)/|k_1 - k_2|$ (orthogonal to u on the shell) and $w = u \times v$,*

$$\boxed{A_{2,ij} = F_S(t, k_0, \mu)\Big(u_i u_j - \tfrac{1}{3}\delta_{ij}\Big) + F_T(t, k_0, \mu)\,(v_i v_j - w_i w_j), \qquad -1 < \mu < 1,} \tag{16}$$

with $F_S = h_S$ and F_T the single TT amplitude. The zero-total-momentum pair ($\mu = -1$) is the separately constructed $K = 0$ kernel, and the collinear pair ($\mu = 1$) has no v.

Proof. A parity-even quadratic vector source has the form $c_M = A\,k_1 + B\,k_2$, with $A(k_1,k_2) = B(k_2,k_1)$ by exchange symmetry. On the shell the scalar invariants are exchange-symmetric, so $A = B$ and $c_M \parallel K$. With no vector seed, $w = \frac{2i}{K^2}\int P_\perp c_M = 0$. For the tensor, the reflection $w \to -w$ removes the $vw + wv$ polarisation, and exchange ($v \to -v$) is compatible with $vv - ww$. `solution/check_shell_reduction.py` verifies $c_{M,z} = c_{T,xz} = c_{T,yz} = 0$ for general momenta, and $K \times c_M = 0$ on the shell identically in t and μ, with an off-shell negative control. □

This is an exact finite-time reduction. On the shell, the second-order metric is two scalar functions of (t,μ) instead of a component matrix.

The tensor amplitude is the known radiation-era kernel plus a gauge correction. With $x = k_0 t/\sqrt{3}$, $s(x) = 2\sin x/x - \cos x$, $K = k_0\sqrt{2(1+\mu)}$, $r = k_0/K$ and $X = Kt$,

$$\boxed{F_T(t,k_0,\mu) = \frac{1-\mu}{2(1+\mu)}\,I_{\mathrm{RD}}(r,r,X) + \frac{3}{2}(1-\mu)\Big[3\mu\big(s(x)-1\big)^2 - s'(x)^2\Big].} \tag{17}$$

Here I_{RD} is the unaveraged finite-time radiation-era induced-tensor kernel of Kohri and Terada [15], and the bracket is an elementary gauge correction. The correction arises because H is the matrix logarithm of the conformally rescaled spatial metric in temperature-frame coordinates, not the Newtonian metric tensor. For the first-order coordinate change $\eta = t + \alpha$, $y^i = X^i + \partial^i\beta$ (with $\beta = \frac{3}{k_0^2}[s(x)-1]g$ and $\alpha = \dot\beta$), the mixed scalar–Hessian terms cancel only after all three steps: the pullback, division by the conformal factor, and the matrix logarithm. What remains is $H^{\mathrm{TT}}_{[2]} = \frac{1}{2}h^{\mathrm{KT}} - \Lambda(\beta_{,ac}\beta_{,bc} + \alpha_{,a}\alpha_{,b})$. In entire functions, with $d = 2r^2 - 3$ and $a_\pm = 1 \pm 2r/\sqrt{3}$,

$$I_{\mathrm{RD}}(r,r,X) = \frac{3}{4r^6 X}\Big\{\mathcal{E} + d^2\big[\sin X\,\mathcal{C} + \cos X\,\mathcal{S}\big]\Big\},$$

with $\mathcal{C} = \mathrm{Cin}(a_-X) + \mathrm{Cin}(a_+X) - 2\,\mathrm{Cin}(X)$, $\mathcal{S} = \mathrm{Si}(a_-X) + \mathrm{Si}(a_+X) - 2\,\mathrm{Si}(X)$, and

$$\mathcal{E} = -4r^2 d\sin X + \frac{24r^2\cos^2 x}{X} - \frac{48\sqrt{3}\,r\cos x\sin x}{X^2} + \frac{12(6+dX^2)\sin^2 x}{X^3}.$$

At the resonance $\mu = -\frac{1}{3}$ one has $a_- = 0$, and $\mathrm{Cin}(0) = \mathrm{Si}(0) = 0$.

The proof has three parts.

1. *The equation.* (17) satisfies the same complete finite-time equation $F_T'' + \frac{2}{t}F_T' + K^2 F_T = \mathcal{S}_T$ as our solution, for symbolic t and μ.

2. *The initial data.* It has the same regular initial data, $F_T = k_0^2(1-\mu)t^2/6 + O(t^4)$, with no constant and no decaying tensor.

3. *Uniqueness.* The difference of the two solves the homogeneous equation with zero regular data, so it vanishes identically.

The source $\mathcal{S}_T$ is rebuilt independently, and the TT part of this repository's own c_T equals $-\frac{1}{2}\mathcal{S}_T(vv - ww)$ identically. The derivation and its convention dictionary, which covers KT's $h/2$ metric normalisation, Fourier measure and polarisation norm, are in `report/tensor_newtonian_crosscheck.md`. The checks are `solution/check_tensor_newtonian.py` and `solution/check_shell_reduction.py`. This validates the analytical tensor solution, not every stored coefficient of the expanded kernel file. At $\mu = -1$ the forced homogeneous $K = 0$ kernel is kept separately.

Third order [explicit nested integrals]. For $C^{(13)}$ only the configuration $(k, q, -q)$ is needed. The source $c^{(3)}$ is explicit (`solution/results/third_order_sources.txt`) in the first-order closed forms and in $K_2(k, q)$, $K_2(k, -q)$ and $K_2(q, -q)$. The last of these is the exact $K = 0$ kernel, never a limit. The third-order kernels are the formulas above applied to $c^{(3)}$, so they are double integrals with closed-form inner integrals. Three levels should be distinguished:

- the general cubic response at arbitrary (k_1, k_2, k_3), which is *defined* constructively by the triangular scheme and the multilinear rule (Thm. 7 covers it);
- the configuration $(k, q, -q)$, which is the only one generated explicitly, because it is the only one $C^{(13)}$ needs;
- the checks of that configuration: an exact series test of all field equations through $O(t^6)$, and a finite-k numerical Ward test (App. F). Neither is a substitution of the assembled cubic kernel at arbitrary times.

5.3.1 Nine-moment reduction of the contracted cubic fields

These field-level shell reductions are used only in the monochromatic spectrum of §8.4, after the complete observable of §6; nothing in that general construction relies on them. For the 13 contraction, the intermediate field equations admit a stronger reduction than constructing the component kernel A_3 and subsequently averaging it. Put

$$k = k_0, \quad c = k/\sqrt{3}, \quad x = ct, \qquad \mathsf{a} = \cos x - \frac{\sin x}{x}, \qquad \mathsf{b} = -6\left(\frac{2\sin x}{x} - \cos x - 1\right).$$

Here a, b are dimensionless first-order profiles, not the background scale factor. For a shell mode p, $\nu_p = \mathsf{a}$ and $H_p = \mathsf{b}(\hat{p}\hat{p} - \delta/3)$, and exactly

$$L_k \mathsf{a} = 0, \qquad L_k = 3\partial_t^2 + k^2 - 6/t^2, \qquad \mathsf{b}' = 6(\mathsf{a}' - \mathsf{a}/t). \tag{18}$$

Write $n_2(t, \mu) = \nu_2(k, q)$, with F_S, F_T the amplitudes of (16), and $n_h(t) = \nu_2(q, -q)$ for the exact homogeneous lapse. Define

$$\langle f \rangle_\mu = \tfrac{1}{2}\int_{-1}^{1} f(\mu)\mathrm{d}\mu, \qquad q_S = (1 + 3\mu)/6, \quad q_T = (1 - \mu)/2,$$

$$N_r = \langle \mu^r n_2 \rangle_\mu, \qquad S_r = \langle \mu^r F_S \rangle_\mu, \qquad T_r = \langle \mu^r F_T \rangle_\mu, \qquad r = 0, 1, 2.$$

The moment subscripts here are angular powers, not perturbative orders.

Proposition 11 (Reduced cubic sources)**.** *On the shell, the longitudinal cubic momentum source has no contribution made solely of three first-order fields, pointwise in μ and at arbitrary finite time. The homogeneous second-order tensor cancels from both angle-averaged cubic scalar sources; n_h does not. Those sources require only the nine moments above, their first time derivatives and n_h:*

$$\bar{c}_N = B_0 + \frac{12\mathsf{a}}{t^2} n_h + 2\left\langle D_0 n_2 + \mathsf{a}k^2\left(\frac{3+\mu}{6}F_S - q_T F_T\right) + \frac{\mathsf{b}'}{2}(q_S F_S' + q_T F_T')\right\rangle_\mu, \tag{19}$$

$$\bar{m} \equiv \overline{ik \cdot c_M} = -2\mathsf{a}k^2 n_h' + 2k^2\left\langle \alpha_S F_S' + \alpha_T F_T' + 2\mu\mathsf{a}n_2' + \beta_S F_S + \beta_T F_T + \beta_n n_2\right\rangle_\mu, \tag{20}$$

where

$$B_0 = \mathsf{a}k^2(16 + 8\mathsf{a} + 2\mathsf{b}^2/9) - 24\mathsf{a}^3/t^2, \qquad D_0 = 12\mathsf{a}/t^2 - k^2(1+\mu)[5 + 4\mathsf{a} + \mathsf{b}(\mu - 1/3)],$$

$$\begin{aligned} \alpha_S &= (3/2 + \mathsf{a})(3+\mu)/6 - \mu\mathsf{b}q_S/4, & \alpha_T &= -(3/2+\mathsf{a})q_T - \mu\mathsf{b}q_T/4, \\ \beta_S &= (1+\mu)\mathsf{b}'q_S/4, & \beta_T &= (1+\mu)\mathsf{b}'q_T/4, \\ \beta_n &= -(1+\mu)[\mathsf{b}'(\mu - 1/3) + 2\mathsf{a}']. \end{aligned}$$

The factor two counts the pairs (k, q) and $(k, -q)$ after $\mu \mapsto -\mu$; it is not an additional Wick or factorial normalisation.

Proof. For $M = e^H$, let $C_i = M^{-1}\partial_i M$. Since $\det M = 1$ and $\operatorname{tr} S = 0$,

$$D_j S^j{}_i = \partial_j S^j{}_i + 3\zeta_{,j} S^j{}_i - \tfrac{1}{2}\operatorname{tr}(C_i S).$$

On the shell $H_1 = \mathsf{b}(t)B(X)$, so $[H_1, H_1'] = 0$. The quadratic mixed-shear commutator and the purely first-order cubic double commutator vanish. The remaining cubic first–second-order commutator is antisymmetric, so its contraction with $k_i k_j$ vanishes. The quadratic commutator in C_i has zero trace against the symmetric first-order shear. This removes every direct 111 term in the longitudinal momentum source, without assuming $[H_1, H_2] = 0$. For the homogeneous tensor, the scalar contractions are proportional to $\mu^2 - 1/3$, whose mean is zero. The remaining contractions give (19)–(20). All weights have degree at most two. The check `solution/check_shell_cubic_sources.py--compare-engine` rebuilds these identities for arbitrary time profiles and arbitrary pair amplitudes, and compares them with the original Christoffel-based field engine at arbitrary finite time. It uses no time expansion or quadrature. □

Derivative-free lapse normal form. Set

$$\Upsilon = \mathsf{a}\langle q_S F_S + q_T F_T\rangle_\mu, \qquad \eta_3 = \bar\nu_3 + \Upsilon, \qquad L_k\eta_3 = -\bar r_N. \tag{21}$$

Then

$$\begin{aligned} \bar r_N &= B_R + \frac{12\mathsf{a}}{t^2} n_h \\ &\quad + 2\left\langle D_R n_2 + \frac{\mathsf{a}k^2}{6}(2 - 3\mu - 3\mu^2)F_S + \frac{\mathsf{a}k^2}{2}(2 + \mu - 3\mu^2)F_T \right\rangle_\mu , \\ D_R &= 12\mathsf{a}/t^2 - k^2(1+\mu)[5 + 2\mathsf{a} - 6\mathsf{a}\mu + \mathsf{b}(\mu - 1/3)], \\ B_R &= k^2(8\mathsf{a} - 8\mathsf{a}^2 - \tfrac{8}{3}\mathsf{a}^2\mathsf{b} + \tfrac{4}{9}\mathsf{a}\mathsf{b}^2) + 4\mathsf{a}\mathsf{a}'\mathsf{b}' - 24\mathsf{a}^3/t^2. \end{aligned} \tag{22}$$

There are no derivatives of any second-order response in this source. Since $\Upsilon = O(t^4)$, the constrained regular t^2 coefficient is unchanged. The second-order equations used to obtain this normal form are

$$L_K n_2 = -C_N, \quad F_S'' + 2F_S'/t - K^2 F_S/3 = -4K^2 n_2 + J_S, \quad F_T'' + 2F_T'/t + K^2 F_T = J_T, \tag{23}$$

with $K^2 = 2k^2(1+\mu)$ and the complete elementary sources

$$C_N = k^2[\tfrac{2}{3}\mathsf{a}\mathsf{b} - 4\mathsf{a} + \mu(4\mathsf{a}^2 - \tfrac{4}{3}\mathsf{a}\mathsf{b} + 2\mathsf{a}) - 2\mu^2\mathsf{a}\mathsf{b}] + \tfrac{1}{2}\mathsf{b}'^2(\mu^2 - 1/3) + 12\mathsf{a}^2/t^2, \tag{24}$$

$$\begin{aligned} J_S = k^2\big[&2(3+\mu)\mathsf{a}^2 + (15\mu^2 + 13\mu - 8)\mathsf{a}\mathsf{b}/3 + 16(1+\mu)\mathsf{a} \\ &+ (5 - 5\mu - 15\mu^2 - 9\mu^3)\mathsf{b}^2/36 + (-10 + 7\mu + 15\mu^2)\mathsf{b}/6 + 5 + 7\mu\big] - (1+3\mu)\mathsf{a}'\mathsf{b}', \end{aligned} \tag{25}$$

$$J_T = \frac{1-\mu}{36}\big\{k^2[\mathsf{b}^2(9\mu^2 + 12\mu + 1) - 6\mathsf{b}(6\mu\mathsf{a} + 3\mu + 2) - 72\mathsf{a}^2 + 36] - 36\mathsf{a}'\mathsf{b}'\big\}. \tag{26}$$

In particular

$$\langle q_S J_S + q_T J_T\rangle_\mu = \tfrac{2}{27}[k^2(12\mathsf{ab} + 72\mathsf{a} - \mathsf{b}^2 + 36) - 18\mathsf{a}'\mathsf{b}'],$$

which evaluates the elementary forcing part in B_R.

The derivative cancellation also holds at general momenta. For a partition $p|(q,r)$, put $P = q+r$, $K = p + P$, $Q_p = pp - \delta p^2/3$, $J = H_{qr}$ and $\phi_{p|qr} = \nu_p Q_p : J/(2p^2)$. The metric-dependent source $-\nu_p K_i J_{ij} p_j + \frac{1}{2} H'_p : J'$ equals $L_K \phi_{p|qr}$ plus

$$-\nu_p K_i J_{ij} p_j - \frac{\nu_p}{2p^2}[(K^2 - p^2)Q_p : J + 3Q_p : (J'' + 2J'/t)].$$

Use the second-order spatial equation for the acceleration and sum the three partitions. There is then no derivative of J. At $p = 0$ use the regular continuation of ν_p/p^2, since the regular scalar has $\nu_p = O(p^2)$ at fixed time. The change of variable must be undone when reconstructing the observable.

An acoustic scalar representation. Use the entire functions

$$\mathsf{T}_j(a,d) = \sum_{m\ge 0} \frac{(-a)^m d^{2m+j}}{(2m+j)!}.$$

The momentum constraint gives the scalar amplitude directly, without the hyperbolic scalar Green function of the unconstrained Cartesian evolution.

Proposition 12 (Acoustic form of the shell scalar amplitude)**.** *On the shell, $c_M \parallel K$ and the longitudinal source is the polynomial*

$$\mathfrak{m} \equiv \frac{3iK\cdot c_M}{K^2} = \mathsf{b}'\Big[(1-3\mu)\big(\tfrac{3}{2} + \mathsf{a}\big) + \tfrac{3\mu^2-1}{4}\mathsf{b}\Big] - 6\mathsf{a}\mathsf{a}'.$$

With $a_K = K^2/3 = 2k^2(1+\mu)/3$, $c_r = k^2(4-\mu)/9$ and C_N from (24),

$$\boxed{\begin{aligned} F_S(t,\mu) = 18c_r\Big[\mathsf{T}_2(a_K,t) - \frac{2\mathsf{T}_3(a_K,t)}{t}\Big] - \int_0^t \mathcal{G}(t,r)C_N(r,\mu)\mathrm{d}r \\ + (1-3\mu)\big[\tfrac{3}{2}\mathsf{b} + 3\mathsf{a}^2 - 6\mathcal{I}\big] + \tfrac{3\mu^2-1}{8}\mathsf{b}^2 - 3\mathsf{a}^2, \end{aligned}} \tag{27}$$

$$\mathcal{G}(t,r) = 2\mathsf{T}_1(a_K,d) + \frac{2[(d-r)\mathsf{T}_2(a_K,d) - 2\mathsf{T}_3(a_K,d)]}{tr}, \qquad \mathcal{I} = \tfrac{1}{2}\,\mathrm{Cin}(2x) + \frac{\sin 2x}{2x} - \frac{\sin^2 x}{2x^2} - \frac{1}{2},$$

with $d = t - r$, $x = k_0 t/\sqrt{3}$ and $\mathcal{I} = \int_0^t \mathsf{a}^2/s\,\mathrm{d}s$. In addition $F'_S = 6(n'_2 - n_2/t) + \mathfrak{m}$ is local in n_2.

Proof. Write $v_K = 9u_1(Kt/\sqrt{3})/K^2$ and let G_N^+ be the retarded inverse of L_N, with sign opposite to the source kernel in the triangular scheme. Insert $n_2 = c_r v_K - \int G_N^+ C_N$ into h_S of Prop. 35. The inner integral is elementary,

$$\int_r^t G_N^+(s,r)s^{-1}\mathrm{d}s = \frac{[\mathsf{T}_3 + r\mathsf{T}_2](a_K, t-r)}{3tr}, \qquad 6v_K - 6\int_0^t \frac{v_K}{s}\mathrm{d}s = 18\Big[\mathsf{T}_2 - \frac{2\mathsf{T}_3}{t}\Big](a_K,t).$$

Then $\int_0^t \mathfrak{m}$ is integrated using $\mathsf{b}' = 6(\mathsf{a}' - \mathsf{a}/t)$. □

Every term in (27) is entire in μ; its Green frequencies are acoustic, not hyperbolic. The scalar moments S_r need one time integration and single-variable entire angular kernels of the type (89), plus elementary moments of the polynomial part. The alternative Cartesian construction, with its nested hyperbolic kernel before constraint cancellation, is retained in App. B.

The nine relative-angle moments have explicit entire-function representations in App. B. Their time integrals remain.

6 The observable through third order

Proposition 13 (Tetrad)**.** *Fermi–Walker transport along u is parallel transport along ∂_t in $\tilde{g}$, that is, $\mathrm{d}E_a{}^i/\mathrm{d}t = -S^i{}_j E_a{}^j$. With $E(t_I) = \gamma(t_I)^{-1/2}$, the transported frame satisfies $E = \gamma_o^{-1/2} R$ at observation, where $R = 1 + \epsilon^2\Omega/2 + O(\epsilon^3)$ and Ω is antisymmetric. Consequently:*

- *$\Theta^{(1)}$ and $\Theta^{(2)}$ are those of the symmetric frame;*
- *the specified Fermi–Walker map is completed by*

$$\Theta^{(3)}_{\mathrm{FW}} = \Theta^{(3)}_{\mathrm{sym}} + 3(\Omega\hat{n})\cdot\nabla_{S^2}\Theta^{(1)}, \qquad \Omega = \frac{1}{4}\int_{\bar{t}_I}^{\bar{t}_o} [H_1(t,0), H_1'(t,0)]\,\mathrm{d}t; \tag{28}$$

- *$C_\ell^{(22)}$ and $\mathrm{Re}\, C_\ell^{(13)}$ are individually unchanged.*

Proof. Let $F = \gamma^{-1/2} = e^{-\zeta}e^{-H/2}$. From $E = FR$ and $E' = -SE$, $R' = -F^{-1}(SF + F')R$. Expanding the matrix exponentials gives $R' = \epsilon^2[H_1, H_1']/8 + O(\epsilon^3)$, with $R(t_I) = 1$. This proves the integral in (28); replacing its endpoints by their background values changes R only at $O(\epsilon^3)$, beyond the map order at issue. Since $\Theta_{\mathrm{FW}}(\hat{n}) = \Theta_{\mathrm{sym}}(R\hat{n})$, the Taylor convention gives the factor 3. The rotation acts on $a^{(1)}_{\ell m}$ by an anti-Hermitian generator, so $\mathrm{Re}\sum_m a^{(1)*}_{\ell m}(D(\Omega)a^{(1)})_{\ell m} = 0$ in every realisation. □

For a monochromatic seed, $H_1(t,0) = f(t)B$ with a time-independent matrix B, and hence $[H_1, H_1'] = 0$. In that specialisation even the maps agree through third order. The following ray and map formulas use the symmetric frame; (28) completes the map for general spectra after the clock composition.

Take the background ray $\bar{X} = \hat{n}(t_o - t)$, $\bar{e} = -\hat{n}$, and write $X = \bar{X} + \epsilon X_1 + \epsilon^2 X_2/2$. All fields are evaluated on $(t, \bar{X})$, and $\sigma(a,b) = \sigma_{ij}a^i b^j$. Because σ, S and Γ vanish on the background, X_3 and e_3 never enter. The exact consequences of Lemma 2 are as follows.

Integrand.

$$\begin{aligned} F_1 &= \sigma_{1,nn}, \qquad F_2 = \sigma_{2,nn} + 2X_1\cdot\partial\sigma_{1,nn} - 4\sigma_1(n,e_1),\\ F_3 &= \sigma_{3,nn} + 3X_1\cdot\partial\sigma_{2,nn} - 6\sigma_2(n,e_1) + 3X_2\cdot\partial\sigma_{1,nn} + 3(X_1\cdot\partial)^2\sigma_{1,nn}\\ &\quad - 12(X_1\cdot\partial)\sigma_1(n,e_1) - 6\sigma_1(n,e_2) + 6\sigma_1(e_1,e_1). \end{aligned}$$

Here ∂ acts only on the field coefficients σ_r, evaluated at the background point $(t,\bar{X})$; X_r and e_r are ray coefficients already computed, and are not differentiated. For example, in $(X_1\cdot\partial)\sigma_1(n,e_1)$ the derivative hits σ_1 alone. Equivalently, in the orthonormal directions of Prop. 3, replace $\sigma_r \to \mathcal{B}_r$ and $e_r \to v_r$. Then $F_2 = \frac{1}{2}H'_{2,nn} + X_1\cdot\partial H'_{1,nn} - 2H_1'(n,v_1)$ contains no local products.

Ray. In this paragraph only, A_r denotes ray acceleration, not the trace-free metric kernel used in §5. The displacements are

$$X_r = e_r(t_o)(t - t_o) + \int_t^{t_o}(t' - t)A_r(t')\mathrm{d}t',$$

with

$$\begin{aligned} A_1 &= 2S_1 n - \Gamma_1(n,n) - n\sigma_{1,nn},\\ A_2 &= [2S_2 n - \Gamma_2(n,n) - n\sigma_{2,nn}] + 2(X_1\cdot\partial)[2S_1 n - \Gamma_1(n,n) - n\sigma_{1,nn}]\\ &\quad + 2[-2S_1 e_1 + 2\Gamma_1(n,e_1) + e_1\sigma_{1,nn} + 2n\sigma_1(n,e_1)]. \end{aligned}$$

The initial directions are the Taylor coefficients of $e(t_o) = -e^{-\zeta}\exp(-H/2)\,\hat{n}$, evaluated at $(t_o, 0)$.

Emission and endpoint. Let φ_r denote ν along the ray, with $\varphi_1 = \nu_1$, $\varphi_2 = \nu_2 + 2X_1 \cdot \partial\nu_1$, and so on. Then

$$d_1 = -\bar{t}_\star \varphi_1, \qquad d_2 = \bar{t}_\star[u_1^2 - \varphi_2 - 2\varphi_1' d_1], \qquad u_1 = d_1/\bar{t}_\star,$$
$$I_1 = \int F_1, \qquad I_2 = \int F_2 - 2F_1 d_1, \qquad I_3 = \int F_3 - 3F_1 d_2 - 3F_2 d_1 - 3F_1' d_1^2.$$

All unlabelled integrals here run from $\bar{t}_\star$ to t_o; endpoint factors are evaluated at $\bar{t}_\star$. Primes on ray-evaluated quantities denote total derivatives along the background ray.

Normalisation in centred form. At fixed coordinate observation time, put $J_c = I - \langle I\rangle_\Omega$, with coefficients $J_{c,r} = I_r - \langle I_r\rangle_\Omega$, and define $\mathsf{P}_0 f = f - \langle f\rangle_\Omega$. A sky-independent shift of I cancels exactly, so

$$\begin{aligned}
\Theta_c &= \frac{e^{-J_c}}{\langle e^{-J_c}\rangle_\Omega} - 1, \\
\Theta_c^{(1)} &= -J_{c,1}, \\
\Theta_c^{(2)} &= -J_{c,2} + \mathsf{P}_0(J_{c,1}^2), \\
\Theta_c^{(3)} &= -J_{c,3} + 3\mathsf{P}_0(J_{c,1}J_{c,2}) - \mathsf{P}_0(J_{c,1}^3) + 3J_{c,1}\langle J_{c,1}^2\rangle_\Omega.
\end{aligned} \tag{29}$$

This is exactly the expanded normalisation formula, with all random-monopole products retained. The dipole is still present inside every product. It is removed only after the complete map is constructed; this final projection does not change $C_{\ell\geq 2}$ and is not a deboost.

Clock. With $\omega(t) = \nu(t, 0)$, define separately

$$W_1(T) = \int_0^T t\,\omega_1(t)\,\mathrm{d}t, \qquad W_2(T) = \int_0^T t\,[\omega_2(t) + \omega_1(t)^2]\,\mathrm{d}t.$$

The initial-time shifts d_{I1}, d_{I2} are the emission-root formulas with $\bar{t}_\star \to \bar{t}_I$ and $\varphi \to \omega$. The observation-time shifts are

$$\begin{aligned}
D_1 &= [\bar{t}_I d_{I1} - W_1(\bar{t}_o) + W_1(\bar{t}_I)]/\bar{t}_o, \\
D_2 &= \big[\bar{t}_I d_{I2} + d_{I1}^2 - W_2(\bar{t}_o) - 2\bar{t}_o\omega_1(\bar{t}_o)D_1 \\
&\qquad + W_2(\bar{t}_I) + 2\bar{t}_I\omega_1(\bar{t}_I)d_{I1} - D_1^2\big]/\bar{t}_o.
\end{aligned}$$

Compose with $t_o = \bar{t}_o + \epsilon D_1 + \epsilon^2 D_2/2 + \cdots$:

$$\begin{aligned}
\Theta_{\mathrm{sym}}^{(2)} &= \Theta_c^{(2)} + 2D_1\partial_{t_o}\Theta_c^{(1)}, \\
\Theta_{\mathrm{sym}}^{(3)} &= \Theta_c^{(3)} + 3D_2\partial_{t_o}\Theta_c^{(1)} + 3D_1\partial_{t_o}\Theta_c^{(2)} + 3D_1^2\partial_{t_o}^2\Theta_c^{(1)}.
\end{aligned}$$

All right-hand sides are evaluated at $t_o = \bar{t}_o$. Equivalently, compose I with this same random time first, centre the resulting exponent, and use (29) once. These are alternative procedures, not two clock corrections to be applied successively.

Cubic product identity, before any angular average. In orthonormal directions the full third-order integrand is

$$\begin{aligned} F_3 = {} & \tfrac{1}{2}H'_{3,nn} + \tfrac{1}{8}[H_1,[H_1,H'_1]]_{nn} + \tfrac{3}{2}X_1\cdot\partial H'_{2,nn} - 3H'_2(n,v_1) \\ & + \tfrac{3}{2}X_2\cdot\partial H'_{1,nn} + \tfrac{3}{2}X_1^iX_1^j\partial_i\partial_j H'_{1,nn} - 6(X_1\cdot\partial)H'_1(n,v_1) - 3H'_1(n,v_2) + 3H'_1(v_1,v_1). \end{aligned} \tag{30}$$

Here H'_r is a partial time derivative before pullback; spatial derivatives act on fields only. The direction coefficients are those of the *pulled-back* v, so the displaced frame is already included. Unit norm gives $n\cdot v_1 = 0$, $n\cdot v_2 = |v_1|^2$; thus the last two terms can also be written $-3H'_1(n,v_{2\perp}) + 3[H'_1(v_1,v_1) - H'_{1,nn}|v_1|^2]$. On the shell the double commutator is zero, but not at general momenta.

For a compact identity including both moving boundaries, set $e = \bar{t}_\star$, $o = \bar{t}_o$, $U_r(o) = \int_e^o F_r(t;o)\mathrm{d}t$, and let $\widehat{I}$ be the exponent after clock composition. Then

$$\boxed{\begin{aligned} \widehat{I}_1 &= U_1, \\ \widehat{I}_2 &= U_2 - 2[F_1d_1]_e + 2D_1\partial_oU_1, \\ \widehat{I}_3 &= U_3 - 3[F_1d_2 + F_2d_1 + F'_1d_1^2]_e \\ &\qquad + 3D_1\partial_o\{U_2 - 2[F_1d_1]_e\} + 3D_2\partial_oU_1 + 3D_1^2\partial_o^2U_1. \end{aligned}} \tag{31}$$

The boundary prime is the total derivative along the background ray. The ∂_o derivatives include the upper integration limit and every dependence on the background observation time, with e fixed; they do not differentiate the clock coefficients outside the braces. Centre $\widehat{I}$ and apply (29) once. In particular the mixed clock–emission product $-6D_1\partial_o[F_1d_1]_e$ must not be omitted. These identities hold at finite times and arbitrary momenta. They follow by expanding the exact orthonormal driver and composing the moving-lower-limit integral, and are checked in `solution/check_cubic_product_identity.py` with generic non-commuting matrices and independent boundary jets, including omission controls. They close the algebraic cubic product identity, not the harder problem of evaluating or cancelling its remaining time integrals. The Fermi–Walker term (28) is still added for a general spectrum.

Fourier dictionary. For a single mode a, with phase $e^{i\lambda k\cdot\hat{n}}$ and $\lambda = t_o - t$, $\mu_a = \hat{k}\cdot\hat{n}$, and $H_{a,ij} = Q_a(t)(k_ik_j - \delta_{ij}k^2/3)$, where $Q_a = -6[2\sin x/x - \cos x - 1]/k^2$:

- $\sigma_{a,nn} = \tfrac{1}{2}Q'_ak^2(\mu_a^2 - \tfrac{1}{3})$;
- $\Gamma_a(n,n) = i[2(k\cdot n)n - k] + \tfrac{i}{2}[2(k\cdot n)H_an - kH_{a,nn}]$;
- $e_a(t_o) = (1 + \tfrac{1}{2}H_a(t_o))\hat{n}$;
- $d_a = -\bar{t}_\star\nu_a(\bar{t}_\star)e^{i\lambda_\star k\cdot\hat{n}}$.

The second-order shear is $\sigma_{ab} = \tfrac{1}{2}H'_{ab} + \tfrac{1}{4}(H_aH_b + H_bH_a)' + H'_a + H'_b$, and

$$\begin{aligned} I_{ab} &= \int_{\bar{t}_\star}^{t_o}\mathrm{d}t\,\{\sigma_{ab,nn} + i(k_b\cdot X_a)\sigma_{b,nn} + i(k_a\cdot X_b)\sigma_{a,nn} - 2\sigma_b(n,e_a) - 2\sigma_a(n,e_b)\} - [F_ad_b + F_bd_a]_{\bar{t}_\star} \\ &= \int_{\bar{t}_\star}^{t_o}\mathrm{d}t\,\{\tfrac{1}{2}H'_{ab,nn} + i(k_b\cdot X_a)\sigma_{b,nn} + i(k_a\cdot X_b)\sigma_{a,nn} - H'_b(n,v_a) - H'_a(n,v_b)\} - [F_ad_b + F_bd_a]_{\bar{t}_\star}, \\ \Theta_{ab} &= I_aI_b - I_{ab} - \langle I_a\rangle_\Omega I_b - \langle I_b\rangle_\Omega I_a + 2\langle I_a\rangle_\Omega\langle I_b\rangle_\Omega - \langle I_aI_b\rangle_\Omega + \langle I_{ab}\rangle_\Omega + D_a\partial\Theta_b + D_b\partial\Theta_a. \end{aligned}$$

In the second form, $v_a = e_a - (1 + \frac{1}{2}H_a)\hat{n}\, e^{i\lambda k_a \cdot \hat{n}}$ vanishes at the observer. The local shear products $\frac{1}{4}(H_a H_b + H_b H_a)' + H_a' + H_b'$ cancel identically against $e_a - v_a$ (Prop. 3). A compact expression for *all* map kernels is obtained by taking $\mathcal{J}_a$, $\mathcal{J}_{ab}$ and $\mathcal{J}_{abc}$ to be the centred-exponent kernels *after* clock composition. In the symmetric frame, (29) polarises to

$$\begin{aligned} \mathcal{K}_1(a) &= -\mathcal{J}_a, \\ \mathcal{K}_2(a,b) &= -\mathcal{J}_{ab} + \mathsf{P}_0(\mathcal{J}_a \mathcal{J}_b), \\ \mathcal{K}_3(a,b,c) &= -\mathcal{J}_{abc} + \mathsf{P}_0\left(\sum_{\text{cyc}} \mathcal{J}_a \mathcal{J}_{bc} - \mathcal{J}_a \mathcal{J}_b \mathcal{J}_c\right) + \sum_{\text{cyc}} \mathcal{J}_a \langle \mathcal{J}_b \mathcal{J}_c \rangle_\Omega. \end{aligned} \tag{32}$$

Each cyclic sum has three terms. The coefficient convention of §2 introduces no further factorials. For the specified general-spectrum map, add the multilinear coefficient of (28); for the spectra below the symmetric-frame kernels suffice. Thus the lengthy expanded kernels can remain in the ancillary files without obscuring the finite-time structure of the final observable.

The independent construction from the exact ray, root, clock and normalisation equations is described in App. E; its comparison with these formulas is separate from the covariance arguments of the next section.

The propagation adjoint eliminates the part of the second ray driven by H_2, with its emission, clock and normalisation terms retained (App. C, §C.1).

7 Gauge invariance and Ward identities

Every coordinate and label is a physical scalar or frame: t, X, the surface $T = T_\star$, the fluid line $X = 0$, proper time from $T = T_I$, and $\gamma^{-1/2}$ in the labels. So $\Theta(\hat{n})$ is invariant under any diffeomorphism acting on fields and labels together. The only residual freedom is the conformal group of the primordial chart (§4). Its dilation generator must act as $\zeta \to \zeta + c \Leftrightarrow k \to e^{-c}k$. We verify this exactly:

- *Fields:* $K_2(k,0) = -k \cdot \partial_k K_1$ and $K_3(k,0,0) = (k \cdot \partial_k)^2 K_1$ solve every equation in closed form. Perturbed predictions fail.
- *Observable:* $\Theta_{sL} = -\kappa \partial_\kappa \Theta_s$ and $\Theta_{sLL} = (\kappa \partial_\kappa)^2 \Theta_s$ hold through $O(\kappa^4)$, where κ scales the momenta. They are verified by the SymPy generator and, independently, by its fast exact port (`solution/observable_fast.py`). These identities test the fields and the dilation covariance of the map. They do *not* test the ray deflection, emission shift, clock or normalisation terms. A zero-momentum long mode neither deflects the ray nor shifts the endpoint, and each piece of the map is separately covariant. Mutating any of those pieces leaves the identities intact. Dropping the $e^{-\zeta}$ normalisation of the initial photon direction, which breaks covariance, makes them fail. Those terms are verified instead by `solution/check_observable_expansion.py`, which checks the map formulas against their exact definitions for arbitrary fields.
- At first order, the result equals the independent Newtonian-gauge computation (SW, Doppler and ISW) for every ℓ.

7.1 Gradient-mode tests and relation to the literature

Mirbabayi and Zaldarriaga, *CMB Anisotropies from a Gradient Mode* [1], provide two especially useful comparisons. Their §§3–4 show why a physical emission surface, the observer's position and

velocity, the observer clock, and the measured temperature normalisation must be retained for a long-gradient cancellation. Their §5 distinguishes an uncompensated primordial gradient, which generates a second-order quadrupole, from a nonlinearly completed adiabatic gradient, which is locally a coordinate transformation. Appendix C demonstrates cancellation of inverse-Laplacian terms in the long-wavelength observable, including boundary contributions.

The symmetry arguments are relevant here, but their numerical transfer coefficients cannot be imported. Their explicit dynamics are matter dominated; the long-long calculation also neglects η_e/η_o, and the squeezed calculation neglects terms of order $(\lambda_S/\lambda_L)^2$. This report instead retains radiation dynamics and finite emission and observation times. The following checks are derived directly in the temperature frame, without making those matter-era approximations.

Proposition 14 (Exact adiabatic-gradient null test)**.** *Let b be a constant vector, and in a nonsingular patch containing the observer define*

$$D_b(X) = 1 - 2b\cdot X + b^2X^2, \qquad Y_b(X) = \frac{X - bX^2}{D_b(X)}, \qquad \zeta_{\rm ad}(X) = -\ln D_b(X). \tag{33}$$

Then the radiation solution $\nu = H = 0$, $\gamma = D_b^{-2}\delta$ has $\Theta(\hat n) = 0$ exactly, at arbitrary finite observation and emission times within the patch.

Proof. The special conformal transformation satisfies $(\partial_X Y_b)^T(\partial_X Y_b) = D_b^{-2}\mathbf{1}$. Thus γ is a flat metric written in different spatial labels, and $g = (a_1 t)^2(-\mathrm{d}t^2 + \mathrm{d}Y_b^2)$ is the unperturbed radiation spacetime. In particular $R_{ij}[\gamma] = 0$, $S = 0$ and all four exact field equations hold. The observer remains at $Y_b(0) = 0$, with $\partial_X Y_b|_0 = \mathbf{1}$, so the specified initial orientation and clock are unchanged. Since $\sigma = 0$, the exact redshift exponent is $I = 0$ for every ray. Normalisation gives $\Theta = 0$, including before dipole subtraction. □

With $b = \epsilon p/2$, the completed primordial profile is

$$\zeta_{\rm ad} = \epsilon\, p\cdot X + \epsilon^2\left[\frac{1}{2}(p\cdot X)^2 - \frac{1}{4}p^2X^2\right] + O(\epsilon^3). \tag{34}$$

The bracket is the coefficient of ϵ^2, not $\zeta^{(2)}$; the latter is twice the bracket in our convention. The completed seed has vanishing curvature, so it also passes (62). An explicit local check is $e^{-\zeta_{\rm ad}} = D_b$, whose Hessian is $2b^2\delta_{ij}$ and has zero trace-free part.

By contrast, for the *uncompleted* local profile $\zeta = \epsilon p\cdot X$ at $X_o = 0$, the leading-gradient formula gives

$$\Theta_{\rm grad,lead}(\hat n) = \epsilon^2\tau^2\left[(p\cdot\hat n)^2 - \frac{1}{3}p^2\right]. \tag{35}$$

It vanishes at first order but not at second order. To compare the primordial completion with [1], use *their* matter-era notation $\phi_1 = q\cdot X = -3\zeta_1/5$, hence $p = -5q/3$. Multiplying the quadratic bracket in (34) by $-3/5$ gives exactly their $\phi_2 = 5q^2X^2/12 - 5(q\cdot X)^2/6$. This agreement concerns the adiabatic initial profile, not equality of the matter and radiation quadrupole coefficients. In particular, adding that quadratic profile would change the Gaussian initial-data prescription of Def. 6; it must not be used to cancel the physical contribution in this report.

Proposition 15 (Soft-gradient identity for the finite-time map)**.** *For the kernels of the complete map at fixed physical labels, assuming differentiability in the soft momentum at zero,*

$$\boxed{\partial_{q_i}\mathcal{K}_2(k,q;\hat n)|_{q=0} = -\mathcal{D}_i\mathcal{K}_1(k;\hat n), \qquad \mathcal{D}_i = k_j\partial_{k_i}\partial_{k_j} - \frac{1}{2}k_i\Delta_k.} \tag{36}$$

Momentum derivatives keep $\hat{n}$, $T_\star$, T_I and τ_o fixed. This identity is valid at general short-mode momentum and finite times; it is not restricted to the superhorizon expansion of the short mode.

Proof. For an arbitrary regular scalar seed ζ_s, its pullback by (33) is the seed $\zeta_s(Y_b(X)) - \ln D_b(X)$. The same observer and tetrad are selected at the origin. Diffeomorphism invariance therefore makes the complete measured map identical for these two seed descriptions. To first order in an auxiliary parameter λ, set $b = \lambda p/2$ and write

$$Y_b = X + \lambda\xi_p(X) + O(\lambda^2), \qquad \xi_p(X) = (p\cdot X)X - \tfrac{1}{2}X^2 p.$$

For a short seed $sh(X)$ the pulled-back seed is $sh + \lambda p\cdot X + s\lambda\xi_p\cdot\nabla h + \cdots$. If $\mathcal{L}$ and $\mathcal{B}$ denote the linear and symmetric bilinear map responses (with the Taylor prefactor $1/2$ on $\mathcal{B}$), equality of the $s\lambda$ coefficients gives

$$\mathcal{B}[h, p\cdot X] = -\mathcal{L}[\xi_p\cdot\nabla h].$$

For $h = e^{ik\cdot X}$, the left-hand side is $-ip_i\partial_{q_i}\mathcal{K}_2|_{q=0}$. The initial deformation on the right is $\xi_p\cdot\nabla e^{ik\cdot X} = -ip_i\mathcal{D}_i e^{ik\cdot X}$, with the displayed k_j coefficients outside the derivatives. Linearity then gives (36). □

Combining this with the dilation identity isolates a soft remainder:

$$\mathcal{R}_2(k,q;\hat{n}) \equiv \mathcal{K}_2(k,q;\hat{n}) + k\cdot\partial_k\mathcal{K}_1(k;\hat{n}) + q_i\mathcal{D}_i\mathcal{K}_1(k;\hat{n}) = O(q^2). \tag{37}$$

Thus both the constant and gradient parts of a soft quadratic response are fixed by the linear transfer function; genuinely new soft information starts at two gradients. This is a useful organisation and consistency condition for the large ancillary kernels. The symmetry proof is exact, and the leading-gradient quadratic kernel independently satisfies (36); a direct arbitrary-time substitution of the full assembled quadratic observable into this identity has not been performed.

What this does and does not cancel. The no-modulation result for a gradient in [1] includes the appropriate transformation of short-mode initial data (or the corresponding inflationary consistency condition). An exactly Gaussian g with an arbitrary spectrum in a fixed marked chart does not by itself impose that condition. Equations (36) and (37) therefore do not license dropping the soft pieces of $\mathcal{K}_2$, or the gradient-squared term in (35), from this problem's spectrum. Nor do long-wavelength cancellations prove that the kernels at comparable, finite momenta reduce to local curvature polynomials. In the notation of §9.1, the positive m_2^2 contribution is retained.

7.2 Further literature

The following work bears directly on particular steps. None of it is used as an input: every result above is derived independently in the temperature frame. Each item states what it corroborates and what has *not* been compared.

- *Adiabatic modes and their Ward identities* [2, 3, 4]. The dilation and special-conformal identities of §7 and Prop. 15 are the lowest members of this tower. The dilation factorisation (65) is the Gaussian average of the first member.
- *Local observers and curvature* [5, 6]. Conformal Fermi coordinates make long modes appear locally only through curvature. This matches the seed-Ricci form (62) and the statement that only the trace-free curvature quadrupole survives at leading gradient order.

- *Superhorizon modes and the CMB quadrupole* [7, 8]. The $C_2 \propto \Xi^2 \propto k_0^4$ behaviour of §9.1 is the Grishchuk–Zel'dovich effect, here in the radiation era, with finite emission time and exactly in ζ.

- *Nonlinear Sachs–Wolfe effect* [9, 10]. These papers show that the second-order large-scale temperature is far simpler than its metric ingredients. Lemma 2 ($T_{\rm obs} \propto e^{-I}$ exactly) is the temperature-frame version of that simplification. Their matter-era coefficients are not comparable.

- *Nonlinear ζ and the gradient expansion* [11, 12, 13]. These support Def. 6: they give the conservation of $\zeta = \frac{1}{6}\ln\det\gamma$ and the uniform-density (δN) interpretation, and (5) is the radiation-era leading term of their expansion.

- *Scalar-induced tensors in the radiation era* [14, 15, 16]. These give closed-form Si /Ci kernels for the second-order tensor sourced by two first-order scalars, including the monochromatic spectrum and its resonance at $K = 2k_0/\sqrt{3}$. That resonance is the sound-cone surface $\mu = -1/3$ of §5, where the entire-function kernels are manifestly finite. Our shell TT amplitude has been compared with the finite-time (unaveraged) Kohri–Terada kernel $I_{\rm RD}$ through the nonlinear gauge map, and the two agree exactly at all t and μ (Prop. 10, (17)). The comparison does not depend on this repository's source generators. The repository's own source is then matched to the source used in the comparison.

8 The spectrum

8.1 Gaussian reduction

With $\mathcal{K}_r(k_1..k_r;\hat{n})$ the kernel of $\Theta^{(r)}$ (observer at the origin) and $A^{(r)}_{\ell m} = \int \mathrm{d}\Omega\, Y^*_{\ell m}\mathcal{K}_r$:

$$C_\ell^{(11)} = \tfrac{1}{2\ell+1}\textstyle\sum_m \int_k P|A^{(1)}_{\ell m}|^2, \qquad C_\ell^{(22)} = \tfrac{2}{2\ell+1}\sum_m \int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2,$$
$$C_\ell^{(13)} = \tfrac{3}{2\ell+1}\textstyle\sum_m \int_k\int_q P(k)P(q)A^{(1)}_{\ell m}(k)A^{(3)}_{\ell m}(k,q,-q)^*.$$

We write $C_\ell^{(4)} \equiv \frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)}$ for the ϵ^4 coefficient. The disconnected part of 22 is proportional to $\delta_{\ell 0}$. In 13 all three pairings survive; the internal $(q,-q)$ pairing is a pairing of the fourth moment and is not removed by normal ordering. Odd orders vanish.

Compact finite-time spectrum. For $\ell \geq 2$, define the corrected linear response

$$A^{\rm eff}_{\ell m}(k) = A^{(1)}_{\ell m}(k) + \frac{\epsilon^2}{2}\int_q P(q)A^{(3)}_{\ell m}(k,q,-q). \tag{38}$$

Then the complete result through fourth order can be written as two norms:

$$\boxed{C_\ell = \frac{1}{2\ell+1}\sum_m\left\{\epsilon^2\int_k P(k)|A^{\rm eff}_{\ell m}(k)|^2 + \frac{\epsilon^4}{2}\int_k\int_q P(k)P(q)|A^{(2)}_{\ell m}(k,q)|^2\right\} + O(\epsilon^6).} \tag{39}$$

The square is understood only through $O(\epsilon^4)$: its $O(\epsilon^6)$ part is not a prediction. To see the factor $1/2$ in (38), Wick-decompose the cubic Gaussian product into its normal-ordered part plus its three single contractions. The cubic prefactor $1/6$ turns these into a $1/2$ correction to the linear response; the remaining normal-ordered cubic part is orthogonal to the linear part. The quadratic

contribution has two connected pairings and its prefactor is $(1/2)^2$, giving the other 1/2 in (39). No 13 interference term has been discarded or absorbed into a redefinition of the primordial variance. The formula applies whenever the displayed integrals exist, in particular to the monochromatic case.

Equations (32)–(39), together with the factored field sources in §5.2, are a compact organisation of the *general finite-time* answer. They retain the same integrals as the expanded result; they do not establish an additional cancellation of the cubic response or evaluate its momentum and time integrals.

Linear transfer function. For $\ell \geq 2$, the convention is

$$\begin{aligned} A^{(1)}_{\ell m}(k) &= 4\pi i^\ell T_\ell(k) Y^*_{\ell m}(\hat{k}), \\ T_\ell(k) &= -3\int_{x_\star}^{x_o} s'(x)\, w_\ell\Big(\sqrt{3}(x_o - x)\Big)\, \mathrm{d}x, \\ s(x) &= \frac{2\sin x}{x} - \cos x, \qquad w_\ell(y) = \frac{1}{3} j_\ell(y) + j_\ell''(y). \end{aligned} \tag{40}$$

The prime on s differentiates x, and those on j_ℓ differentiate its full argument. Thus $C_\ell^{(11)} = 4\pi \int \mathcal{P}(k) T_\ell(k)^2 \,\mathrm{d}\ln k$.

Closed form of T_ℓ for every ℓ [closed form]. The integrand is a product of a sound wave in the acoustic phase x and a spherical wave in the photon phase $\sqrt{3}(x_o - x) = k(t_o - t)$. Split both into their Hankel (outgoing) parts,

$$\begin{aligned} s'(x) &= \mathrm{Re}\left[e^{ix} S(x)\right], \quad S(x) = -i + \frac{2}{x} + \frac{2i}{x^2}, \\ w_\ell(y) &= \mathrm{Re}\left[e^{iy} V_\ell(y)\right], \quad V_\ell(y) = e^{-iy}\Big(\frac{1}{3} + \partial_y^2\Big) h^{(1)}_\ell(y), \end{aligned} \tag{41}$$

and put $V_\ell^+ = V_\ell$, $V_\ell^- = \overline{V_\ell}$ (conjugate coefficients), $\omega_\varsigma = 1 - \varsigma\sqrt{3}$ and $\Delta = x_o - x_\star$. Both S and V_ℓ are Laurent polynomials; for example

$$\begin{aligned} V_2(y) &= -\frac{2i\,(y^4 + 6iy^3 - 24y^2 - 54iy + 54)}{3y^5}, \\ V_3(y) &= -\frac{2\,(y^5 + 9iy^4 - 54y^3 - 204iy^2 + 450y + 450i)}{3y^6}. \end{aligned}$$

Proposition 16 (Entire-function form of the transfer function)**.** *Use the subtracted kernels of §5,*

$$J_m(\omega, t) = \int_0^t s^{-m}\Big[e^{i\omega s} - \sum_{j<m} \frac{(i\omega s)^j}{j!}\Big]\mathrm{d}s,$$

each of which is t^{1-m} times an entire function of ωt. Then

$$\boxed{\begin{aligned} T_\ell = -\frac{3}{2}\,\mathrm{Re}\sum_{\varsigma=\pm1}\Big\{ & e^{i\varsigma\sqrt{3}x_o}\sum_{m=1}^{2} a_m^\varsigma\left[J_m(\omega_\varsigma, x_o) - J_m(\omega_\varsigma, x_\star)\right] \\ & + e^{ix_o}\sum_{m=1}^{\ell+3} b_m^\varsigma\, J_m(-\omega_\varsigma, \Delta)\Big\}, \end{aligned}} \tag{42}$$

where the coefficients depend on x_o only:

$$a_2^\varsigma = 2iV_\ell^\varsigma(\sqrt{3}x_o), \quad a_1^\varsigma = 2V_\ell^\varsigma(\sqrt{3}x_o) - 2\sqrt{3}\,i\,V_\ell^{\varsigma\prime}(\sqrt{3}x_o), \quad b_m^\varsigma = \left[u^{-m}\right] S(x_o-u)\,V_\ell^\varsigma(\sqrt{3}u),$$

with $[u^{-m}]$ denoting the Laurent coefficient at $u=0$.

Proof. By (41), $-3s'w_\ell = -\frac{3}{2}\,\mathrm{Re}\sum_\varsigma e^{i\varsigma\sqrt{3}x_o}e^{i\omega_\varsigma x}R_\varsigma(x)$, where $R_\varsigma(x) = S(x)V_\ell^\varsigma(\sqrt{3}(x_o-x))$ is rational and vanishes at infinity. Its only poles are at $x=0$ (order two, from S) and $x=x_o$ (order $\ell+3$, from V_ℓ), so R_ς is the sum of its two principal parts. These are $a_2^\varsigma/x^2 + a_1^\varsigma/x$ and, in $u=x_o-x$, $\sum_m b_m^\varsigma u^{-m}$. Write each $e^{i\omega x}x^{-m}$ as the subtracted integrand of J_m plus the principal part of $e^{i\omega x}x^{-m}$. The discarded pieces add up to the principal part, at $x=0$ and at $x=x_o$, of the complex integrand. The real part of that sum is the principal part of $-3s'(x)w_\ell(\sqrt{3}(x_o-x))$, which is regular at both points ($s'(x) = x/3 + O(x^3)$, and w_ℓ is entire). Hence the discarded pieces vanish under Re, and term-by-term integration gives (42). □

Remark 17 (The superhorizon series determines T_ℓ). T_ℓ is entire in k_0 at fixed times, so its superhorizon series determines it even inside the horizon. Once the frequencies and pole orders of the integrand fix a finite function class, recovering the transfer becomes exact linear algebra. For $\ell=2$ this reconstruction is unique and predicts further series coefficients without using Prop. 16 (App. E). The same idea is attractive at fourth order, but the appropriate finite function class is not yet known.

The representation has no logarithms, no Ci and no elementary remainder. It uses the same entire functions as the second-order kernels, and the only frequencies are the beats $\sqrt{3}\mp 1$ between the photon and acoustic phases. Every term is manifestly finite as $x_\star \to x_o$ and as $x_\star \to 0$. If preferred, the recursion for J_M reduces everything to $J_1 = -\,\mathrm{Cin} + i\,\mathrm{Si}$ together with elementary terms. Expressed instead in Ci and Si, the same function carries removable $\ln(\sqrt{3}\pm 1)$ constants. `solution/transfer_entire.py` proves (42) exactly for $\ell = 1,\dots,6$, as an identity in $x_\star$ together with its value at $x_\star = x_o$. The case $\ell=1$, the dipole of $\Theta^{(1)}$, is needed in §8.2.

8.2 Log-map decomposition of the fourth-order spectrum

By Lemma 2, the centred *log-temperature map*

$$L(\hat{n}) \equiv \ln T_{\rm obs}(\hat{n}) - \langle \ln T_{\rm obs}\rangle_\Omega = -\big(I(\hat{n}) - \langle I\rangle_\Omega\big) \tag{43}$$

is an observable. It is linear in the redshift exponent I and contains no normalisation nonlinearity; the clock composition and the frame of §6 are included. The map is recovered from it exactly by $\Theta = e^L/\langle e^L\rangle_\Omega - 1$. Write $L = \epsilon\theta + \epsilon^2 L_2/2 + \epsilon^3 L_3/6$. Then $\theta = \Theta^{(1)}$ is Gaussian and isotropic, with spectrum $C_\ell \equiv C_\ell^{(11)}$ and correlation function $\xi(\hat{n}\cdot\hat{n}') = \sum_{\ell\ge 1}\frac{2\ell+1}{4\pi}C_\ell P_\ell(\hat{n}\cdot\hat{n}')$.

Theorem 18 (Log-map decomposition)**.** *For $\ell \ge 1$, exactly,*

$$\boxed{\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} = C_\ell^{L,4} + \mathcal{N}_\ell[C] + \mathcal{M}_\ell,} \tag{44}$$

where, writing $C_\ell[F,G] = \frac{1}{2\ell+1}\sum_m \langle F_{\ell m}G^*_{\ell m}\rangle$, the three terms are the fourth-order spectrum of the log map, the lognormal normalisation and the collapsed log-map bispectrum:

$$C_\ell^{L,4} = \tfrac{1}{4}C_\ell[L_2,L_2] + \tfrac{1}{3}\,\mathrm{Re}\,C_\ell[\theta,L_3], \tag{45}$$

$$\mathcal{N}_\ell[C] = \frac{1}{2}\sum_{\ell_1,\ell_2\ge 1}\frac{(2\ell_1+1)(2\ell_2+1)}{4\pi}\begin{pmatrix}\ell_1 & \ell_2 & \ell\\ 0 & 0 & 0\end{pmatrix}^2 C_{\ell_1}C_{\ell_2} - \frac{C_\ell^2}{2\pi}, \tag{46}$$

$$\mathcal{M}_\ell = \tfrac{1}{2}\,\mathrm{Re}\,C_\ell[L_2,\theta^2] + \mathrm{Re}\,C_\ell[\theta,\theta L_2]. \tag{47}$$

In real space, with $w^{(4)}$ the ϵ^4 coefficient of $\langle\Theta(\hat n)\Theta(\hat n')\rangle$ (for $\ell\ge 1$) and $(\xi\star\xi)(\hat n\cdot\hat n')=\int \mathrm{d}\Omega_m\,\xi(\hat n\cdot\hat m)\,\xi(\hat m\cdot\hat n')$,

$$w^{(4)}=\tfrac14\langle L_2L_2'\rangle+\tfrac13\langle\theta L_3'\rangle+\tfrac12\xi^2-\tfrac{1}{2\pi}\,\xi\star\xi+\tfrac12\langle\theta^2L_2'\rangle+\langle\theta\,\theta' L_2'\rangle, \tag{48}$$

where primes denote evaluation at $\hat n'$, and the right-hand side is symmetrised in $\hat n\leftrightarrow\hat n'$.

Proof. With $J_c=-L$, (29) gives $\Theta^{(2)}=L_2+\mathsf{P}_0\theta^2$ and $\Theta^{(3)}=L_3+3\mathsf{P}_0(\theta L_2)+\mathsf{P}_0\theta^3-3\theta\langle\theta^2\rangle_\Omega$; the projections P_0 affect only $\ell=0$. Insert these into the bilinear spectra and collect terms. For the terms in θ alone, Wick's theorem gives $\langle\theta^2\theta'^2\rangle_c=2\xi^2$, $\langle\theta\,\theta'^3\rangle=3\xi\,\xi(1)$ and $\langle\theta\theta'\langle\theta^2\rangle_\Omega\rangle=\xi\,\xi(1)+\frac{2}{4\pi}\xi\star\xi$. The variance terms $\xi(1)$ cancel between the last two. The harmonic form follows from $P_{\ell_1}P_{\ell_2}=\sum_L(2L+1)\left(\begin{smallmatrix}\ell_1&\ell_2&L\\0&0&0\end{smallmatrix}\right)^2P_L$ and from the fact that $\xi\star\xi$ has Legendre coefficients C_ℓ^2. □

Four features make (44) the natural organisation of the answer.

1. *The lognormal term is explicit.* $\mathcal{N}_\ell$ depends on $C^{(11)}$ alone. By Prop. 16 it is an explicit convergent harmonic series for every ℓ; it is not a finite closed expression. Equivalently, it is a single angular integral that we do not evaluate:

$$\mathcal{N}_\ell=\pi\int_{-1}^{1}P_\ell(u)\,\xi(u)^2\,\mathrm{d}u-\frac{C_\ell^2}{2\pi},\qquad \ell\ge 1. \tag{49}$$

   The map variance $\xi(1)$ cancels identically. The dipole ($\ell_1=1$ or $\ell_2=1$) enters explicitly, because it is removed only after the nonlinear normalisation, as the problem requires. The term $-C_\ell^2/2\pi$ is the effect of normalising by the *random* sky monopole. On the monochromatic shell,

$$\boxed{\mathcal{N}_\ell=2\pi\sigma^4\Big[\sum_{\ell_1,\ell_2\ge1}(2\ell_1+1)(2\ell_2+1)\begin{pmatrix}\ell_1&\ell_2&\ell\\0&0&0\end{pmatrix}^2T_{\ell_1}^2T_{\ell_2}^2-4T_\ell^4\Big].} \tag{50}$$

   The sum converges absolutely. T_ℓ is an integral of $w_\ell(y)$ over $y\in[0,\sqrt3(x_o-x_\star)]$, so it decays faster than any power once ℓ exceeds the maximal argument.

2. *The mixed term needs only a three-point function.* $\mathcal{M}_\ell$ involves only the leading bispectrum $\langle\theta\theta L_2\rangle$ of the log map, in two collapsed configurations. L_3 does not enter it.

3. *All cubic response sits in the log-map spectrum.* L_3 appears only in $C_\ell^{L,4}$. The quadratic dynamics (L_2) enters both $C_\ell^{L,4}$ and $\mathcal{M}_\ell$. $\mathcal{N}_\ell$ is pure measurement nonlinearity (exponentiation and normalisation). $\mathcal{M}_\ell$ is the interference of that nonlinearity with the quadratic dynamics.

4. *Gradient counting.* In the superhorizon limit $\theta=O(\kappa^2)$, so $\mathcal{N}_\ell=O(\kappa^8)$. Power counting alone gives $\mathcal{M}_\ell=O(\kappa^6)$, but that order cancels. At leading gradient order $\theta=-\tau^2A_{nn}$ and $L_2=\tau^2(4gA+2B)_{nn}$, with $A=(\partial\partial g)_{\mathrm{TF}}$ and $B=(\partial g\,\partial g)_{\mathrm{TF}}$. At a point, g, A and ∇g are independent, and $\langle g\rangle=\langle B\rangle=0$. Hence $\langle\theta(\hat n)\theta(\hat n')L_2(\hat m)\rangle=0$ for arbitrary directions at $O(\kappa^6)$; spatial parity removes $O(\kappa^7)$. Therefore

$$\mathcal{M}_\ell=O(\kappa^8),\qquad \mathcal{N}_\ell=O(\kappa^8),$$

   and the fourth-order temperature and log-temperature spectra agree through total gradient order six. This order contains the leading quadrupole ($O(k_0^4)$, §9.1) and the leading octupole ($O(k_0^6)$, §9.2).

For a pure linear quadrupole, the normalisation contribution is $\mathcal{N}_2=-9C_2^2/(28\pi)$: the random-monopole term outweighs the positive quadratic lognormal term. Exact finite-sky checks of the decomposition are in App. E.

8.3 Observer-centred dilation decomposition

Split the primordial field into its value at the observer and the observer-centred remainder, $g = g_o + g_c$, with $g_o = g(0)$ and $g_c = g - g(0)$. A constant shift of ζ is exactly a momentum dilation (§7). Hence $\Theta[\epsilon g] = \exp(-\epsilon g_o \mathfrak{D})\,\Theta[\epsilon g_c]$ exactly, where $\mathfrak{D} = \sum_a k_a \cdot \partial_{k_a}$ acts on the kernels at fixed $\hat{n}$ and fixed physical labels. Expanding gives the kernels of the observer-centred field:

$$\begin{aligned}\mathcal{K}_2^c(k,q) &= \mathcal{K}_2(k,q) + \mathfrak{D}\mathcal{K}_1(k) + \mathfrak{D}\mathcal{K}_1(q),\\ \mathcal{K}_3^c(k,q,-q) &= \mathcal{K}_3(k,q,-q) + \mathfrak{D}\big[\mathcal{K}_2^c(k,q) + \mathcal{K}_2^c(k,-q) + \mathcal{K}_2^c(q,-q)\big]\\ &\quad - \mathfrak{D}^2\big[\mathcal{K}_1(k) + \mathcal{K}_1(q) + \mathcal{K}_1(-q)\big].\end{aligned}$$

Write $A^c_{r,\ell m}$ for their harmonic coefficients and $D = k\partial_k$.

Proposition 19 (Observer-dilation decomposition)**.** *For $\ell \geq 1$ and any spectrum for which the integrals exist, with $\sigma^2 = \langle g^2\rangle = \int_k P$,*

$$\boxed{\begin{aligned}\frac{1}{4}C_\ell^{(22)} + \frac{1}{3}\,\mathrm{Re}\,C_\ell^{(13)} &= C_\ell^{c,4} + \frac{\sigma^2}{2(2\ell+1)}\int_k P(k)\,D^2\sum_m |A^{(1)}_{\ell m}(k)|^2\\ &\quad - \frac{2}{2\ell+1}\int_k\!\!\int_q P(k)P(q)\,\mathfrak{D}\,\mathrm{Re}\sum_m A^c_{2,\ell m}(k,q)A^{(1)*}_{\ell m}(k).\end{aligned}} \tag{51}$$

Here $C_\ell^{c,4}$ is the fourth-order spectrum built from the observer-centred kernels,

$$C_\ell^{c,4} = \frac{1}{2(2\ell+1)}\sum_m \iint PP\,|A_2^c|^2 + \frac{1}{2\ell+1}\,\mathrm{Re}\sum_m \iint PP\,A^{(1)}A_3^{c*}.$$

On the monochromatic shell, with σ^2 held fixed, the last two terms become $\frac{1}{2}\sigma^2 D_{k_0}^2 C_\ell^{(11)}$ and a total derivative $-2D_{k_0}X_\ell$, where $D_{k_0} = k_0\,\mathrm{d}/\mathrm{d}k_0$ at fixed times and $X_\ell = \frac{1}{2\ell+1}\,\mathrm{Re}\sum_m \iint PP\,A_2^c A^{(1)*}$.

Proof. Insert the kernel relations into the Gaussian reductions of §8. Several terms are independent of $\hat{k}$ and vanish against $A^{(1)}_{\ell m}(k)$ after the $\hat{k}$ integral for $\ell \geq 1$: $\mathfrak{D}\mathcal{K}_2^c(q,-q)$ and $\mathfrak{D}^2\mathcal{K}_1(\pm q)$ in $\mathcal{K}_3$, and the cross term $\mathfrak{D}\mathcal{K}_1(k)\,\mathfrak{D}\mathcal{K}_1(q)^*$ in $|\mathcal{K}_2|^2$. The purely dilation parts give $\sigma^2\int P\,[\,|DA^{(1)}|^2 + \mathrm{Re}\,A^{(1)}D^2A^{(1)*}] = \frac{\sigma^2}{2}\int P\,D^2|A^{(1)}|^2$. In the dilation–centred interference, the 22 term contributes $-2\,\mathrm{Re}\,A_2^c\,DA^{(1)*}$ (by $k \leftrightarrow q$ symmetry) and the 13 term contributes $-2\,\mathrm{Re}\,A^{(1)}(\mathfrak{D}A_2^c)^*$ (by $q \to -q$). Their sum is the total dilation derivative $-2\,\mathfrak{D}\,\mathrm{Re}\,A_2^cA^{(1)*}$, because $A^{(1)}(k)$ does not depend on q. On the shell, $\iint PP\,\mathfrak{D}F = D_{k_0}\iint PP\,F$ at fixed σ^2. □

The following consequences place this result.

- *Relation to the infrared theorem.* Integrating by parts in $\ln k$, the dilation term equals $\frac{\sigma^2}{2(2\ell+1)}\int_k \sum_m |A^{(1)}_{\ell m}|^2(D+3)^2P$. This is Thm. 32 with the long-mode variance σ_L^2 replaced by the full variance. The observer's own value of the field acts as a long mode for every wavelength.

- *Superhorizon limit.* At leading gradient order, $\mathcal{K}_1 = \tau^2\,\mathrm{TF}(kk)_{nn}$, and two exact simplifications follow: $\mathcal{K}_2^c = -\tau^2\,\mathrm{TF}(kq+qk)_{nn}$ is pure gradient-squared, and $\mathcal{K}_3^c(k,q,-q) = 0$ identically. The cross term vanishes after the $\hat{q}$ average. Hence $\frac{29}{3} = 8 + \frac{5}{3} + 0 + 0$: the dilation term is exactly the 8, the centred 22 part is the $\frac{5}{3}$, and the centred 13 part and the cross term vanish. The decomposition (51) is the finite-k_0 extension of (65).

- *Next gradient order.* Every term of (51) is evaluated at $O(\epsilon^4 k_0^6)$ for $\ell = 2$. In this paragraph set $\sigma = k_0 = 1$ and write $e = \bar{t}_\star$, $o = \bar{t}_o$, $\iota = \bar{t}_I$, $\delta = o^2 - e^2$ (the Δ of §9.3). The four-gradient map, linear transfer and (85) give

$$\tfrac{\sigma^2}{2} D^2 C_2^{(11)}\big|_{k_0^6} = 18\, C_2^{(11)}\big|_{k_0^6}, \qquad C_2^{c,22}\big|_{k_0^6} = \tfrac{28}{3}\, C_2^{(11)}\big|_{k_0^6}, \qquad C_2^{c,13}\big|_{k_0^6} = \tfrac{4\pi}{25}\,\tfrac{\delta}{9}\,\langle \mathcal{K}_3^c \rangle_{k_0^4}.$$

The cross term is

$$[-2D_{k_0} X_2]_{k_0^6} = \frac{8\pi\delta}{91125}\big(204e^4 - 280e^3 o + 210e^2 o^2 - 89o^4 - 45\iota^4\big). \tag{52}$$

The observer-local ι^4 dependence occurs only in centred 13 and this cross term. Summing all four pieces gives

$$[C_2^{(4)}]_{k_0^6} = \frac{2\pi\delta}{1913625}\big(38856e^4 - 64640e^3 o + 48480e^2 o^2 - 18601o^4 - 4095\iota^4\big), \tag{53}$$

in agreement with §9.3. Restore $\sigma^4 k_0^6$ for general shell variance and radius. The signs use $\delta = o^2 - e^2 = -(e-o)(e+o)$.

- *Why the centred-22 coefficient is* 28/3. Put $\delta\mathcal{K}_1 = \mathcal{K}_1(k+q) - \mathcal{K}_1(k) - \mathcal{K}_1(q)$. At degree two, $\mathcal{K}_2^c = -\delta\mathcal{K}_1$ pointwise. At degree four the overlap of $\mathcal{K}_2^c$ with $\delta\mathcal{K}_1^{[2]}$ equals that of $-2\delta\mathcal{K}_1^{[4]}$: in this projection it acts as $-\mathfrak{D}\delta\mathcal{K}_1/2$. Consequently the degree-six centred-22 norm is fixed by the linear transfer alone.

  The mechanism splits into two exact identities. The emission endpoint, clock, sky normalisation and dilation subtraction are orthogonal to the leading template. The local four-gradient shear carries $o^4 - e^4$; the field phase, ray position and ray direction carry $\int_e^o t(o-t)^2 \mathrm{d}t$, in the overlap ratio $7:1:14$. In each of these two time structures separately, the projected quadratic correction is -2 times the corresponding linear polarisation. Their sum explains both the coefficient 28/3 and its independence of the physical initial event.

  No extension to higher orders is established. At degree eight both $|\mathcal{K}_2^{c,[4]}|^2$ and the interference of $\mathcal{K}_2^{c,[2]}$ with $\mathcal{K}_2^{c,[6]}$ enter; the existing map does not determine their sum. Numerical suggestions about higher-order failure are kept separately in App. F.2. The finite-wavelength size and further simplification of the centred and cross terms remain open.

8.4 Monochromatic spectrum

After the complete observable is constructed, set

$$\mathcal{P}(k) = \sigma^2 \delta(\ln k/k_0), \qquad P(k) = \frac{2\pi^2\sigma^2}{k_0^2}\delta(k - k_0), \qquad \langle g^2 \rangle = \sigma^2 \text{ held fixed.}$$

The background and the labels $T_\star$, T_I and τ_o are also held fixed. Using joint rotation invariance, with $\hat{k} = \hat{z}$ and $\hat{q}_\mu = (\sqrt{1-\mu^2}, 0, \mu)$:

$$C_\ell^{(11)} = 4\pi\sigma^2 T_\ell(k_0)^2, \tag{54}$$

$$C_\ell^{(22)} = \frac{\sigma^4}{4\pi}\int_{-1}^{1}\mathrm{d}\mu \int \mathrm{d}\Omega \int \mathrm{d}\Omega'\, P_\ell(\hat{n}\cdot\hat{n}')\, \mathcal{K}_2(\hat{n})\mathcal{K}_2(\hat{n}')^*, \tag{55}$$

$$C_\ell^{(13)} = \frac{3\sigma^4}{2}\, i^\ell T_\ell(k_0) \int_{-1}^{1}\mathrm{d}\mu \Big[\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, \mathcal{K}_3(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu; \hat{n})\Big]^*. \tag{56}$$

Primary form of the answer. Put $v = \epsilon^2\sigma^2$ and define two dimensionless response functions of k_0 at fixed times:

$$Q_\ell = \frac{1}{4(2\ell+1)} \int_{-1}^{1} \mathrm{d}\mu \sum_m \left| A^{(2)}_{\ell m}(k_0\hat{z}, k_0\hat{q}_\mu) \right|^2 \geq 0, \tag{57}$$

$$R_\ell = \frac{1}{8\pi i^\ell} \int_{-1}^{1} \mathrm{d}\mu \int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, \mathcal{K}_3(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu; \hat{n}). \tag{58}$$

Then, exactly through fourth order,

$$\boxed{C_\ell = 4\pi v\, T_\ell^2 + v^2\left[Q_\ell + 4\pi T_\ell \,\mathrm{Re}\, R_\ell\right] + O(v^3).} \tag{59}$$

This is a reorganisation, not an evaluation, but it exposes the physics directly:

- $Q_\ell \geq 0$: the quadratic response always generates power;
- R_ℓ: the cubic response interferes with the linear response;
- at a linear node, $C_\ell = v^2 Q_\ell + O(v^3)$ (Prop. 23);
- a negative fourth-order correction away from nodes is not inconsistent.

The decompositions of Thm. 18 and Prop. 19 explain how Q_ℓ and R_ℓ are built.

What the final integrals contain. For auditability, here is every ingredient of (59), in one place:

1. *Linear transfer.* T_ℓ: Prop. 16, closed form.
2. *Second-order map kernel.* $\mathcal{K}_2 = A^{(2)}$ in harmonic space. It is the map of §6, centred as in (29) and composed with the clock.
   - Its fields are the first-order closed forms and, on the shell, the two amplitudes F_S, F_T of (16) together with ν_2, in closed form (§5).
   - Its time integrals are the ray integral over $[\bar{t}_\star, \bar{t}_o]$, which contains the nested displacement integral over $[t, \bar{t}_o]$, and the emission-endpoint and clock terms.
3. *Cubic map kernel.* $\mathcal{K}_3(k, q, -q)$.
   - The new third-order field enters linearly through $\sigma^{(3)} \ni \frac{1}{2}A_3'$. Only its $\hat{q}$-average is needed, and by Prop. 21 that average is a single scalar function. The sources reduce to nine second-order moments and the homogeneous lapse (§5.3.1). Their relative-angle integrals are evaluated; (60) gives the transfer without constructing the cubic field or differentiating a second-order response.
   - Everything else is a product of lower-order fields: $\sigma^{(3)}$ products, X_2, e_2, the emission shift d_2, the clock D_2 and the normalisation. These contain the second-order tensor and ν_2, which are kept in full.
   - The pair $(q, -q)$ uses the exact $K = 0$ kernel.
4. *Sky integrals.* Every term has the form $c\,\Pi(\hat{n})e^{i\hat{n}\cdot V}$, and the sky integrals are evaluated in closed form below.
5. *Remaining integrals.* In general, relative-angle and nested time integrals remain. The relative-angle integral of the top-order cubic field is evaluated in §5.3.1; lower-order propagation products still require additional work. The point $\mu = -1$ is a null set (Thm. 25).

Where the angular integral is non-elementary. Substitute $\mu = 2u^2 - 1$, so that $\mathrm{d}\mu = 4u\,\mathrm{d}u$ with $u \in [0,1]$. On the shell $K = k_0\sqrt{2(1+\mu)} = 2k_0 u$ exactly.

- Every frequency in the field kernels ($k_0/\sqrt{3}$ and $K/\sqrt{3}$, K, and their combinations) is then linear in u.
- The Green functions, such as $\sin K(t-s)/K$, and the projector factors $1/K^2$ are elementary in u.
- The angular factors of the amplitudes are polynomial in u and $\sqrt{1-u^2}$.
- In the sky integrals, terms whose two modes share one line-of-sight time have $|V| = K\lambda = 2k_0 u\lambda$. This covers σ_{ab}, the emission endpoint and the clock, and these become entire functions of u as well.

A further complication comes from modes at *different* line-of-sight times, in the ray-displacement terms and in the products $\theta_a\theta_b$. There

$$|V| = k_0\big|\lambda_1\hat{k} + \lambda_2\hat{q}\big| = k_0\sqrt{(\lambda_1-\lambda_2)^2 + 4\lambda_1\lambda_2 u^2},$$

and it enters through $j_\ell(|V|)$. These mixed-time sky factors are one obstruction, not the only unevaluated part: angular moments of the field amplitudes and nested time integrals remain even for single-time sky factors. Entire dependence on a variable proves regularity, not existence of an elementary antiderivative. The following separation is a structural reorganisation, not an evaluation.

Proposition 20 (Separated form of the mixed-time sky factors)**.** *For $V = \lambda_1 k + \lambda_2 q$ with $\hat{k} = \hat{z}$, $\hat{k}\cdot\hat{q} = \mu$ and $|k| = |q| = k_0$,*

$$\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\, e^{i\hat{n}\cdot V} = 4\pi \sum_{L_1,L_2} (2L_1+1)(2L_2+1)\, i^{L_1+L_2}\, j_{L_1}(k_0\lambda_1)\, j_{L_2}(k_0\lambda_2) \begin{pmatrix} \ell & L_1 & L_2 \\ 0 & 0 & 0 \end{pmatrix}^2 P_{L_2}(\mu).$$

Proof. Expand each plane wave, $e^{i\lambda k\cdot\hat{n}} = \sum_L (2L+1) i^L j_L(k_0\lambda) P_L(\hat{k}\cdot\hat{n})$. Then combine $P_\ell P_{L_1}$ at the common direction $\hat{z}$ by the Legendre product rule, and integrate against $P_{L_2}(\hat{q}\cdot\hat{n})$ by the addition theorem. The series converges absolutely, since $j_L(x)$ decays faster than any power once L exceeds x, and for fixed ℓ the triangle rule leaves a single infinite sum. `solution/check_bessel_translation.py` checks the identity exactly against the Funk–Hecke form, order by order through total degree 10 in (λ_1, λ_2), for $\ell \le 4$ and symbolic μ, with a negative control. □

The square root disappears: each Bessel function has a single time argument, and μ enters only through the polynomial $P_{L_2}(\mu)$. Polynomial prefactors in $\hat{n}\cdot\hat{k}$ and $\hat{n}\cdot\hat{q}$ follow by differentiating with respect to $k_0\lambda_1$ and $k_0\lambda_2$. Other prefactors add further Gaunt couplings, and the double sky integral of $C^{(22)}$ separates copy by copy. The remaining angular integral is therefore a set of Legendre moments $\int_{-1}^{1} P_L(\mu) F(\mu,\ldots)\,\mathrm{d}\mu$ of the field amplitudes, and with $\mu = 2u^2 - 1$ their integrands are entire in u. This moves the difficulty into a well-defined set of scalar moments without evaluating them.

Proposition 21 (The contracted cubic response is scalar)**.** *For an isotropic spectrum, define the angle-averaged cubic metric response*

$$\bar{A}_{3,ij}(k,q;t) = \frac{1}{4\pi}\int \mathrm{d}\Omega_{\hat{q}}\, A_{3,ij}(k,q,-q;t).$$

Then

$$\bar{A}_{3,ij} = F_3(k,q;t)\Big(\hat{k}_i\hat{k}_j - \tfrac{1}{3}\delta_{ij}\Big),$$

so the vector and TT parts of the angle-averaged cubic metric response vanish.

Proof. The third-order operator depends on the total momentum k only, so averaging over $\hat{q}$ commutes with solving. After the average, $\hat{k}$ is the only preferred direction. A symmetric trace-free tensor built from $\hat{k}$ and δ alone is proportional to $\hat{k}\hat{k} - \delta/3$. □

The new cubic field work for $C^{(13)}$ is therefore one scalar problem with averaged sources, now reduced explicitly to the angle-evaluated nine-moment construction of §5.3.1. Two caveats: the lower-order tensor contributions inside the cubic sources and propagation products must be kept, and the uncontracted map $\Theta^{(3)}$ still needs all three sectors. The exploratory evaluation in App. F confirms numerically that keeping only the scalar part reproduces the full result.

The scalar third-order field need not be solved explicitly: its contribution to the observable passes through the Green operator onto the sources. Write $\mathcal{T}_\ell[h](s) = \frac{1}{2}\int_{\max(s,\bar{t}_\star)}^{t_o} \partial_t h(t)\, w_\ell(k_0(t_o - t))\,\mathrm{d}t$ for the partial transfer integral of a scalar metric profile h, so that $\mathcal{T}_\ell[h_{S1}](0) = T_\ell$ with $h_{S1} = Q_1 k^2$. Let $\mathcal{H}[\nu] = \frac{3}{2K^2}(4K^2\nu + 12\nu'/t + 12\nu/t^2)$ be the lapse-to-metric map of the triangular scheme, and $c = k_0/\sqrt{3}$.

Proposition 22 (Adjoint form of the top-order cubic response)**.** *On the shell, the part of* $\frac{1}{3}\operatorname{Re} C_\ell^{(13)}$ *that is linear in the third-order field kernels is*

$$\boxed{\begin{aligned}\tfrac{1}{3}\operatorname{Re} C_\ell^{(13)}\big|_{\mathrm{top}} &= 4\pi\sigma^4\, T_\ell\, \mathcal{A}_\ell,\\ \mathcal{A}_\ell &= 14\, T_\ell + \int_0^{t_o} \mathcal{W}_\ell^0(s)\,\bar{c}_N(s)\,\mathrm{d}s\\ &\quad + \frac{3}{2k_0^2}\int_{\bar{t}_\star}^{t_o} w_\ell(k_0(t_o-t))\,\overline{iK\cdot c_M}(t)\,\mathrm{d}t.\end{aligned}}$$

where $\bar{c} = \frac{1}{2}\int_{-1}^{1}\operatorname{Re} c^{(3)}(k,q,-q)\,\mathrm{d}\mu$ *are the shell-averaged third-order sources (for the momentum source, the real part of* $iK\cdot c_M$ *as a whole),* $14 = c_3/c_1$ *is the regular-lapse ratio on the shell, and the adjoint kernel is*

$$\mathcal{W}_\ell^0(s) = -\frac{1}{3c}\Big[u_1(cs)\,\mathcal{T}_\ell[\mathcal{H}u_2](s) - u_2(cs)\,\mathcal{T}_\ell[\mathcal{H}u_1](s)\Big].$$

Here $\mathcal{T}_\ell[\mathcal{H}u_1](s) = -\mathcal{T}_\ell[h_{S1}](s)$ *is minus the partial first-order transfer function, and* $\mathcal{T}_\ell[\mathcal{H}u_2](s)$ *is the same integral for the decaying lapse mode* u_2*. Both are constant for* $s < \bar{t}_\star$*. The adjoint kernel is thus the Wronskian of the growing- and decaying-mode partial transfer functions. No third-order field equation is solved. The identity holds at transfer nodes; for* $T_\ell \neq 0$ *it reads* $\frac{1}{3}\operatorname{Re} C_\ell^{(13)}|_{\mathrm{top}} = \sigma^2 C_\ell^{(11)}\,\mathcal{A}_\ell/T_\ell$*.*

Proof. At order three the field kernels (ν_3, A_3) enter the map exactly as a first-order field of momentum $K = k$ would. They do not enter through the third-order emission or clock shifts: d_3 multiplies $F_0 = 0$ and D_3 multiplies $\partial_{t_o}\Theta_0 = 0$. They do not enter the ray, since X_3 and e_3 multiply background derivatives that vanish. (The lower-order shifts d_1, d_2, D_1, D_2 do remain in the cubic map; they are part of the non-top terms.) What remains is $-\int \frac{1}{2}A'_{3,nn}e^{iK\cdot n\lambda}\mathrm{d}t$. By Prop. 21 only h_{S3} survives the shell average, and it carries the first-order angular structure, so its contribution is $4\pi\sigma^4 T_\ell\,\mathcal{T}_\ell[h_{S3}](0)$. Insert $h_{S3} = \mathcal{H}[\nu_3] + \frac{3}{2K^2}c_{\mathrm{Ham}}^{(3)}$ and the Green formula for ν_3

of §5. The homogeneous term $\frac{9c_3}{K^2}u_1$ gives $14T_\ell$, because $\nu_1 = -u_1$ and $\mathcal{H}[u_1] = 6 - h_{S1}$. In the particular term, $\partial_t\mathcal{H}$ contains ν_3'', whose Green function has the jump $\partial_t G_N(t,t) = -\frac{1}{3}$. This gives a local term $-\frac{3}{k_0^2 t}w_\ell\, c_N$. The scalar Hamiltonian and longitudinal momentum constraints of the solved lower-order fields imply the source compatibility

$$c'_{\mathrm{Ham}} - \tfrac{4}{t}c_N = 2iK\cdot c_M.$$

Indeed, differentiating h_S and using the lapse equation gives $h'_S = 6(\nu' - \nu/t) + \frac{3}{2K^2}(c'_{\mathrm{Ham}} - \frac{4}{t}c_N)$, while the momentum constraint contracted with K gives $h'_S = 6(\nu' - \nu/t) + \frac{3i}{K^2}K\cdot c_M$. So the local term and the c'_{Ham} term combine pointwise into the momentum term. Finally exchange the t and s integrals. For $s < \bar{t}_\star$ the partial transfer integrals are constants, $u_1(cs) = O(s^2)$, $u_2(cs) = O(s^{-1})$, and $c_N(s) = O(s^2)$, so the integrand is $O(s)$ near $s = 0$. The rest of the domain is compact and regular, which justifies Fubini. The sources are local in time, so the μ average passes through. □

Exact checks of the operators, diagonal jump, factor 14, source compatibility and endpoint combinatorics are in `solution/check_adjoint_exact.py`. The identity treats the top-order field response. Products of lower-order fields still contribute to the full $C^{(13)}$; their combination is Prop. 33.

Nine-moment, derivative-free form of the top-order adjoint. Using the notation of §5.3.1, put

$$\begin{aligned}
\mathcal{P} =& \; -2\mathsf{a}n_h + 2\langle \alpha_S F_S + \alpha_T F_T + 2\mu\mathsf{a}n_2\rangle_\mu,\\
\mathcal{Q} =& \; 2\mathsf{a}'n_h + 2\left\langle \left[\frac{(1+2\mu)\mathsf{b}'}{4}q_S - \frac{\mathsf{a}'(3+\mu)}{6}\right]F_S\right.\\
&+\left[\frac{(1+2\mu)\mathsf{b}'}{4}q_T + \mathsf{a}'q_T\right]F_T\\
&\left.+\left[-(1+\mu)\mathsf{b}'(\mu-1/3) - 2\mathsf{a}'(1+2\mu)\right]n_2\right\rangle_\mu .
\end{aligned}$$

Then (20) is exactly $\bar{m} = k^2(\mathcal{P}' + \mathcal{Q})$. Set $e = \bar{t}_\star$, $o = \bar{t}_o$, $w(t) = w_\ell(k(o-t))$ and $\mathcal{Z} = \mathcal{P} - 2\Upsilon$. Proposition 22 becomes

$$\boxed{\begin{aligned}
\mathcal{A}_\ell =& \; 14T_\ell + \int_0^o \mathcal{W}_\ell^0(s)\bar{r}_N(s)\mathrm{d}s\\
&+\frac{3}{2}[w\mathcal{Z}]_e^o + \frac{3}{2}\int_e^o\left[w\mathcal{Q} - w'\mathcal{Z} + \frac{2w}{t}\Upsilon\right]\mathrm{d}t.
\end{aligned}} \tag{60}$$

Here w' differentiates t. All functions on the right require only the nine moments, n_h, and known linear transfer profiles. Their relative-angle integrals are evaluated in (89)–(92); no cubic kernel or derivative of a second-order response is needed. The boundary term includes the observer and must be kept.

To prove the identity, use $\bar{h}'_{S3} = 6(\bar{\nu}'_3 - \bar{\nu}_3/t) + 3\bar{m}/k^2$ and solve $L_k\eta_3 = -\bar{r}_N$ with regular homogeneous part $14\mathsf{a}$. Recover $\bar{\nu}_3 = \eta_3 - \Upsilon$. Its contribution beyond the lapse Green integral is $-3\int_e^o w(\Upsilon' - \Upsilon/t)\mathrm{d}t$; the momentum contribution is $\frac{3}{2}\int_e^o w(\mathcal{P}' + \mathcal{Q})\mathrm{d}t$. Integrating the two derivatives by parts gives (60). The regular coefficient is unchanged since $\Upsilon = O(t^4)$. The script `solution/check_shell_cubic_sources.py` checks the source identities and the combined boundary coefficients on arbitrary jets. This evaluates the relative-angle part of the top-order field response, not the complete R_ℓ: the lower-order propagation and measurement products must still be added.

Proposition 23 (At linear-transfer nodes, fourth order reduces to the quadratic term)**.** *On the shell, $C_\ell^{(13)} \propto T_\ell(k_0)$, and the cubic kernel is finite at finite times. Hence, at every nontrivial zero of $T_\ell(k_0)$, $C_\ell^{(11)} = C_\ell^{(13)} = 0$ and*

$$\boxed{C_\ell = \frac{\epsilon^4}{4} C_\ell^{(22)} + O(\epsilon^6) \qquad \text{when } T_\ell(k_0) = 0.}$$

At these points the fourth-order spectrum is determined by second-order quantities alone, namely the closed-form kernels of §5 and the map of §6. This is a reduction, not an evaluation: $C_\ell^{(22)}$ is still an unevaluated finite-dimensional integral, and the node positions are roots of T_ℓ, not closed-form numbers. In terms of Thm. 18, $\frac{1}{3}\operatorname{Re} C_\ell[\theta, L_3]$, the term $\operatorname{Re} C_\ell[\theta, \theta L_2]$ and the $-C_\ell^2/2\pi$ part of $\mathcal{N}_\ell$ all vanish there. What remains is $\frac{1}{4}C_\ell[L_2, L_2] + \frac{1}{2}\operatorname{Re} C_\ell[L_2, \theta^2] + \frac{1}{4}C_\ell[\theta^2, \theta^2]$.

Near such a node a large ratio $C_\ell^{(22)}/C_\ell^{(11)}$ is expected: it is nonlinear power filling a linear spectral node. The condition $\epsilon^4 C_\ell^{(22)} \ll \epsilon^2 C_\ell^{(11)}$ is a condition for linear dominance in that multipole, not for the validity of the perturbative expansion. Exploratory values at the first nodes of T_2 and T_3 are listed in App. F.

All sky integrals in closed form. Every term of $\mathcal{K}_r$ has the form $c(\lambda, \ldots)\,\Pi(\hat{n})\, e^{i\hat{n}\cdot V}$. Here $V = \sum_a k_a \lambda_a$ collects the line-of-sight and nested times, and Π is a polynomial. By Funk–Hecke and $n_i \to -i\partial_{V_i}$:

$$\int \mathrm{d}\Omega\, P_\ell(\hat{z}\cdot\hat{n})\,\Pi(\hat{n}) e^{i\hat{n}\cdot V} = \Pi(-i\partial_V)\big[4\pi i^\ell j_\ell(|V|) P_\ell(\hat{z}\cdot\hat{V})\big],$$

$$\iint P_\ell(\hat{n}\cdot\hat{n}')\Pi(\hat{n}) e^{i\hat{n}\cdot V}\Pi'(\hat{n}') e^{-i\hat{n}'\cdot V'} = \Pi(-i\partial_V)\Pi'(i\partial_{V'})\big[(4\pi)^2 j_\ell(|V|) j_\ell(|V'|) P_\ell(\hat{V}\cdot\hat{V}')\big].$$

The sky *integrals* at $\ell = 0$ give $4\pi j_0$; the sky *means* in the normalisation terms include the factor $1/(4\pi)$ and therefore give j_0. The remaining integrals are therefore:

- for $C^{(22)}$: μ, plus per copy one line-of-sight time, one ray time and one field-kernel time;
- for $C^{(13)}$: μ and up to five nested times in this unreduced construction. The combined representation of §10 lowers the bound to three independent times per distributed term.

All integrands are elementary or closed-form.

Shell locality and the real-space form. On the shell, g obeys $\nabla^2 g = -k_0^2 g$ exactly, and $\langle g(X) g(Y)\rangle = \sigma^2 j_0(k_0|X - Y|)$. Every first-order field is therefore a fixed time profile multiplying a *local* derivative of g:

$$\nu_1 = -u_1(x_0)\, g, \qquad H_{1,ij} = -Q(t)\Big(\partial_i\partial_j + \tfrac{1}{3}\delta_{ij} k_0^2\Big) g, \qquad x_0 = k_0 t/\sqrt{3},$$

with Q the first-order kernel of §6 at $k = k_0$. For a general spectrum the time profiles depend on $|k|$, and this locality is lost.

Proposition 24 (Shell locality)**.** *On the monochromatic shell:*

1. *The first-order map is a local line integral of the primordial field:*

$$\theta(\hat{n}) = \mathsf{P}_0\Big[\tfrac{1}{2}\int_{\bar{t}_\star}^{\bar{t}_o} Q'(t)\,\mathcal{D}_{\hat{n}}\,g\big(\hat{n}(\bar{t}_o - t)\big)\,\mathrm{d}t\Big], \qquad \mathcal{D}_{\hat{n}} = (\hat{n}\cdot\partial)^2 + \tfrac{1}{3}k_0^2.$$

Its correlation function is a double line-of-sight integral of an elementary function,

$$\xi(\hat{n}\cdot\hat{n}') = \frac{\sigma^2}{4}\iint \mathrm{d}t\,\mathrm{d}t'\,Q'(t)Q'(t')\,\mathcal{D}^X_{\hat{n}}\mathcal{D}^Y_{\hat{n}'}\,j_0(k_0|X-Y|)\Big|_{X=\hat{n}\lambda,\,Y=\hat{n}'\lambda'} - (\ell=0\ \textit{part}), \quad (61)$$

with $\lambda = \bar{t}_o - t$ and $\lambda' = \bar{t}_o - t'$. Its Legendre coefficients are $4\pi\sigma^2 T_\ell^2$. Consequently the real-space lognormal term $\frac{1}{2}\xi^2 - \frac{1}{2\pi}\xi\star\xi$ of (48) has elementary integrands. Its multipoles still require the viewing-angle projection (49).

2. *At second order, the* only *non-local input to the log map is the order-two trace-free metric along the background ray:*

$$L_2 = L_2^{\mathrm{loc}} - \mathsf{P}_0\Big[\tfrac{1}{2}\int_{\bar{t}_\star}^{\bar{t}_o} A'_{2,nn}\big(t, \hat{n}(\bar{t}_o - t)\big)\,\mathrm{d}t\Big].$$

Here L_2^{loc} is given by the formulas of §6 with $A_2 \to 0$. It is a local quadratic form in derivatives of g along the ray and at the observer, with fixed time profiles.

3. *At third order, the non-local inputs are A_2, A_3 and ν_2. They enter through σ_2, X_2, e_2, the emission shift d_2 and the clock shift D_2.*

Proof. (1) For a single mode, $\frac{1}{2}Q'\mathcal{D}_{\hat{n}}e^{ik\cdot X} = -\frac{1}{2}Q'k^2(\mu^2 - \frac{1}{3})e^{ik\cdot X} = -\sigma_{1,nn}$, so $\theta = -\mathsf{P}_0 I_1$, as in §6. Then (61) follows from the shell correlation of g. (2) Since $\gamma = e^{2\zeta}e^H$, the second-order shear is $\sigma^{(2)} = \frac{1}{2}A'_2 + \frac{1}{2}(H_1^2)' + 2gH'_1$. Every other ingredient of I_2 and of the second-order clock composition is first order: X_1, e_1, d_1, D_1, and the frame. The lapse ν_2 enters only through d_2 and D_2, which first appear in $\Theta^{(3)}$. (3) Inspect F_3, the ray source of X_2, I_3 and D_2 in §6. □

By Wick's theorem, every contraction between local quantities is a derivative of $\sigma^2 j_0(k_0|X-Y|)$, evaluated at points on the two rays or at the observer. Therefore all of $\mathcal{N}_\ell$, and every *local–local* part of $C_\ell^{L,4}$ and $\mathcal{M}_\ell$, have correlation functions that are line-of-sight integrals of elementary functions. These parts need no integral over the primordial relative-momentum angle μ and no Green-function times. Two different angular integrations must be kept apart, however. The *viewing-angle* Legendre projection $C_\ell = 2\pi\int_{-1}^{1} P_\ell(\hat{n}\cdot\hat{n}')\,w(\hat{n}\cdot\hat{n}')\,\mathrm{d}(\hat{n}\cdot\hat{n}')$ remains, and it is not evaluated here. The relative-momentum integral over μ (equivalently the total momentum $K \in [0, 2k_0]$), and the nested Green-function times, are confined to the terms that contain A_2, A_3 or ν_2: the resonant second- and third-order field response. This is a reorganisation of the same exact integrals; the local line-of-sight integrals are not evaluated.

Theorem 25 (Existence)**.** *For all finite $0 < x_\star < x_o$, the integrals for $C_\ell^{(22)}$ and $C_\ell^{(13)}$ converge absolutely. The restriction to the shell is well defined.*

Proof. The momentum, angle and time domains are compact. The integrands are finite sums of products of phases, first-order closed forms, and the higher-order kernels and their first time derivatives. It therefore suffices to show that each kernel and its time derivative is bounded, and continuous outside a null set. The entire-function representation removes every frequency-resonance denominator, but not the inverse powers of the total momentum K introduced by the projectors. These are handled sector by sector.

1. *Lapse.* The Green function $-\frac{1}{3c}[u_2(ct)u_1(cs) - u_1(ct)u_2(cs)]$ and the homogeneous term $\frac{9c_r}{K^2}u_1(ct)$ are even entire functions of $c = K/\sqrt{3}$ (the latter equals $c_r t^2$ times one). Hence ν_r and ν_r' are continuous in K including $K = 0$, where they reduce to the $K = 0$ formulas.

2. *Scalar.* The Hamiltonian form of h_S carries K^{-2}. Use instead the equivalent longitudinal momentum-constraint form $h_S(t) = \int_0^t \left[6(\nu_r' - \nu_r/s) + 3i\, K\cdot c_M(s)/K^2\right]\mathrm{d}s$. The two agree for $K \neq 0$ by Thm. 7. Since c_M is analytic in the momenta and vanishes at $K = 0$, $|K\cdot c_M|/K^2 = O(1)$, so h_S and h_S' are bounded as $K \to 0$.

3. *Vector.* $P_\perp c_M = O(K)$, so $w = \frac{2i}{K^2}\int P_\perp c_M = O(K^{-1})$, and the metric component $K \otimes w$ is $O(1)$. On the shell $w \equiv 0$ (Prop. 10).

4. *Tensor.* The TT projector is bounded, and the Green function $\sin K(t-s)/K$ is entire and even in K.

The projected sectors depend separately on $\hat{K}$, but their sum does not. Prop. 8 shows that the recombined Cartesian kernel A_r is analytic in K, with the exact $K = 0$ kernel as its limit. So the integrand is continuous on the whole compact domain, including $\mu = -1$, and not merely bounded. Where an exact zero total momentum occurs inside the cubic sources, namely the pair $(q, -q)$, the exact $K = 0$ kernel is used; it is also the limit. The ray and line-of-sight integrals run over finite intervals, and the emission and observation times are positive. □

The theorem and the spectrum formulas therefore apply to finite times and to the monochromatic spectrum, and to any spectrum that is a nonnegative finite measure with compact support in $0 < k < \infty$ (so the point variance is finite; compact support alone does not exclude a non-integrable infrared tail at $k \to 0$). Arbitrary spectra without compact support are not covered: the ultraviolet behaviour is open (§9).

9 Limits

9.1 Superhorizon limit, exact in ζ

Theorem 26. *Let all momenta scale as $\kappa \to 0$ at fixed times. Then, to all orders in ϵ,*

$$\Theta(\hat{n}) = -\tau^2\, e^{-2\zeta}\ \mathrm{TF}\left[\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta\right]_o \hat{n}^i\hat{n}^j + O(\kappa^3), \qquad \tau^2 = \tfrac{1}{6}(\bar{t}_o^2 - \bar{t}_\star^2).$$

Proof. At zeroth gradient order the local seed is constant, $\zeta = \zeta_o$, and the ray direction is $e = -e^{-\zeta_o}\hat{n}$, not $-\hat{n}$. Spatial variation of ζ changes this direction by $O(\kappa)$, while H and σ are $O(\kappa^2)$. Their spatial variation along the finite ray adds one further gradient. Since $e^{-2\zeta_o}\sigma = \frac{1}{2}(e^H)'$ locally, the constant conformal factor cancels against the two direction vectors, giving

$$I_{\mathrm{lead}} = \frac{1}{2}n^i n^j[H_{ij}(\bar{t}_o, 0) - H_{ij}(\bar{t}_\star, 0)].$$

Endpoint and clock shifts are $O(\kappa^2)$ and multiply $O(\kappa^2)$ quantities. The Fermi–Walker correction also enters beyond the displayed order. Substitution of (5) gives the claimed I_{lead}. It is trace-free, so $\langle I_{\mathrm{lead}}\rangle_\Omega = 0$ and $\Theta = -I_{\mathrm{lead}} + O(\kappa^3)$.

At each fixed amplitude order the remainder is uniform in the angles on compact momentum and time domains, so the gradient expansion may be taken inside the compact integrals. This statement does not exchange an infinite amplitude series with the full nonlinear solution or its ensemble average.

□

Geometric form. Let $f = e^{-\zeta}$ and $\gamma^{(0)} = e^{2\zeta}\delta$ be the primordial seed metric. Since $f\,\partial_i\partial_j f = -e^{-2\zeta}(\partial_i\partial_j\zeta - \partial_i\zeta\partial_j\zeta)$, the theorem has the equivalent forms

$$\boxed{\Theta_{\rm lead}(\hat{n}) = \tau^2 f(\partial_i\partial_j f)_{\rm TF} n^i n^j = \tau^2 R_{\langle ab\rangle}[\gamma^{(0)}]n^a n^b.} \tag{62}$$

Here $R_{\langle ab\rangle}$ is the trace-free spatial Ricci tensor in the orthonormal frame $e^{-\zeta}\partial_a$ of the *seed* metric. It is not the Ricci tensor of the physical metric at the observation event. The leading observable is a local curvature quadrupole.

Proposition 27 (Gaussian average of the leading-gradient map)**.** *Let $m_j = \int k^j \mathcal{P}(k)\,\mathrm{d}\ln k$ for $j = 0, 2, 4$, with these moments finite. For $\zeta = \epsilon g$, the exact Gaussian average of the leading-gradient expression is*

$$\boxed{C_{2,\rm lead} = \frac{16\pi}{225}\tau^4\epsilon^2 e^{8\epsilon^2 m_0}\left(m_4 + \frac{5}{3}\epsilon^2 m_2^2\right).} \tag{63}$$

For a general spectrum this is the leading-gradient coefficient when all momenta scale to zero at fixed times, not a formula for finite-wavelength modes.

Proof. Put $A_{ij} = (\partial_i\partial_j g)_{\rm TF}$ and $B_{ij} = (\partial_i g\,\partial_j g)_{\rm TF}$. The leading quadrupole tensor is $Q = -\tau^2 e^{-2\epsilon g}(\epsilon A - \epsilon^2 B)$. Isotropy and Gaussianity imply that g, A and ∇g are mutually independent: the g–Hessian covariance is pure trace and all odd-derivative covariances vanish. Thus g is independent of B and

$$\langle A:A\rangle = \frac{2}{3}m_4, \qquad \langle B:B\rangle = \frac{10}{9}m_2^2, \qquad \langle A:B\rangle = 0, \qquad \langle e^{-4\epsilon g}\rangle = e^{8\epsilon^2 m_0}.$$

The ensemble mean of Q vanishes. For a trace-free tensor Q, $\sum_m |a_{2m}|^2 = \frac{8\pi}{15}Q:Q$, so $C_2 = \frac{8\pi}{75}\langle Q:Q\rangle$, proving (63). □

On the monochromatic shell, $m_0 = \sigma^2$, $m_2 = k_0^2\sigma^2$ and $m_4 = k_0^4\sigma^2$. With $\Xi = x_o^2 - x_\star^2 = k_0^2(\bar{t}_o^2 - \bar{t}_\star^2)/3$,

$$\boxed{C_{2,\rm lead} = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2 e^{8\epsilon^2\sigma^2}\left(1 + \frac{5}{3}\epsilon^2\sigma^2\right).} \tag{64}$$

This resums the Gaussian average of the leading-gradient map only. It is not a convergence claim for the full amplitude expansion and supplies no arbitrary-$k_0 t_o$ spectrum. Its expansion, $e^{8v}(1+5v/3) = 1 + 29v/3 + O(v^2)$ with $v = \epsilon^2\sigma^2$, reproduces the fourth-order answer. Separately, exact Wick moments give the leading $O(k_0^4)$ coefficients:

$$C_2^{(11)} = \frac{4\pi}{225}\sigma^2\Xi^2, \qquad C_2^{(22)} = \frac{272\pi}{675}\sigma^4\Xi^2, \qquad C_2^{(13)} = \frac{16\pi}{75}\sigma^4\Xi^2,$$

$$C_2 = \frac{4\pi}{225}\epsilon^2\sigma^2\Xi^2\left[1 + \frac{29}{3}\epsilon^2\sigma^2\right] + O(\epsilon^6) + O(k_0^6), \qquad C_{\ell\geq 3} = O(k_0^6) \text{ (the octupole is given in §9.2).}$$

Two checks: $C_2^{(11)}$ agrees with the line-of-sight transfer function $T_2 \to -\frac{2}{15}k_0^2\tau^2$, and the direct ϵ-expansion of $\langle\sum_m |a_{2m}|^2\rangle$ reproduces $\frac{1}{4}C^{(22)} + \frac{1}{3}C^{(13)}$. The fourth-order correction is a finite, positive renormalisation of the quadrupole by the factor $1 + \frac{29}{3}\epsilon^2\sigma^2$. It is independent of T_I at this order.

Dilation factorisation: what the 29/3 is made of. The map (62) is exactly covariant under the residual dilation: $\Theta_{\rm lead}[\zeta + c] = e^{-2c}\,\Theta_{\rm lead}[\zeta]$, and it carries two spatial derivatives. A constant shift of ζ is therefore *exactly* the momentum dilation $k \to e^{-c}k$. In the Gaussian average, the observer's own value $\zeta_o = \epsilon g(0)$ is independent of the trace-free Hessian and the gradient at the observer, so it acts as a dilation of everything else:

$$\boxed{\begin{aligned} C_{2,\rm lead} &= \underbrace{\langle e^{-4\zeta_o}\rangle}_{\text{observer dilation}} \frac{16\pi}{225}\tau^4\epsilon^2\Big(\underbrace{m_4}_{\text{Hessian}} + \underbrace{\tfrac{5}{3}\epsilon^2 m_2^2}_{\text{gradient}^2}\Big), \\ \langle e^{-4\zeta_o}\rangle &= \langle e^{-\zeta_o D}\rangle\big|_{D=4} = e^{\frac{1}{2}\epsilon^2 m_0 D^2}\big|_{D=4} = e^{8\epsilon^2 m_0}. \end{aligned}} \tag{65}$$

Here $D = k\partial_k$ acts on the degree-four momentum norms. Three consequences follow.

1. *Origin of the coefficients.* The fourth-order coefficient splits as $\frac{29}{3} = 8 + \frac{5}{3}$. The $8 = \frac{1}{2}D^2|_{D=4}$ is the observer dilation; it is shared equally between $\frac{1}{4}C^{(22)}$ (4) and $\frac{1}{3}C^{(13)}$ (4). The $\frac{5}{3}$ is pure $\frac{1}{4}C^{(22)}$. It is exactly the Gaussian average of the gradient-squared quadrupole (35), the effect isolated by [1].

2. *Same operator as in the infrared.* The $O(\epsilon^2)$ term of $\langle e^{-\zeta_o D}\rangle$ is $\frac{1}{2}\epsilon^2 m_0 D^2$. This is the operator of Thm. 32, $\frac{1}{2}\sigma_L^2 D^2$ acting on the short-mode spectrum, which is equivalent to acting on the response after integration by parts in $\ln k$. In the superhorizon limit every mode is "long" relative to the horizon, and the field at the observer dilates its own gradients.

3. *What is, and is not, a convention.* A genuine constant relabelling $X' = e^c X$, $\zeta'(X') = \zeta(e^{-c}X') - c$ changes the mean, $\mu' = \mu - c$, and rescales the moments, $m_j' = e^{-jc}m_j$, but it leaves the variance $m_0' = m_0$ unchanged. The full spectrum is invariant: the deterministic factor $e^{-4\mu}$ compensates the rescaled moments. The stochastic factor $e^{8\epsilon^2 m_0}$ is the average of the observer's *random* local dilation, and it survives every such relabelling. For the ensemble specified in the problem (Gaussian ζ with the chart-shell k_0 and σ^2 held fixed), the fourth-order coefficient is therefore $\frac{29}{3}$, observer dilation included. The ratio

$$\frac{C_{2,\rm lead}}{\frac{16\pi}{225}\tau^4\epsilon^2\langle e^{-4\zeta_o}\rangle m_4} = 1 + \frac{5}{3}\epsilon^2\frac{m_2^2}{m_4} \xrightarrow{\text{shell}} 1 + \frac{5}{3}\epsilon^2\sigma^2 \tag{66}$$

   is a useful *dilation-factored* ratio, valid to all orders in ϵ at leading gradient order. It is not a replacement answer.

Equivalently, write $k_{\rm phys} = e^{-\zeta_o}k$ for the wavenumber in the seed's orthonormal frame at the observer, and put $\Xi_{\rm phys} = e^{-2\zeta_o}\Xi$. Then, as an identity on the shell,

$$C_{2,\rm lead} = \frac{4\pi}{225}\epsilon^2\sigma^2\,\langle\Xi_{\rm phys}^2\rangle\Big(1 + \frac{5}{3}\epsilon^2\sigma^2\Big). \tag{67}$$

Holding a locally inferred scale fixed instead of the chart-shell k_0 would be a *different*, conditional specification of the ensemble. It is not the one posed. `solution/check_dilation_factorisation.py` verifies the covariance exactly in ζ, the mean-shift/dilation equivalence and the Gaussian dilation operator.

9.2 Leading octupole: a curvature-gradient multipole

For every $\ell \geq 2$, the lowest powers of $s'(x) = x/3 + O(x^3)$ and $w_\ell(y) = \ell(\ell-1)y^{\ell-2}/(2\ell+1)!! + O(y^\ell)$ in (40) give

$$T_\ell(k) = -\frac{k^\ell(\bar{t}_o - \bar{t}_\star)^{\ell-1}\big[\bar{t}_o + (\ell-1)\bar{t}_\star\big]}{3\,(2\ell+1)!!} + O(k^{\ell+2}). \tag{68}$$

For $\ell = 2$ this is $T_2 \to -\frac{2}{15}k^2\tau^2$. For $\ell = 3$ it is $T_3 \to -\frac{2}{35}k^3\beta_3$, with

$$\beta_3 = \frac{1}{3}\int_{\bar{t}_\star}^{\bar{t}_o} t\,(\bar{t}_o - t)\,\mathrm{d}t = \frac{(\bar{t}_o - \bar{t}_\star)^2(\bar{t}_o + 2\bar{t}_\star)}{18}.$$

Theorem 28 (Leading octupole, exact in ζ)**.** *Let all momenta scale as $\kappa \to 0$ at fixed times. To every order in ϵ, the $\ell = 3$ part of the map is*

$$\boxed{\Theta_{3,\mathrm{lead}}(\hat{n}) = \beta_3\,\big[\nabla_{\langle a}R_{bc\rangle}\big]_o\,n^a n^b n^c + O(\kappa^4), \qquad \nabla_{\langle a}R_{bc\rangle} = \frac{f}{2}\big[\partial_a\partial_b\partial_c(f^2)\big]_{\mathrm{STF}},} \tag{69}$$

where $f = e^{-\zeta}$. The curvature, the covariant derivative and the orthonormal components are those of the seed metric $\gamma^{(0)} = e^{2\zeta}\delta$, and $\langle\cdots\rangle$ denotes the symmetric trace-free part.

Proof. 1. *Coordinates.* Use seed Riemann normal coordinates at the observer. This is a time-independent relabelling of the comoving labels, so it is allowed at fixed physical labels. The Christoffel symbols vanish at the observer and are $O(\kappa^2)$ along the ray. Ray deflection and displacement therefore enter only at $O(\kappa^4)$, and so do the endpoint, the clock, the frame rotation and the nonlinear normalisation terms.

2. *Shear.* The leading shear is covariant, $\sigma_{ab} = -\frac{t}{3}R_{\mathrm{TF},ab}$, by (5).

3. *Expansion along the ray.* Expanding along the radial ray,

$$\Theta = \frac{1}{3}\int t\,\big[R_{\mathrm{TF}} + (\bar{t}_o - t)\,n^c\nabla_c R_{\mathrm{TF}}\big]_{nn}\,\mathrm{d}t + O(\kappa^4).$$

   The first term is the quadrupole (62). The second term is $\beta_3\nabla_{\langle a}R_{bc\rangle}n^a n^b n^c$ plus a dipole, which by the contracted Bianchi identity is proportional to $n\cdot\nabla R$. The dipole is removed by the final projection and does not affect $\ell = 3$.

4. *Conformal identity.* The conformal identity in (69) is proved for generic ζ in `solution/check_gradient_octupole.py`.

□

With the moments m_j of Prop. 27 (now including m_6), the same independence argument gives an exact Gaussian average. Write $\Theta_{3,\mathrm{lead}} = \beta_3 Q_{abc}n^a n^b n^c$ with $Q_{abc} = e^{-3\epsilon g}\big[-\epsilon g_{abc} + 6\epsilon^2 g_{(ab}g_{c)} - 4\epsilon^3 g_a g_b g_c\big]_{\mathrm{STF}}$, by (69). The three coefficients come from three facts:

- $\langle|g_{\langle abc\rangle}|^2\rangle = \frac{2}{5}m_6$;
- the $\ell = 3$ part of the trace-free Hessian $\otimes$ gradient carries $\frac{7}{15}$ of $\langle A{:}A\rangle\langle|\nabla g|^2\rangle$;
- $\langle|(\nabla g)^{\otimes 3}_{\mathrm{STF}}|^2\rangle = \frac{2}{5}\langle|\nabla g|^6\rangle = \frac{14}{9}m_2^3$.

With $\sum_m |a_{3m}|^2 = \frac{8\pi}{35} Q{:}Q$, the result is

$$\boxed{C_{3,\mathrm{lead}} = \frac{16\pi}{1225}\beta_3^2\epsilon^2 e^{18\epsilon^2 m_0}\Big[m_6 + 28\epsilon^2 m_2 m_4 + \frac{560}{9}\epsilon^4 m_2^3\Big].} \tag{70}$$

On the shell, with $v = \epsilon^2\sigma^2$,

$$\boxed{C_{3,\mathrm{lead}} = \frac{4\pi}{3675}(x_o - x_\star)^4(x_o + 2x_\star)^2\epsilon^2\sigma^2 e^{18v}\Big(1 + 28v + \frac{560}{9}v^2\Big),} \tag{71}$$

and, through the requested order,

$$C_3 = \epsilon^2 C_3^{(11)}\big[1 + 46\epsilon^2\sigma^2\big] + O(\epsilon^6) + O(k_0^8), \qquad \frac{C_3^{(22)}}{C_3^{(11)}} = 148\sigma^2, \quad \frac{C_3^{(13)}}{C_3^{(11)}} = 27\sigma^2.$$

The linear term agrees with $4\pi\sigma^2 T_3^2$ from (68). The structure parallels the quadrupole:

- the observer dilation is $e^{\frac{1}{2}\epsilon^2 m_0 D^2}$ at degree $D = 6$, giving e^{18v} (compared with $D = 4$ and e^{8v} for the quadrupole);
- $46 = 18 + 28$, the second term being the Hessian–gradient interference;
- the v^2 term is the cubed gradient.

9.3 Next-gradient quadrupole and general spectra

The complete map through four spatial gradients determines the next quadrupole coefficient, with symbolic emission, observation and physical initial times. Write $o = \bar{t}_o$, $e = \bar{t}_\star$, $\iota = \bar{t}_I$ and $\Delta = o^2 - e^2$. On the shell,

$$\begin{aligned}
C_2^{(11)} &= \frac{4\pi}{2025}\sigma^2 k_0^4\Delta^2 \\
&\quad - \frac{4\pi}{637875}\sigma^2 k_0^6 (o-e)^2(o+e)\big(59o^3 + 59o^2 e - 106oe^2 + 114e^3\big) + O(k_0^8), \\
\tfrac{1}{4}C_2^{(22)} &= \frac{68\pi}{6075}\sigma^4 k_0^4\Delta^2 \\
&\quad - \frac{8\pi}{637875}\sigma^4 k_0^6\Delta\big(105\iota^4 + 719o^4 - 1920o^2e^2 + 2560oe^3 - 1464e^4\big) + O(k_0^8), \\
\tfrac{1}{3}\operatorname{Re} C_2^{(13)} &= \frac{16\pi}{2025}\sigma^4 k_0^4\Delta^2 \\
&\quad - \frac{2\pi}{1913625}\sigma^4 k_0^6\Delta\big(2835\iota^4 + 9973o^4 - 25440o^2e^2 + 33920oe^3 - 21288e^4\big) + O(k_0^8).
\end{aligned}$$

The leading ratios $\frac{1}{4}C^{(22)}/C^{(11)} = \frac{17}{3}\sigma^2$ and $\frac{1}{3}\operatorname{Re} C^{(13)}/C^{(11)} = 4\sigma^2$ reproduce the factor $\frac{29}{3}$ of Thm. 26 from the generator's own kernels.

Three-tensor form. The polynomials have a compact geometric origin. Let $\mathfrak{a} = -R[\gamma_0]/36$ and $\mathsf{A}_{ab} = -\frac{1}{3}R_{\langle ab\rangle}[\gamma_0]$ be the curvature scalar and trace-free Ricci tensor of the seed metric $\gamma_0 = e^{2\zeta}\delta$, so that $\nu = t^2\mathfrak{a} + t^4\mathfrak{b} + \cdots$ and $H = t^2\mathsf{A} + t^4\mathsf{B} + \cdots$. The field equations give $30\mathfrak{b} = \nabla^2\mathfrak{a} - 6\mathfrak{a}^2 - \operatorname{tr}\mathsf{A}^2$ and

$$\mathsf{B} = \tfrac{3}{20}(\nabla\nabla\mathfrak{a})_{\mathrm{TF}} + \tfrac{3}{20}(\mathsf{A}^2)_{\mathrm{TF}} - \tfrac{7}{10}\mathfrak{a}\mathsf{A}.$$

There is a useful stronger statement before multipole projection. Put $M = \int_e^o t(o-t)\mathrm{d}t = 3\beta_3$, $L = \int_e^o t(o-t)^2\mathrm{d}t$ and $E = o^4 - e^4$. In a seed-orthonormal frame at the observer the full map through four gradients is

$$\begin{aligned}\Theta = \mathsf{P}_0\Big[&-\tfrac{\Delta}{2}\mathsf{A}_{nn} - M(\nabla_n\mathsf{A})_{nn} - \tfrac{L}{2}(\nabla_n\nabla_n\mathsf{A})_{nn} - \tfrac{E}{2}\mathsf{B}_{nn} \\ &- \tfrac{\Delta^2}{4}(\mathsf{A}^2)_{nn} + \tfrac{3\Delta^2}{8}\mathsf{A}^2_{nn} + \Big(\tfrac{o^4+3\iota^4}{4} - e^4\Big)\,\mathfrak{a}\mathsf{A}_{nn}\Big] + O(\partial^5).\end{aligned} \tag{72}$$

This retains the dipole until the final projection. Covariant Taylor expansion along the seed radial geodesic gives the M, L terms; the time-dependent direction response gives the A^2 terms, with exponentiation contributing to A^2_{nn}; the last coefficient retains the physical clock and emission surface. This is exact in amplitude at the stated gradient order, not at general wavelength.

This local formula provides the complete four-gradient map directly from the seed curvature, without solving a photon ray separately. Its kernels agree component by component with the independent field-and-ray construction at general $(k, q, -q)$; the exact comparisons and their scope are recorded in App. E.

Projecting (72), exactly in ζ and through fourth gradient order, the quadrupole is a combination of only three local correction tensors:

$$\boxed{\Theta_{\ell=2} = Q_{ab}n^a n^b, \qquad Q = -\tfrac{\Delta}{2}\mathsf{A} + \beta_H(\nabla\nabla\mathfrak{a})_{\mathrm{TF}} + \beta_S(\mathsf{A}^2)_{\mathrm{TF}} + \beta_a\,\mathfrak{a}\mathsf{A} + O(\partial^6),} \tag{73}$$

with

$$\begin{aligned}\beta_H &= -\tfrac{11}{14}L - \tfrac{3}{40}E = -\tfrac{o-e}{420}\big(59o^3 + 59o^2e - 106oe^2 + 114e^3\big),\\ \beta_S &= \tfrac{27}{14}L - \tfrac{3}{40}E - \tfrac{1}{28}\Delta^2 = \tfrac{o-e}{140}\big(7o^3 + 7o^2e - 118oe^2 + 62e^3\big),\\ \beta_a &= \tfrac{27}{7}L + \tfrac{7}{20}E + \tfrac{o^4+3\iota^4}{4} - e^4 = \tfrac{3}{140}\big(43o^4 - 90o^2e^2 + 120oe^3 - 108e^4 + 35\iota^4\big).\end{aligned}$$

The line-of-sight Hessian, the direction response, the map normalisation, the emission surface $(-e^4)$ and the observer clock $(\frac{1}{4}(o^4 + 3\iota^4))$ are all inside these three coefficients.

Origin of the coefficients. With $R_{ab} = -3\mathsf{A}_{ab} - 12\mathfrak{a}\delta_{ab}$, the contracted Bianchi identity reads $\nabla^b\mathsf{A}_{ab} = 2\nabla_a\mathfrak{a}$. Conformal flatness (vanishing Cotton tensor) makes $\nabla_c(\mathsf{A}_{ab} + \mathfrak{a}\delta_{ab})$ symmetric in b, c. With the three-dimensional Riemann tensor, the trace-free parts then obey

$$\nabla^c\nabla_a\mathsf{A}_{cb} = 2\nabla_a\nabla_b\mathfrak{a} - 9\mathsf{A}^2_{ab} - 18\mathfrak{a}\mathsf{A}_{ab}, \qquad \nabla^2\mathsf{A}_{ab} = 3\nabla_a\nabla_b\mathfrak{a} - 9\mathsf{A}^2_{ab} - 18\mathfrak{a}\mathsf{A}_{ab}.$$

Equation (T) at $O(t^2)$ then gives B. For locally constant A the canonical photon momentum is conserved, so $e^{-\dot{I}} = [n^T e^{\Delta\mathsf{A}} n]^{-1/2}$, which gives both Δ^2 terms of (72). The $\ell = 2$ part of $T_{abcd}n^a n^b n^c n^d$, for symmetric T, is $\frac{6}{7}[T_{abcc}]_{\mathrm{TF}}n^a n^b$. This sends $\mathsf{A}^2_{nn} \to \frac{4}{7}(\mathsf{A}^2)_{\mathrm{TF}}$ and, after the trace of the symmetrised second derivative, $(\nabla_n\nabla_n\mathsf{A})_{nn} \to \frac{6}{7}[\frac{11}{6}\nabla\nabla\mathfrak{a} - \frac{9}{2}\mathsf{A}^2 - 9\mathfrak{a}\mathsf{A}]_{\mathrm{TF}}$. These fix every L, E and Δ^2 coefficient above.

The $(\mathsf{A}^2)_{\mathrm{TF}}$ tensor drops out of the fourth-order spectrum after the Gaussian average: its interference terms are proportional to fourth-order contractions of a Gaussian Hessian with a trace-free gradient square, which vanish. So two time coefficients determine every k_0^6 term:

$$\boxed{\begin{aligned}[C_2^{(11)}]_{k_0^6} &= \tfrac{16\pi}{6075}\sigma^2 k_0^6\Delta\,\beta_H, & [\tfrac{1}{4}C_2^{(22)}]_{k_0^6} &= \tfrac{32\pi}{18225}\sigma^4 k_0^6\Delta\,(30\beta_H - \beta_a),\\ [\tfrac{1}{3}\,\mathrm{Re}\,C_2^{(13)}]_{k_0^6} &= \tfrac{8\pi}{18225}\sigma^4 k_0^6\Delta\,(110\beta_H - 9\beta_a), & [C_2^{(4)}]_{k_0^6} &= \tfrac{8\pi}{18225}\sigma^4 k_0^6\Delta\,(230\beta_H - 13\beta_a).\end{aligned}}$$

The complete fourth-order quadrupole through $O(k_0^6)$ is therefore

$$\boxed{C_2^{(4)} = \frac{29}{3}\cdot\frac{4\pi}{2025}\sigma^4 k_0^4\Delta^2 + \frac{8\pi}{18225}\sigma^4 k_0^6\Delta\,(230\beta_H - 13\beta_a) + O(k_0^8),}$$

whose second term expands to $-\frac{2\pi}{1913625}\sigma^4 k_0^6\Delta\big(4095\iota^4 + 18601o^4 - 48480o^2e^2 + 64640oe^3 - 38856e^4\big)$. The observer-dilation part $\frac{1}{2}\sigma^2 D^2 C_2^{(11)}$ is $18\sigma^2 C_2^{(11)}$ at this order (against 8 at leading order), that is $\frac{8\pi}{18225}\sigma^4 k_0^6\Delta\cdot 108\beta_H$. What remains after subtracting it is $\frac{8\pi}{18225}\sigma^4 k_0^6\Delta(122\beta_H - 13\beta_a)$. A longer derivation is in `report/curvature_and_adjoint_derivations.md`. Four features of these results:

- *Observer clock.* The initial event $\bar{t}_I$ first enters at this order, through the observer clock.
- *Endpoint and clock pieces.* These are separately nonzero. For $\frac{1}{4}C_2^{(22)}$ they are $\frac{32\pi}{18225}\sigma^4 k_0^6\Delta e^4$ and $-\frac{8\pi}{18225}\sigma^4 k_0^6\Delta(o^4 + 3\iota^4)$. For $\frac{1}{3}\operatorname{Re} C_2^{(13)}$ they are $\frac{8\pi}{2025}\sigma^4 k_0^6\Delta e^4$ and $-\frac{2\pi}{2025}\sigma^4 k_0^6\Delta(o^4 + 3\iota^4)$. The third-order shifts d_3, D_3 do not enter, but the lower-order ones do.
- *Both gradient orderings matter.* The degree-six cubic interference is $K_1^{[2]}K_3^{[4]} + K_1^{[4]}K_3^{[2]}$, not just its first term. The curvature and field-and-ray constructions retain both and agree exactly (App. E).
- *Locality.* The temperature-frame field kernels, and hence the map kernels, are polynomial in the momenta and times through this order. The generator carries $1/|k\pm q|^2$ as formal symbols. After substitution, both $\mathcal{K}_2(k,q)$ and $\mathcal{K}_3(k,q,-q)$ have only numerical denominators. The non-local ∇^{-2} terms familiar from Poisson gauge do not arise.

General spectra at the next gradient order. The curvature construction does not require a shell. With $m_j = \int k^j\mathcal{P}(k)\mathrm{d}\ln k$, write $[\cdot]_{[6]}$ for the coefficient of total gradient degree six when all momenta are scaled together at fixed times. Exact sphere moments followed by Wick contraction give

$$\boxed{\begin{aligned}
[C_2^{(11)}]_{[6]} &= \tfrac{16\pi}{6075}\Delta\,\beta_H m_6,\\
[\tfrac{1}{4}C_2^{(22)}]_{[6]} &= \tfrac{32\pi}{18225}\Delta\big[12\beta_H m_0 m_6 + (18\beta_H - \beta_a)m_2 m_4\big],\\
[\tfrac{1}{3}\operatorname{Re} C_2^{(13)}]_{[6]} &= \tfrac{8\pi}{18225}\Delta\big[60\beta_H m_0 m_6 + (50\beta_H - 9\beta_a)m_2 m_4\big],\\
[C_2^{(4)}]_{[6]} &= 18m_0[C_2^{(11)}]_{[6]} + \tfrac{8\pi}{18225}\Delta(122\beta_H - 13\beta_a)m_2 m_4.
\end{aligned}} \tag{74}$$

The entire $m_0 m_6$ term is the observer dilation, since $D^2/2 = 18$ at degree six. After subtracting it, only one spectral shape moment, $m_2 m_4$, remains. This is a simplification of the *general-spectrum* result, not just a factorisation on the shell. The β_S contraction vanishes here too. Substituting $m_j = \sigma^2 k_0^j$ recovers all three shell coefficients above. The same unprojected map independently gives the leading octupole contributions $C_3^{(22)}/4 = B_3(9m_0m_6 + 28m_2m_4)$ and $\operatorname{Re} C_3^{(13)}/3 = 9B_3 m_0 m_6$, where $B_3 = 16\pi\beta_3^2/1225$. These polynomial contractions are derived and checked by `solution/curvature_kernels.py --spectrum`. Compact spectral support justifies taking the expansion inside the integrals. Finiteness of the displayed moments defines the formal coefficients more generally; it is not by itself a uniform remainder estimate for unbounded spectra.

Bounds for general spectra. For any nonnegative spectrum with finite, nonzero moments, the leading-gradient fourth-order factors are

$$\frac{C_{2,\mathrm{lead}}}{\epsilon^2 C_{2,\mathrm{lead}}^{(11)}} = 1 + \epsilon^2 b_2, \quad b_2 = m_0\Big(8 + \tfrac{5}{3}r_2\Big), \qquad \frac{C_{3,\mathrm{lead}}}{\epsilon^2 C_{3,\mathrm{lead}}^{(11)}} = 1 + \epsilon^2 b_3, \quad b_3 = m_0(18 + 28r_3),$$

up to $O(\epsilon^4)$, with shape parameters $r_2 = m_2^2/(m_0 m_4)$ and $r_3 = m_2 m_4/(m_0 m_6)$. The Cauchy–Schwarz inequalities $m_2^2 \le m_0 m_4$ and $m_4^2 \le m_2 m_6$ give $0 \le r_3 \le r_2 \le 1$. Hence

$$\boxed{8\,m_0 \le b_2 \le \tfrac{29}{3}\,m_0, \qquad 18\,m_0 \le b_3 \le 46\,m_0,} \tag{75}$$

with equality at the upper ends exactly for the monochromatic shell. At fixed primordial variance, the shell therefore maximises both relative leading-gradient corrections. The lower ends are approached by very broad spectra, for which only the observer dilation survives.

9.4 Late time, first order

Proposition 29. *For $\ell \ge 2$ and fixed $x_\star$, $T_\ell(x_\star, x_o) = O(\ln x_o/x_o)$ as $x_o \to \infty$. Hence $C_\ell^{(11)} \to 0$.*

Proof. Write $T_\ell = -3\int_0^{x_o - x_\star} s'(x_o - u) w_\ell(\sqrt{3}u)\mathrm{d}u$, with $w_\ell = j_\ell/3 + j_\ell''$. Two integrations by parts give $\int_0^\infty e^{iu} w_\ell(\sqrt{3}u)\mathrm{d}u = 0$ exactly for $\ell \ge 2$, because $j_\ell(0) = j_\ell'(0) = 0$. This is the observer-local sound-cone cancellation. Since $s'(x) = \sin x + O(1/x)$, the $O(1)$ oscillation therefore vanishes. The remaining pieces are oscillatory with distinct frequencies $\sqrt{3} \pm 1$ and amplitudes $O(1/x_o)$, giving $O(\ln x_o/x_o)$ by integration by parts. □

At fourth order the late-time limit is non-uniform at the resonances $\mu = \pm 1$ and $\mu = -1/3$. A pointwise late-time limit therefore cannot simply be interchanged with the angular integral without a uniform estimate or an explicit treatment of the resonant regions. Matched asymptotics in angle and time, or an analytically subtracted representation, would be legitimate routes. The evaluation is [open].

9.5 Subhorizon limit, first order

The late-time proposition fixes $x_\star$. The subhorizon limit proper is $k_0 \to \infty$ at fixed physical times, so emission is also deep inside the horizon.

Proposition 30 (Subhorizon transfer function)**.** *Let $x_o \to \infty$ with $x_\star = x_o/R$ for fixed $R = \bar{t}_o/\bar{t}_\star > 1$, and put $y = \sqrt{3}(x_o - x_\star) = k_0(\bar{t}_o - \bar{t}_\star)$, the comoving distance to the emission surface in units of $1/k_0$. Then, for $\ell \ge 2$,*

$$\boxed{T_\ell = -\big[\cos x_\star\, j_\ell(y) + \sqrt{3}\,\sin x_\star\, j_\ell'(y)\big] + O(x_o^{-2}),}$$

and hence $C_\ell^{(11)} = 4\pi\sigma^2 T_\ell^2 = O(k_0^{-2})$. All growing, constant and logarithmic terms cancel exactly. The observer enters first at $O(x_o^{-2})$, at its own acoustic frequency. For $\ell = 2$ that term is exactly $2\cos x_o/x_o^2$, the local fluid shear ($w_2(0) = \frac{2}{15} \ne 0$). For $\ell > 2$ it carries the constants π and $\ln(2+\sqrt{3})$.

Proof. Start from (42). The recursion $J_m = -t^{1-m} g_m/(m-1) + i\omega J_{m-1}/(m-1)$, with $g_m = e^{i\omega t} - \sum_{j<m}(i\omega t)^j/j!$, reduces every J_m to J_1 and elementary terms. Exactly, $J_1(\omega, t) = -\mathrm{Ein}(-i\omega t) = -\gamma - \ln(|\omega| t) + \frac{i\pi}{2}\operatorname{sgn}\omega - E_1(-i\omega t)$. The only asymptotic input is the standard expansion of E_1 on the imaginary axis. Put $x_\star = x_o/R$ and collect by frequency. The result is a finite sum of phases $e^{i(\pm\sqrt{3}(x_o - x_\star) \pm x_\star)}$ and $e^{\pm i x_o}$, each with a Laurent series in x_o. The coefficients are exact elements of $\mathbb{Q}(\sqrt{3}, i)[R, \pi, \gamma, \log\ldots]$, and the cancellations and the leading term are identities among them. With $j_\ell(y) = \sin(y - \ell\pi/2)/y + O(y^{-2})$ and $j_\ell'(y) = \cos(y - \ell\pi/2)/y + O(y^{-2})$, the four emission phases combine into the boxed form. `solution/transfer_subhorizon.py` does this exactly and asserts the cancellations, the leading form for $\ell = 2, 3, 4$, and the $\ell = 2$ observer term. □

The leading term has the familiar acoustic structure, here derived rather than assumed. At emission the first-order temperature perturbation is $-\nu_1 = \sin x_\star/x_\star - \cos x_\star \to -\cos x_\star$, the acoustic monopole. The $\sin x_\star$ term is its time derivative, the fluid-velocity phase. So the two terms correspond to the emission-surface Sachs–Wolfe term $\propto j_\ell$ and the Doppler term $\propto j'_\ell$. The relative factor $\sqrt{3} = 3c_s$ is the standard ratio of velocity to monopole amplitude for a radiation fluid. The identification is by structure; the formula itself does not depend on it. In the temperature frame both come out of the single line-of-sight integral I through its emission endpoint. The observer end cancels at $O(1)$, the subhorizon form of the cancellation in the late-time proposition. The next asymptotic orders follow from the same entire-function representation; they are not included in the displayed leading law. Supplementary comparisons are in App. F.

9.6 Infrared and ultraviolet behaviour of general spectra

Remark 31 (Short-mode generation of superhorizon power). At fixed finite time, a regular short-mode input cannot generate a dimensionless spectrum falling more slowly than K^3 under the following sufficient assumptions: the input is Gaussian with a smooth power spectrum of compact support bounded away from zero, and the generated scalar field is considered at fixed perturbative order using the recombined Cartesian kernels. For example, the connected quadratic spectrum is

$$P_{Y^{(2)}}(K) = 2\int_q P(q)P(|K-q|)\,|\mathcal{Y}_2(q, K-q; t)|^2.$$

The kernel is continuous and bounded on the compact integration domain, including zero total momentum (Prop. 8). Dominated convergence gives finite $P_{Y^{(2)}}(0)$. More generally, Wick decomposition at any fixed order leaves finitely many convolution integrals and compact internal pair integrals with the same bounds. The linear Wick component has no support near zero for the stated input. Therefore the connected generated spectrum obeys

$$\Delta^2_Y(K) = \frac{K^3}{2\pi^2}\,[P_Y(0) + o(1)].$$

If the constant vanishes, the spectrum is more suppressed. This is a finite-time statement about regular short-mode inputs, not a general consequence of causality without statistical assumptions. Entireness alone does not imply compact real-space support, and the weaker condition $\int \xi^2 \mathrm{d}^3 r < \infty$ does not suffice for arbitrary higher-order Wick components.

An exactly monochromatic seed is different: $\xi \propto \sin(k_0 r)/(k_0 r)$ has long-range correlations. The shell-overlap integral $\int d^3q\,\delta(|q|-k_0)\delta(|K-q|-k_0) = 2\pi k_0^2/K$ for $0 < K < 2k_0$ allows $P_{Y^{(2)}} \propto K^{-1}$ and $\Delta^2_Y \propto K^2$ when the zero-pair response is nonzero. The lapse provides an example: its regular pair starts with $5q^2t^2/9$. This is not a claim about every field or about the observed angular spectrum. A smooth finite-width replacement satisfies the compact-spectrum argument, but the width and $K \to 0$ limits need not commute. Nor is the above bound uniform as $t \to \infty$; late-time infrared logarithms require a separate analysis.

Theorem 32 (Infrared). *For spectra smooth on $(0, \infty)$, consider the $q \to 0$ region. By the Ward identities, $K_2(k, q) = -k \cdot \partial_k K_1(k) + O(q)$ and $K_3(k, q, -q) = (k \cdot \partial_k)^2 K_1(k) + O(q^2)$. Here the kernels are analytic in q at fixed finite times, and the absence of an $O(q)$ term in K_3 follows from its symmetry under $q \to -q$. The $O(q)$ cross term in $|K_2|^2$ is odd in $\hat{q}$ and integrates to zero.*

Both the q-long and the k-long regions of $C^{(22)}$ contribute. Let $\sigma_L^2 = \int_{q<\Lambda} P$ and $D = k \cdot \partial_k$.

The infrared-sensitive part of $\frac{1}{4}C^{(22)} + \frac{1}{3}\operatorname{Re} C^{(13)}$ *is then*

$$\frac{\sigma_L^2}{2\ell+1}\int_k P(k)\Big[\sum_m |DA^{(1)}_{\ell m}|^2 + \operatorname{Re}\sum_m A^{(1)}_{\ell m} D^2 A^{(1)*}_{\ell m}\Big] = \frac{\sigma_L^2}{2(2\ell+1)}\int_k \Big[\sum_m |A^{(1)}_{\ell m}|^2\Big](D+3)^2 P(k).$$

The equality uses two integrations by parts in $\ln k$ *and assumes vanishing radial boundary terms; for spectra with explicit cutoffs those boundary terms must be retained. Under that assumption the expression vanishes for scale-invariant* $P \propto k^{-3}$*. Otherwise it is the physical rescaling of the short-mode spectrum by the long-mode variance. The remainder is* $O(q^2)P(q)$*, which is integrable for* $\mathcal{P}(q)$ *with* $\int_0 \mathcal{P}(q) q^2 \,\mathrm{d}\ln q < \infty$*.*

Ultraviolet [power counting only]. At fixed finite times:

- the first-order kernels are bounded uniformly in k;
- each order of the sources carries at most two spatial derivatives;
- each ray displacement carries one.

Counting powers on this basis gives $|\mathcal{K}_2(k,q)| \lesssim q^2$ and $|\mathcal{K}_3(k,q,-q)| \lesssim q^4$ as $q \to \infty$. That suggests absolute convergence of $C^{(22)}$ and $C^{(13)}$ whenever $\int^\infty \mathcal{P}(q) q^4 \,\mathrm{d}\ln q < \infty$.

This is *not* a proof. A rigorous bound needs uniform estimates of the lapse Green function near $s \to 0$ (where $|G_N|$ grows like $1/(K^3 t s)$) against sources whose small-s Taylor coefficients grow with q. The true large-q behaviour for a scale-invariant spectrum, including possible cancellations, is [open].

10 Reduced representation of the complete cubic response

The complete map can be evaluated without an independent third-order field solve or a second-order photon ray. Its three ingredients are a measurement on first-order fields, a measurement Hessian on the known second-order fields, and a linear response to the complete cubic source. The Einstein constraints make the last response acoustic and wave-like: the hyperbolic scalar Green functions cancel before integration.

Angular and time conventions. Write $e = \bar{t}_\star$, $o = \bar{t}_o$, $\iota = \bar{t}_I$, with $0 < \iota < e < o$, and $X_0 = n(o-t)$. For a quadrupole covector C (symmetric and trace free), define

$$w = \frac{15}{2} n^T \mathsf{C} n, \qquad W = \frac{w}{4\pi}, \qquad J = \langle w\Theta\rangle_\Omega = \int W\Theta \,\mathrm{d}\Omega, \qquad C_K(t) = \int \mathrm{d}\Omega\, W\ \mathrm{TF}(nn) e^{iK\cdot n(o-t)}. \tag{76}$$

Thus $\langle w\ \mathrm{TF}(nn)\rangle_\Omega = \int W\ \mathrm{TF}(nn)\mathrm{d}\Omega = \mathsf{C}$. Every full-solid-angle objective, source pairing and multiplier below and in App. C uses W; sky means use w. For other multipoles, W is the corresponding zero-mean harmonic weight. The notation J_{abc} denotes the polarised cubic response of this objective, not a power spectrum.

10.1 The combined response

Proposition 33 (Combined cubic response)**.** *For an arbitrary triple with $P = k_a + k_b + k_c \neq 0$ and any zero-mean harmonic weight,*

$$\boxed{J_{abc} = J_{111,abc} - \sum_{a|bc} S_{\mathrm{meas},a}[\nu_{bc}, A_{bc}] + J_{\mathrm{lin}}[y_{abc}].} \tag{77}$$

Here J_{111} is the complete measurement on the first-field-only family $(\zeta, \nu, H) = \epsilon(g, a, h)$, including its physical roots, clock, frame and normalisation. The three partitions $a|bc$ each have coefficient one. Their fields $y_{bc} = (\nu_{bc}, A_{bc})$ are the closed-form physical second-order fields of §5, with the exact $K = 0$ kernel for a zero pair momentum. The functional S_{meas} is minus the reduced measurement Hessian; its explicit no-probe-ray form is (105). Finally,

$$\begin{aligned} J_{\mathrm{lin}}[y_{abc}] = & -3\int_e^o \ell_P(t)\Big(\nu'_{abc} - \frac{\nu_{abc}}{t}\Big)\mathrm{d}t - i\int_e^o \mathcal{P}_P^T[C_P(t)]\cdot c_M^{(3)}(t)\,\mathrm{d}t \\ & - \frac{1}{2}\int_e^o C_P : h_{abc}^{\mathrm{TT}\prime}\,\mathrm{d}t, \end{aligned} \tag{78}$$

where $\ell_P = \hat{P}\cdot C_P\cdot\hat{P}$, $\mathcal{P}_P^T[C] = 2CP/P^2 - (P\cdot C\cdot P)P/(2P^4)$, and the responses mean the explicit source integrals

$$\nu_{abc}(t) = c_{abc}v_P(t) - \int_0^t G_N^+(t,s)c_N^{(3)}(s)\mathrm{d}s, \qquad h_{abc}^{\mathrm{TT}}(t) = -2\int_0^t \frac{s}{t}\frac{\sin P(t-s)}{P}\,\mathrm{TT}[c_T^{(3)}](s)\mathrm{d}s.$$

Here $v_P = 9u_1(Pt/\sqrt{3})/P^2$, and G_N^+ is the retarded inverse of the lapse operator L_N, with the sign opposite to the source kernel in §5. The complete sources $c^{(3)}$ are local in the first- and second-order fields. No independent third-order field solve, first field adjoint or second ray is needed.

Proof. Apply the third derivative chain rule to the exact field-reduced measurement: $J_{abc} = J_{111,abc} + \sum_{a|bc} D_a(J_{\mathrm{red},y})[y_{bc}] + J_y[y_{abc}]$, where D_a differentiates along the frozen first direction (y_a, g_a). Each middle term is the measurement Hessian on that direction and the second-order state; by definition it is $-S_{\mathrm{meas},a}[y_{bc}]$. The cubic field equation is $\mathcal{L}_P y_{abc} + c^{(3)} = 0$, with its constrained initial row. The source is compatible by Prop. 34. Therefore Prop. 35 replaces its non-TT spatial response by the momentum integral. Contracting its time derivative with $-C_P/2$ uses $\mathcal{P}_P^T[C_P]\cdot P = 3\ell_P/2$ and gives (78). Equivalently, the four background-adjoint blocks of (101) combine into this single pairing with the complete cubic source. □

This supplies the cubic ingredient of (59): $R_\ell = \frac{1}{8\pi i^\ell}\int_{-1}^1 J_{abc}\,\mathrm{d}\mu$, with weight $W = P_\ell(\hat{z}\cdot n)$ and triple $(k_0\hat{z}, k_0\hat{q}_\mu, -k_0\hat{q}_\mu)$.

The direct measurement J_{111} is given without a second ray in (107)–(108). Its frame term is already included and must not be added again. Counting independent times after distributing products, every term of (77) has at most two time integrations except J_{111}, whose ray products have three. On the shell, the top-order response reduces to the acoustic nine-moment form (60). These are exact reductions, not evaluations of the surviving time integrals. At total momentum $P = 0$, replace the projected response by the regular Cartesian $K = 0$ integral of §5; no hyperbolic kernel is needed there either. Implementation comparisons are collected in App. F.1.

10.2 Why the constraints remove hyperbolic kernels

The Cartesian spatial operator has eigenvalue $-K^2/3$ on $\mathrm{TF}(KK)$. Its unconstrained scalar Green function is hyperbolic. Physical sources do not excite an independent growing scalar mode: the momentum rows replace the scalar/vector evolution rows, while the initial Hamiltonian row fixes the regular lapse coefficient.

Proposition 34 (Source compatibility)**.** *Let $(c_N, c_M, c_T, c_{\mathrm{Ham}})$ be the order-r sources of §5, with $L_{\mathrm{Ham}} = 4K^2\nu + 12\nu'/t + 12\nu/t^2 - K\cdot A\cdot K$. If all lower orders solve the field equations, then*

$$c'_{\mathrm{Ham}} - \tfrac{4}{t}c_N = 2iK\cdot c_M, \tag{79}$$

$$c'_M + \tfrac{2}{t}c_M = iK\cdot c_T - \tfrac{iK}{6}(c_{\mathrm{Ham}} + 4c_N). \tag{80}$$

Proof. Matter conservation is built in (Lemma 1), so the contracted Bianchi identity holds exactly. At order r, lower residuals vanish and its action is the background Bianchi operator on $\mathcal{L}_K[y_r]+c^{(r)}$. For arbitrary off-shell ν, A, direct differentiation of the linear rows gives

$$(L_{\mathrm{Ham}})' - \frac{4}{t}L_N - 2iK\cdot L_M = 0, \qquad L'_M + \frac{2}{t}L_M - iK\cdot L_T + \frac{iK}{6}(L_{\mathrm{Ham}} + 4L_N) = 0.$$

Subtracting these identities leaves (79)–(80) for the sources. The lower-order on-shell assumption is essential; an arbitrary mixed residual need not satisfy these source identities on its own. □

Proposition 35 (Constraint-row response)**.** *For $K \neq 0$ and compatible sources, the regular solution satisfies*

$$\boxed{K\cdot A(t) = V(t) = \int_0^t \big[4K(\nu' - \nu/s) + 2ic_M\big]\mathrm{d}s, \qquad A - \mathrm{TT}[A] = \mathcal{P}_K[V],} \tag{81}$$

where $\mathcal{P}_K[V] = \mathrm{TF}(K\otimes V + V\otimes K)/K^2 - (K\cdot V)\,\mathrm{TF}(KK)/(2K^4)$. The lapse ν is the acoustic response with its regular coefficient fixed by the initial Hamiltonian row. Equivalently, $h_S = 6\nu - 6\int_0^t \nu/s\,\mathrm{d}s + (3i/K^2)\int_0^t K\cdot c_M\,\mathrm{d}s$. Only the acoustic kernel, the TT wave kernel and one cumulative integral occur.

Proof. $\mathcal{P}_K[V]$ is trace free, has no TT part, and $K\cdot\mathcal{P}_K[V] = V$. Contract the Cartesian equation with K and use $K\cdot\mathfrak{B}_K\mathcal{P}_K[V] = -(K\cdot V)K/3$. The transverse equation is (80) transverse to K. For the longitudinal part put $E = 12(\nu'/t + \nu/t^2) + c_{\mathrm{Ham}}$. The lapse equation and (79) give $E' = -4K^2\nu/t + 2iK\cdot c_M$; the initial Hamiltonian row gives $E(0) = 36c_r + c_{\mathrm{Ham}}(0) = 0$. The remaining terms are the longitudinal part of (80). Thus $\mathcal{P}_K[V]$ solves the non-TT Cartesian equation with zero regular data, and uniqueness gives the result. □

The inverse powers of K in $\mathcal{P}_K$ are removable only in the recombined physical response. At $K = 0$ use the Cartesian kernel $s(t-s)/t$, which is already non-hyperbolic. On the shell this construction gives the single acoustic integral for F_S in Prop. 12. The weak transpose retains explicit emission and observer point terms ((111)); they cannot be replaced by zero terminal data.

10.3 The stationary explanation

Write the independent residual system of the complete experiment as $F(y, g) = 0$. The state y includes fields, rays, roots, clock, frame and normalisation. Let $A = F_y|_0$ have the regular inverse fixed by the physical boundary conditions and set

$$L = J + \langle\lambda, F\rangle, \quad A^*\lambda_0 = -J_y|_0, \quad Ay_a = -F_g[g_a], \quad A^*\lambda_a = -D_a(L_{0,y}),$$

where D_a differentiates along the frozen first direction (y_a, g_a) and the pairing includes every boundary residual.

Proposition 36 (Polarised stationary identity)**.**

$$\boxed{J_{abc} = D_a D_b D_c L_0 + \sum_{a|bc} \langle \lambda_a, D_b D_c F \rangle.} \tag{82}$$

No second or third state response and no second adjoint appears. Each partition has coefficient one; for a repeated direction, $J''' = D^3 L_0 + 3\langle \lambda_1, D^2 F \rangle$.

Proof. Differentiate along the implicit solution. The background adjoint removes y_{abc}. The remaining second-state terms satisfy $\langle D_a L_{0,y}, y_{bc} \rangle = -\langle \lambda_a, A y_{bc} \rangle = \langle \lambda_a, D_b D_c F \rangle$. □

This identity trades second-order fields for first adjoints. Formula (77) trades back to the already closed-form second-order fields and combines the background blocks before applying compatibility. The two representations serve different purposes: the stationary form exposes boundary duality, while the combined form is the smaller physical source representation. Their complete sources and boundary covectors are in App. C.

10.4 Evaluated consequences

The cubic quadrupole through four gradients. Scale shell momenta by κ, align the external unit vector with $(1, 0, 0)$ and choose $\mathsf{C} = \hat{k}\hat{k} - \mathbf{1}/3$. The complete internally averaged cubic kernel is

$$\overline{J}_{abc} = \kappa^2 \frac{4(o^2 - e^2)}{9} + \frac{\kappa^4}{3402}\left[3984e^4 - 6256e^3 o + 4692e^2 o^2 - 1853o^4 - 567\iota^4\right] + O(\kappa^6). \tag{83}$$

The homogeneous-lapse partition contributes $-\kappa^4(3o^4 + 5\iota^4 - 8e^4)/54$ and does not cancel. By contrast the homogeneous tensor drops out of the entire isotropic contraction: its response is proportional to $Q_{\hat{q}} = \hat{q}\hat{q} - \mathbf{1}/3$, and the remaining linear functional is independent of $\hat{q}$, so $\langle Q_{\hat{q}} \rangle_{\hat{q}} = 0$. The uncontracted map retains that tensor. The uncontracted four-gradient result agrees as a polynomial identity with the curvature map of §9.3; both gradient orderings are needed in $C^{(13)}$.

The exact dilation channel. For one finite momentum and two exactly zero momenta, $Y = e^s X$ maps $g = s + \epsilon e^{ip\cdot X}$ to $\epsilon e^{ie^{-s}p\cdot Y}$ and preserves the whole experiment, including its constrained seed, physical clock and frame. Uniqueness of the regular branch gives

$$\boxed{\Theta^{(3)}(p, 0, 0; n) = (p \cdot \partial_p)^2 \Theta^{(1)}(p; n), \qquad T_\ell^{(3)}(k, 0, 0) = (k\partial_k)^2 T_\ell(k).} \tag{84}$$

The derivative acts on the transfer profiles and the ray phase. By Prop. 16 the right side is explicit. This exact zero-momentum channel is not the nonzero-radius shell contraction and does not authorize exchanges of soft limits with spectral integrals.

The first nonzero observer-centred cubic coefficient. With the same quadrupole weight, isotropy removes the $(q, -q)$ and single-q dilation terms. Thus $\langle \mathcal{K}_3^c \rangle = \overline{J}_{abc} + 2\mathfrak{D}\langle \mathcal{K}_2(k, q) \rangle + \mathfrak{D}^2 \mathcal{K}_1(k)$. Substitution of the four-gradient quadratic map and the linear transfer cancels the degree-two term identically and gives

$$\boxed{\langle \mathcal{K}_3^c(k, q, -q) \rangle_{\hat{q}} = \frac{k^4}{270}\Big[4(e - o)(12e^3 - 8e^2 o + 7eo^2 + 7o^3) + 5(o^4 - \iota^4)\Big] + O(k^6).} \tag{85}$$

The first part vanishes with the photon path; the second is a pure observer-clock term. Its contribution to $\frac{1}{3}C_2^{(13)}$ at degree six is $(4\pi/25)\sigma^4(o^2-e^2)k_0^2\langle\mathcal{K}_3^c\rangle/9$.

Remark 37 (Initial-event dependence at fixed measurement labels). The clock roots are $t_I e^{\nu(t_I,0)} = \bar{t}_I$ and $\int_{t_I}^{t_o} te^{\nu(t,0)}\mathrm{d}t = (\bar{t}_o^2 - \bar{t}_I^2)/2$. Holding $\bar{t}_o$ fixed while varying $\bar{t}_I$ (thus adjusting the prescribed elapsed proper time) gives exactly

$$\left.\frac{\partial t_o}{\partial \bar{t}_I}\right|_{\bar{t}_o} = \frac{\bar{t}_I}{t_o e^{\nu_o}}\left[\frac{e^{-\nu_I}}{1+t_I\nu_I'} - 1\right].$$

At fixed elapsed proper time τ_o, the same formula holds without the -1. For the symmetric-frame coordinate map, initial-event dependence is entirely clock composition: $\partial_{\bar{t}_I}\Theta_{\mathrm{sym}} = (\partial_{t_o}\Theta_{\mathrm{sym}}^{\mathrm{coord}})\,\partial_{\bar{t}_I}t_o$. For the specified Fermi–Walker map on the shell this identity holds through cubic amplitude, with an $O(\epsilon^4)$ remainder, since $[H_1, H_1'] = 0$ there. Mixed higher-order commutators need not vanish. Nevertheless the two frames differ by a global sky rotation at each observation event, so every realization's $\sum_m |a_{\ell m}|^2$ is exactly unchanged. Frame independence of the spectrum does not require all-orders equality of the maps. At fixed $\bar{t}_o$, the first-order clock coefficient obeys $\partial_{\bar{t}_I} D_1 = -\bar{t}_I[\nu_1(\bar{t}_I,0) + \bar{t}_I\nu_1'(\bar{t}_I,0)]/\bar{t}_o$. This accounts for the ι^4 dependence in (85) and the degree-six cross term, localising it in the clock and initial-event sector.

11 What remains unevaluated

The finite-time integral representation is complete under the compact-spectrum assumptions of Thm. 25. The remaining questions concern evaluation, further structure and extension of its domain:

1. *Finite-wavelength shell integrals.* Evaluate the surviving relative-angle and two-/three-time integrals in (77). A focused target is $\langle\mathcal{K}_3^c(k,q,-q)\rangle_{\hat{q}}$, the observer-centred part left after the exact dilation split. Its first nonzero coefficient is (85). The shell-averaged direct measurement J_{111} and its combination with $S_{\mathrm{meas}}[y_{bc}]$ are natural next steps: match their phases and physical boundary terms before integrating. The nine equal-time moments do not automatically close mixed-time propagation products. Even at linear-transfer nodes, where only $C^{(22)}$ survives, the remaining integral has not been evaluated in closed form.

2. *A structural explanation and the next gradients.* Derive the two projected factor-two identities of §8.3 from a geometric or polarisation principle rather than coefficient comparison. No higher-order extension is established. The $O(k_0^8)$ quadrupole requires the six-gradient map as well as the square of the four-gradient map; the next octupole requires the five-gradient map. Squaring the existing result alone is insufficient.

3. *Fourth order inside the horizon and at late times.* Control the non-uniform collinear and tensor-resonant regions before exchanging angular integration with a large-time limit. The exact longitudinal-redshift identity is available, but the nonlinear gauge generators and the fourth-order asymptotics are not. A prospective route is recorded in App. D.

4. *Noncompact spectra.* Prove uniform ultraviolet bounds and control the soft expansion over unbounded external momenta. The current ultraviolet estimate is power counting, and the infrared theorem retains its stated boundary assumptions. The compact-spectrum existence result does not by itself settle either extension.

The expansion is order by order on the regular branch, with a single emission intersection and no caustics. No convergence in ϵ is asserted. The exact Gaussian averages of the leading-gradient maps do not justify exchanging the full nonlinear solution, an infinite amplitude series and an ensemble average.

A Requirements and scope

Requirement	Result (section; status)
Gaussian variable; no higher-order modes	$\zeta = \frac{1}{6}\ln\det\gamma_{ij}$, exactly conserved; Def. 6 (§4)
Constrained initial data at every order	regular solution at any t_i; constraints proved (Thm. 7); forced data nonzero
Field solution through ϵ^3 incl. vectors/tensors	triangular scheme (§5); 2nd order [closed form]; 3rd order [explicit nested integrals]; contracted shell cubic sources: [nine moments, angle integrals evaluated] (§5.3.1); scalar moments hyperbolic-free (Prop. 12)
Observer, clock, tetrad, emission surface	§6; explicit Fermi–Walker rotation, irrelevant to the spectrum at this order (Prop. 13)
Complete $\Theta^{(1)}, \Theta^{(2)}, \Theta^{(3)}$	§6, with an independent generator and Ward checks; propagation adjoint removes the second-order-field ray response, with emission, clock and normalisation retained (§C.1); combined form with no independent third-order field solve, second ray or first field adjoint (Prop. 33); cubic quadrupole [verified through four gradients at arbitrary triples], dilation channel [closed form] (§10)
$C_\ell^{(11)}$	[closed form, every ℓ]: entire-function T_ℓ (Prop. 16); the proof is analytic for every ℓ, and the script checks $\ell = 1, \ldots, 6$
$C_\ell^{(22)}$, $C_\ell^{(13)}$	[exact finite-dimensional integrals], organised as $C_\ell = 4\pi v T_\ell^2 + v^2[Q_\ell + 4\pi T_\ell \operatorname{Re} R_\ell]$ ((59)); sky integrals evaluated (§8); shell metric reduced to two amplitudes, contracted cubic response scalar; existence proved for finite times and finite spectral measures of compact support in $0 < k < \infty$; log-map decomposition with the lognormal part [explicit] (convergent $3j$ series, or one Legendre integral of ξ^2; Thm. 18)
Monochromatic, held-fixed quantity	$\langle g^2\rangle = \sigma^2$ fixed (§8.4)

Requirement	Result (section; status)
Asymptotic limits	superhorizon: [closed form], including an all-amplitude Gaussian average of the leading-gradient map, factorised into the random observer dilation $\times$ a curvature norm ((65)); leading octupole [closed form] (§9.2); complete fourth-order C_2 at the next gradient order $O(k_0^6)$: [closed form], via three local curvature tensors ((73)), checked by two independent routes; late time: [first order]; subhorizon first order: [closed form], Sachs–Wolfe + Doppler (Prop. 30); general k_0 at fourth order: [unevaluated integrals] (§9)
Convergence for general spectra	finite spectral measure of compact support in $0 < k < \infty$: [proved]; IR: leading cancellation [derived] from the Ward identities; UV: [power counting only]
Dipole and normalisations	dipole retained through the nonlinear normalisation and removed only afterwards ((29)); perturbative, Fourier, convolution and angular conventions fixed in §2 and §8
Constraints	proved at every order (Thm. 7); all ten equations checked by exact series through $O(t^8)$ at second order and $O(t^6)$ at third order (§E)
Fluid normalisation; null and geodesic equations	$u = N^{-1}\partial_t$ is unit and Euler holds identically in the exact frame (Lemma 1); exact conformal null-geodesic and redshift laws for arbitrary fields (Lemma 2); the ray expansion is checked against those definitions for arbitrary fields (§E)
Gauge invariance at fixed labels	all labels are physical scalars or frames (§7); dilation Ward identities for fields and map, special-conformal null test and soft-gradient identity (§7.1)
Verification record	§E (exact symbolic, generic-jet proofs)

B Entire-function formulas for the nine angular moments

The acoustic scalar representation of Prop. 12 is the preferred physical form. This appendix retains the Cartesian moment construction as an alternative exact representation, including its hyperbolic terms before constraint cancellation.

Evaluated relative-angle moments. No cubic kernel is needed to compute the nine functions. Put $y = (1+\mu)/2$ and $F_\beta(z) = {}_0F_1(;\beta;z)$. The regular Green kernels, with $d = t - s$, are

$$G_N(t,s;y) = -\tfrac{1}{3}[dF_{3/2}(-c^2d^2y) + d^3F_{5/2}(-c^2d^2y)/(3ts)], \tag{86}$$

$$G_S(t,s;y) = (s/t)dF_{3/2}(c^2d^2y), \qquad G_T(t,s;y) = (s/t)dF_{3/2}(-k^2d^2y). \tag{87}$$

The positive argument in G_S is the scalar eigenvalue $-K^2/3$ of the Cartesian operator; the lapse forcing must be retained for its cancellations. Prop. 12 shows that this cancellation is exact. The momentum constraint gives F_S with acoustic kernels and one time integral, so (91) and (92) can be bypassed. At $y=0$, $G_N=-(t^3-s^3)/(9ts)$, exactly the homogeneous-momentum Green kernel. Expand the explicit polynomials above after $\mu=2y-1$ as

$$C_N=\sum_{j=0}^{2}\mathfrak{c}_j(t)y^j,\qquad J_S=\sum_{j=0}^{3}\mathfrak{s}_j(t)y^j,\qquad J_T=\sum_{j=0}^{3}\mathfrak{t}_j(t)y^j,$$
$$c_2^{\mathrm{init}}=d_0+d_1y,\qquad (d_0,d_1)=(5k^2/9,-2k^2/9).$$

Define entire universal functions

$$\mathcal{A}_m^{\beta}(z)=\frac{{}_1F_2(m+1;\beta,m+2;z)}{m+1},\qquad \mathcal{B}_m^{\beta,\delta}(z,w)=\sum_{r,h\ge 0}\frac{z^r w^h}{(\beta)_r(\delta)_h r!h!(m+r+h+1)}. \tag{88}$$

For $m=0,1,2$, the complete angle-free representations are

$$\mathcal{N}_m(t)=t^2\sum_{j=0}^{1}d_j\mathcal{A}_{m+j}^{5/2}(-c^2t^2)$$
$$+\int_0^t \mathrm{d}s\sum_{j=0}^{2}\mathfrak{c}_j(s)\left[-\frac{t-s}{3}\mathcal{A}_{m+j}^{3/2}(-c^2(t-s)^2)-\frac{(t-s)^3}{9ts}\mathcal{A}_{m+j}^{5/2}(-c^2(t-s)^2)\right], \tag{89}$$

$$\mathcal{T}_m(t)=\int_0^t \mathrm{d}s\,\frac{s(t-s)}{t}\sum_{j=0}^{3}\mathfrak{t}_j(s)\mathcal{A}_{m+j}^{3/2}(-k^2(t-s)^2), \tag{90}$$

$$\mathcal{S}_m(t)=\int_0^t \mathrm{d}s\,\frac{s(t-s)}{t}\sum_{j=0}^{3}\mathfrak{s}_j(s)\mathcal{A}_{m+j}^{3/2}(c^2(t-s)^2)-16k^2\int_0^t \mathrm{d}s\,\frac{s(t-s)}{t}\mathcal{V}_m(t,s), \tag{91}$$

where $z=c^2(t-s)^2$ and

$$\mathcal{V}_m(t,s)=s^2\sum_{j=0}^{1}d_j\mathcal{B}_{m+1+j}^{3/2,5/2}(z,-c^2s^2)$$
$$+\int_0^s \mathrm{d}r\sum_{j=0}^{2}\mathfrak{c}_j(r)\left[-\frac{s-r}{3}\mathcal{B}_{m+1+j}^{3/2,3/2}(z,-c^2(s-r)^2)-\frac{(s-r)^3}{9sr}\mathcal{B}_{m+1+j}^{3/2,5/2}(z,-c^2(s-r)^2)\right]. \tag{92}$$

Then $N_r=\sum_{m=0}^{r}\binom{r}{m}2^m(-1)^{r-m}\mathcal{N}_m$, and likewise for S_r,T_r. The separately retained homogeneous lapse is

$$n_h=5k^2t^2/9-\int_0^t\frac{t^3-s^3}{9ts}C_N(s,-1)\mathrm{d}s. \tag{93}$$

These formulas follow by substituting the Green representations and integrating powers of y times one or two F_β factors. All required universal indices are at most five.

The hypergeometric notation is optional. With $\alpha^2=-4z$,

$$F_{3/2}(-\alpha^2u^2/4)=\frac{\sin\alpha u}{\alpha u},\qquad F_{5/2}(-\alpha^2u^2/4)=\frac{3(\sin\alpha u-\alpha u\cos\alpha u)}{\alpha^3u^3}.$$

Set $y = u^2$, multiply by $2u^{2m+1}$, and expand the sines and cosines. The result is a finite Laurent-exponential sum. Nonnegative powers integrate to $\mathfrak{e}_p(i\omega)$; powers u^{-M} integrate to $J_M(\omega, 1)$. The discarded principal parts sum to zero. The largest pole is u^{-3}, so only J_1, J_2, J_3 are needed, also for imaginary frequencies in G_S. For example

$$\mathcal{B}_0^{3/2,3/2}(-\alpha^2/4, -\gamma^2/4) = \frac{\mathrm{Cin}(\alpha+\gamma) - \mathrm{Cin}(\alpha-\gamma)}{\alpha\gamma}.$$

Every apparent zero-argument division is defined by the entire series (88). At finite times those series converge locally uniformly and absolutely; $C_N(s,\mu) = O(s^2)$ uniformly in μ, while $J_S, J_T = O(1)$. Thus the $1/s$ and $1/r$ factors are integrable and all angular/time exchanges above are justified. No late-time exchange is asserted. The relative-angle integrals for the *top-order cubic fields* are evaluated; the displayed time integrals and the lower-order propagation products in the full map are not.

C Cubic response: adjoints, boundaries and Green kernels

The conventions for w, W and J are those of §10.

C.1 Adjoint reduction of the second-order-field propagation products

The part of the second-order ray driven by H_2 can be eliminated exactly. This subsection isolates *all terms linear in the second-order fields* H_2 and ν_2, excluding the top-order fields H_3, ν_3 and the remaining products made only of first-order fields. It is a decomposition of the already complete map, not an omission or a new initial-state assumption. The adjoint below holds at general momenta and finite times.

Write $h = H_1$, $j = H_2$, $m = \nu_2$, $P = 1 - nn^T$, $e = \bar{t}_\star$, $o = \bar{t}_o$, $\bar{X} = n(o-t)$. In this subsection h', j' denote partial time derivatives before pullback; derivatives of ray and adjoint variables are total derivatives along $\bar{X}$. Spatial derivatives act only on the fields. Let $X = X_1$, $v = v_1$ be the known first-order orthonormal ray, and let X_j, v_j denote only the part of X_2, v_2 linear in j. Since $\zeta^{(2)} = 0$, their equations and observer data are

$$X_j' = v_j + \tfrac{1}{2} jn, \qquad v_j' = G_j = \tfrac{1}{2} P j' n + \tfrac{1}{2}[\nabla j_{nn} - (n\cdot\partial)(jn)], \qquad X_j(o) = v_j(o) = 0. \tag{94}$$

The lapse m does not deflect conformal null rays; it enters this sector only through the physical emission surface and observer clock. Put

$$\mathfrak{B}[h,j] = \int_e^o \left[\tfrac{1}{2} X\cdot\nabla j_{nn}' - j'(n,v) + \tfrac{1}{2} X_j\cdot\nabla h_{nn}' - h'(n,v_j)\right] \mathrm{d}t. \tag{95}$$

The cubic bulk contribution linear in j is $3\mathfrak{B}[h,j]$ in the full-field Taylor convention. Kernel polarisation instead gives the sum over its three single–pair partitions, with no extra factor three.

Proposition 38 (Propagation adjoint)**.** *Define first-order-only functions*

$$r = \tfrac{1}{2}\nabla h_{nn}', \qquad s = -Ph'n, \qquad \boldsymbol{Y} = v + \tfrac{1}{2} P\boldsymbol{K},$$
$$\boldsymbol{R}(t) = \int_e^t r(u)\mathrm{d}u, \qquad \boldsymbol{K}(t) = \int_e^t [s(u) - \boldsymbol{R}(u)]\mathrm{d}u.$$

Then the unknown second-order ray can be eliminated exactly:

$$\mathfrak{B}[h,j] = \int_e^o \left[\tfrac{1}{2} X\cdot\nabla j_{nn}' - j'(n,v) - \tfrac{1}{2}\boldsymbol{R}\cdot jn - \boldsymbol{K}\cdot G_j\right] \mathrm{d}t. \tag{96}$$

A further integration by parts removes all partial time derivatives of j from the bulk:

$$\boxed{\begin{aligned}\mathfrak{B}[h,j] = {}& [\tfrac{1}{2}X\cdot\nabla j_{nn} - \boldsymbol{Y}\cdot jn]_e^o \\ &+ \int_e^o \big\{\tfrac{1}{2}X\cdot\nabla(n\cdot\partial)j_{nn} - \tfrac{1}{2}(X' + P\boldsymbol{K})\cdot\nabla j_{nn} \\ &\qquad -v\cdot(n\cdot\partial)(jn) + (\boldsymbol{Y}' - \tfrac{1}{2}\boldsymbol{R})\cdot jn\big\}\,\mathrm{d}t.\end{aligned}} \tag{97}$$

Only first-order ray and adjoint functions and the second-order metric remain. In particular, $\boldsymbol{R}$ must not be projected transversely in its coupling to jn: the coordinate displacement has a longitudinal part even though v_j is transverse.

Proof. Differentiate $\boldsymbol{R}\cdot X_j + \boldsymbol{K}\cdot v_j$ and use (94). Its derivative is $r\cdot X_j + s\cdot v_j + \boldsymbol{R}\cdot jn/2 + \boldsymbol{K}\cdot G_j$. Its endpoint values vanish, because the adjoints vanish at e and the ray responses at o. This proves (96). In the remaining j' terms use $\partial_t j = \mathrm{d}j/\mathrm{d}t + (n\cdot\partial)j$ on the background ray. The longitudinal $\boldsymbol{K}$ pieces cancel between the two spatial-derivative terms, leaving (97). The resulting metric boundary is *not* zero: at o, $X = v = 0$ but $\boldsymbol{Y} = P\boldsymbol{K}/2$ need not vanish. □

Emission, physical clock and normalisation. Define $F_h = h'_{nn}/2$, $F_j = j'_{nn}/2$, $U_h = \int_e^o F_h\mathrm{d}t$, $U_j = \int_e^o F_j\mathrm{d}t$ and

$$\mathcal{C}[m] = \frac{-\bar{t}_I^2 m(\bar{t}_I,0) - \int_{\bar{t}_I}^o t m(t,0)\mathrm{d}t}{o}.$$

This is the part of D_2 linear in m, not an additional clock correction. The complete composed-exponent contribution of this sector is

$$\widehat{I_3}\big|_2 = 3\mathfrak{D}_2, \qquad \mathfrak{D}_2 = \mathfrak{B}[h,j] + [eF_h m - F_j d_1]_e + D_1\partial_o U_j + \mathcal{C}[m]\partial_o U_h. \tag{98}$$

Here $d_1 = -e\nu_1(e,\bar{X}_e)$, and D_1 is the original physical-clock coefficient. The ∂_o derivatives include all background-ray and upper-limit dependence, with e fixed. After centring, the corresponding full normalised-map contribution is

$$\Theta^{(3)}\big|_2 = 3\mathsf{P}_0\left[-\mathfrak{D}_2 + (U_h - \langle U_h\rangle_\Omega)(U_j - \langle U_j\rangle_\Omega)\right]. \tag{99}$$

Thus the second-order-map normalisation product cannot be discarded. The dipole is still retained in that product. All terms made only of first-order fields, including the mixed clock–emission product in (31), remain outside this isolated sector and must be added as before. The Fermi–Walker correction is also first-order-field-only at this map order; it does not add a j-dependent term here.

An evaluated shell emission contraction. The emission part of this sector already fits the nine-moment construction. Let $\mathcal{C}_S(t) = \langle q_S F_S + q_T F_T\rangle_\mu$ (so $\Upsilon = \mathsf{a}\mathcal{C}_S$), $Q_k = \hat{k}\hat{k} - \delta/3$. After polarising the full cubic expression and averaging over $\hat{q}$, the emission-exponent kernel is exactly

$$\overline{\mathcal{I}_{3,e}\big|_2}(k;n) = \frac{e}{2}\left[\mathsf{b}'(n_h + 3N_2 - N_0) + 3\mathsf{a}\mathcal{C}'_S\right]_e Q_{k,nn}\, e^{ik\cdot n(o-e)}. \tag{100}$$

Indeed the two mixed partitions share one emission time, so their phases combine to the external phase. The azimuthal averages are $\overline{Q_q} = P_2(\mu)Q_k$ and $\overline{H_2(k,q)} = \frac{3}{2}(q_S F_S + q_T F_T)Q_k$. The homogeneous tensor averages to zero and the homogeneous lapse remains. For $\ell \geq 2$, its temperature-transfer contribution is

$$\mathcal{A}_{\ell,e}\big|_2 = \frac{e}{2}w_\ell(k(o-e))[\mathsf{b}'(n_h + 3N_2 - N_0) + 3\mathsf{a}\mathcal{C}'_S]_e.$$

The sign follows from $\Theta = -I$ at linear order in an exponent kernel and the $Q_{k,nn}$ Funk–Hecke projection. This contribution can be added to the explicit emission boundary of (60); no identity setting their sum to zero is asserted.

What this does and does not settle. The bulk reduction and complete sector (98) are checked exactly in `solution/check_propagation_adjoint.py`: matrix-exponential driver, arbitrary plane-wave jets, ray duality, boundary and clock coefficients, centring algebra and shell emission projections, with omission controls. It does not evaluate the full propagated 13 response. In a mixed-time product, for example,

$$(k+q)\cdot n(o-t) - q\cdot n(o-u) = k\cdot n(o-t) + q\cdot n(u-t).$$

The second phase is present unless $u = t$. Consequently the finite nine polynomial moments proved for the top-order fields and shared-time emission terms cannot simply be assumed to close the remaining propagation integrals. No cancellation of the full cubic response or late-time limit is proved. The unevaluated remainder is the combination of (60), (97), the clock terms and the remaining first-order-only products, with the displayed physical boundaries retained.

C.2 Field-reduced stationary functional

Field-reduced form. Eliminating the rays, endpoints, clock, frame and normalisation into an exact reduced observable, and keeping as field rows the lapse equation (N), the five orthonormal trace-free rows $2e^{H/2}(T)e^{-H/2}$, and the initial row $\lim_{t\to 0}\nu/t^2 - c[g]$, gives

$$\begin{aligned} J_{abc} = J_{111,abc} &+ \int_0^o \mathrm{d}t\Big[\Lambda_{N,0}C_{N,abc} + \Lambda_{H,0} : C_{H,abc} + \sum_{a|bc}\big(\Lambda_{N,a}C_{N,bc} + \Lambda_{H,a} : C_{H,bc}\big)\Big] \\ &- \Lambda_{c,0}c_{abc} - \sum_{a|bc}\Lambda_{c,a}c_{bc}. \end{aligned} \tag{101}$$

Here J_{111} is the complete measurement on the first-field-only family $(\zeta,\nu,H) = \epsilon(g,a,h)$, $C_{N,H}$ are the direct residual derivatives along that family, and c_{bc}, c_{abc} are the report's initial kernels (§5). There are three partitions, each with coefficient one; the background adjoint uses the total momentum and the first adjoint the pair momentum $k_b + k_c$, which may vanish. The Hamiltonian and momentum constraints are *not* added as further multiplier rows: they are consequences (Thm. 7), and adding them would make A redundant (§10.2 instead *replaces* rows).

Background transpose and the regular-origin covector. At momentum K the inverse of $L_K\nu = f_N$, $L_H H + 4Q_K\nu = f_H$ is $\nu = b_c v_K + \int_0^t G_N^+ f_N$ and $H = \int_0^t \mathfrak{G}_K[f_H - 4Q_K\nu]$, with $v_K = \frac{9}{K^2}u_1(Kt/\sqrt{3})$ and G_N^+ the retarded lapse kernel (opposite sign to the coefficient of c_N in §5). Its transpose has an initial-data covector that does not vanish:

$$\Lambda_c = 9\lim_{s\to 0} s\Lambda_N(s), \tag{102}$$

from $\lim_{s\to 0} 9sG_N^+(t,s) = v_K(t)$. For the linear quadrupole objective, $J_y[\delta H] = -\frac{1}{2}\int_e^o C_K : \delta H'\mathrm{d}t$ with $C_K(t) = \int \mathrm{d}\Omega\, W\ \mathrm{TF}(nn)e^{iK\cdot n(o-t)}$, and

$$\Lambda_{H,0}(s) = \tfrac{1}{2}\int_{\max(s,e)}^o \partial_t\mathfrak{G}_K(t,s)[C_K(t)]\mathrm{d}t, \quad \Lambda_{N,0}(s) = -4\int_s^o G_N^+(t,s)Q_K : \Lambda_{H,0}(t)\mathrm{d}t.$$

The derivative acts on the kernel before any exchange of integrals, so the observer and emission boundaries are kept. The nonlinear initial matching contributes $-\Lambda_{c,0}c_{abc} - \sum\Lambda_{c,a}c_{bc}$ in (101); dropping these rows loses part of the physical answer (see the gradient check below).

Exact normalisation adjoint. With $F_Z = Z - e^{-I}$, $F_M = M - \langle Z\rangle_\Omega$, $F_\Theta = M(1+\Theta) - Z$ and $\int W\,\mathrm{d}\Omega = 0$, the multiplier of the exponent is

$$\lambda_I = (1+\Theta)\Big[W - \frac{J}{4\pi}\Big], \qquad \lambda_{I,a} = W\Theta_a - \frac{1}{4\pi}\int W\Theta_a\,\mathrm{d}\Omega, \tag{103}$$

so $\int \lambda_I\,\mathrm{d}\Omega = 0$ exactly, and the random monopole has already cancelled from the first adjoint source. The emission and clock multipliers have first variations $\lambda_{\mathrm{em},a} = -e\,W F_a(e)$ and $\lambda_{\mathrm{clock},a} = -\partial_o J^c_a/(a_1 o)$; together they reproduce the second-order clock functional $\mathcal{C}[m] = -[\iota^2 m(\iota,0) + \int_\iota^o tm(t,0)\mathrm{d}t]/o$.

The constant-direction source identities. For the dilation channel of (84), the direct residuals satisfy $C_{p00} = -2C_{p0}$, $c_{p00} = -4p^2/9 = -2c_{p0}$ and $C_{00} = 0$. The reduced constant-direction adjoints are $-p\cdot\partial_p$ of the background ones. These identities fix an exact zero-momentum channel, not a finite-radius shell average.

A physical check. Applied to the leading-gradient map $\Theta_{\mathrm{lead}} = -\tau^2 e^{-2\epsilon g}[\epsilon\mathsf{A} - \epsilon^2\mathsf{B}]$ of §9.1, with the nonlinear seed as a residual, the identity gives $\Theta^{(3)}_{\mathrm{lead}} = -12\tau^2(g^2\mathsf{A} + g\mathsf{B})$, in agreement with direct differentiation. Omitting the seed residual loses part of the $g^2\mathsf{A}$ term.

C.3 The complete first field-adjoint source

The first field adjoint solves $A^*\Lambda_a = \mathcal{S}_a$, with $\mathcal{S}_a[b] = -D_a J_{\mathrm{red},y}[b] - \langle \Lambda_0, D_a E_y[b]\rangle$, on an independent *state* probe $b = (0, m(t), j(t))e^{iq\cdot X}$ (no primordial part). For the first input $(\zeta,\nu,H) = (z,a,h)e^{ip\cdot X}$, $G = 2z\mathbf{1} + h$, $P = p + q$, $d = p\cdot q$, the off-shell mixed residuals are

$$\begin{aligned} C_N &= \big[-q^T hP - z(2q^2 - d) + 4da + 12a/t^2\big]m - a\,p^T jP + \tfrac{1}{2}h' : j', \\ C_H &= 2a'j' + 2m'h' + 2\,\mathrm{TF}\,\big[\mathcal{R}_{p,q}(G,j) - 2\mathcal{U} - 2am(pq^T + qp^T)\big], \end{aligned} \tag{104}$$

with $\mathcal{U} = \frac{1}{2}\{m[P(Gq)^T + (Gq)P^T - dG] + a[P(jp)^T + (jp)P^T - dj]\}$ and $\mathcal{R}_{p,q}$ the finite bilinear orthonormal Ricci coefficient (an explicit polynomial in the momenta; no denominators). There are no second time derivatives of the probe. Two exact controls: a homogeneous lapse probe ($q = 0$, $j = 0$) gives $C_N = 12am/t^2$, $C_H = 2m'h'$; a constant primordial shift gives $C_N = -2zq^2 m$, $C_H = -2z\mathfrak{B}_q j - 8zQ_q m$, the finite-wavelength dilation e^{-2z}.

Measurement Hessian without a probe ray. Let $\phi_q = e^{iq\cdot n(o-t)}$, $N_n = nn^T - \frac{1}{3}$, $F_a = \frac{1}{2}\phi_p h'_{nn}$, $U_a = \int_e^o F_a$, and let X_a, v_a be the known fixed-coordinate-observer first ray, with propagation adjoints $R_a = \int_e^t \frac{1}{2}\nabla(\phi_p h'_{nn})$, $K_a = \int_e^t (s_a - R_a)$, $s_a = -P_n\phi_p h' n$ (as in Prop. 38). With $\lambda_{I,a}$ from (103), the first clock shift $D_a = \mathcal{C}[a]$ and the sky-integrated $A_a = \int W\,\partial_o U_a\,\mathrm{d}\Omega$,

$$\begin{aligned} S_{\mathrm{meas},a}[m,j] = &\int_e^o \mathrm{d}t \int \mathrm{d}\Omega\,\phi_q\Big\{WE_0 : j + \big[WE_1 + \tfrac{1}{2}\lambda_{I,a}N_n + \tfrac{i}{2}WD_a(q\cdot n)N_n\big] : j'\Big\} \\ &+ \int \mathrm{d}\Omega\,\phi_q(e)\Big[e\,WF_a(e)m(e) + \tfrac{e}{2}Wa_e N_n : j'(e)\Big] + \frac{D_a}{2}\int \mathrm{d}\Omega\,WN_n : j'(o) \\ &- \frac{A_a}{o}\Big[\iota^2 m(\iota) + \int_\iota^o tm(t)\mathrm{d}t\Big], \end{aligned} \tag{105}$$

where $E_1 = \frac{i}{2}(q\cdot X_a)N_n - \mathrm{TF}\,\mathrm{sym}(v_a n^T) - \frac{1}{2}\,\mathrm{TF}\,\mathrm{sym}((P_nK_a)n^T)$ and $E_0 = -\frac{1}{2}\,\mathrm{TF}\,\mathrm{sym}(R_a n^T) - \frac{i}{2}(q\cdot K_a)N_n + \frac{i}{2}(q\cdot n)\,\mathrm{TF}\,\mathrm{sym}(K_a n^T)$. The probe ray has been removed by the same duality as in

§C.1; its total derivative vanishes only by $R_a(e) = K_a(e) = 0$ and $X_b(o) = v_b(o) = 0$. The clock line lies on the observer worldline and is not integrated over the sky again. The complete source is $\mathcal{S}_a = S_{\mathrm{meas},a} - \int_0^o (V_\nu m + D_\nu m' + V_H : j + D_H : j')\mathrm{d}t$, where V_ν, D_ν, V_H, D_H are the finite weak transposes of (104) against $\Lambda_{N,0}, \Lambda_{H,0}$ at momentum P.

Frame. With $R = e^{g(0)} e^{H(t,0)/2} E_{\mathrm{FW}}$, $R' = \Omega R$ and $\Omega_{ab} = \frac{1}{8}([h_a, h_b'] + [h_b, h_a'])$, the first rotation vanishes, so the frame does not source the first *field* adjoint. Its first adjoint is $\Lambda_{R,a} = -\mathcal{K}_a$, $\mathcal{K}_a = \int \mathrm{d}\Omega\, \mathrm{skew}[W(\nabla_{S^2}\Theta_a)n^T]$, and the cubic frame pairing $\sum_{\mathrm{cyc}} \int_\iota^o \mathcal{K}_a : \Omega_{bc}\mathrm{d}t$ is retained. It vanishes on the monochromatic shell, where $[h_a, h_b'] = 0$, but not for general triples.

Regular reconstruction and its boundaries. On the causal Green probes $\Phi_H = (0, \mathfrak{G}_q[B])$, $\Phi_N = (G_N^+, -4\int \mathfrak{G}_q Q_q G_N^+)$ and the origin probe Φ_c, the first adjoint is $\Lambda_{H,a}(s) : B = \mathcal{S}_a[\Phi_H]$, $\Lambda_{N,a} = \mathcal{S}_a[\Phi_N]$, $\Lambda_{c,a} = \mathcal{S}_a[\Phi_c]$. The point evaluations in (105) force

$$\Lambda_{H,a}(o^-) = \frac{D_a}{2}\int W N_n \,\mathrm{d}\Omega = \frac{D_a}{2}\mathsf{C}, \qquad \Lambda_{H,a}(e^-) - \Lambda_{H,a}(e^+) = \frac{e}{2}\int \mathrm{d}\Omega\, W a_e \phi_q(e) N_n, \tag{106}$$

and $\Lambda_{c,a} = 9\lim_{s\to 0} s\Lambda_{N,a}(s)$. Imposing zero terminal data or a continuous spatial adjoint would lose the physical clock and emission terms.

C.4 The direct first-field measurement without a second ray

For the first-field-only family write $v_2 = u_2 + n|v|^2$ with $n \cdot u_2 = 0$ (the radial part is fixed by normalisation). Then $X_2' = u_2 + Q_X$ and $u_2' = Q_v$, where $Q_X = n|v|^2 - Gv - \frac{1}{4}G^2 n + (X\cdot\nabla G)n$ and Q_v is the explicit transverse forcing built from first fields and first rays (working note `stationary_cubic_assembly.md`, eq. (6)). The same adjoints R, K give the exact direct cubic line of sight

$$\boxed{U_3 = \int_e^o \left[L_{111} - 3R\cdot Q_X - 3K\cdot Q_v\right]\mathrm{d}t - 3K(o)\cdot R_{\mathrm{FW},2}\, n, \qquad R_{\mathrm{FW},2} = \tfrac{1}{4}\int_\iota^o [h(t,0), h'(t,0)]\mathrm{d}t,} \tag{107}$$

with the local remainder $L_{111} = \frac{1}{8}[h,[h,h']]_{nn} + \frac{3}{2}X^iX^j\partial_i\partial_j h'_{nn} - 6v^T(X\cdot\nabla h')n + 3v^T h' v - 3|v|^2 h'_{nn}$. No second ray remains; the frame enters through the observer value $u_2(o) = -R_{\mathrm{FW},2}n$ and must not be added a second time through Λ_R. With the physical emission shifts $d_1 = -ea_e$, $d_2 = e[a_e^2 - 2X_e\cdot\nabla a_e - 2d_1(D_t a)_e]$, the clock shift $D_1 = \mathcal{C}[a]$ and

$$D_2 = \frac{2\iota^3 a(\iota,0)a'(\iota,0) - \int_\iota^o t a(t,0)^2\mathrm{d}t - 2oa(o,0)D_1 - D_1^2}{o}, \tag{108}$$

the exponent derivatives $I_{1,2,3}$ follow by composition (including $-6D_1[(\partial_o F_{1,e})d_1 + F_{1,e}\partial_o d_1]$), and the exact normalised map is $\Theta^{(3)}_{111} = P_0[-I_3 + 3ub - u^3 + 3\langle u^2\rangle_\Omega u]$ with $u = I_1 - \langle I_1\rangle_\Omega$, $b = I_2 - \langle I_2\rangle_\Omega$. Along the same family $D^3\,\mathrm{tr}\,S^2 = 0$, and the temporal part of the direct cubic spatial residual is $\frac{1}{4}[h,[h,h'']] + [h',[h,h']] + \frac{1}{2t}[h,[h,h']]$. *On the monochromatic shell every temporal commutator here and in* (107) *vanishes pointwise*, so the direct three-first-field Einstein source has no time derivatives of first fields; spatial nonlinearities and transport remain.

The common fourth time. The only four-time term, $-3\int_e^o (n\cdot R)|v|^2\mathrm{d}t$, has its common time integrated exactly (its integrand depends on it only through the integration domain):

$$U_{3,R;abc} = -\sum_{S_3}\int_e^o \mathrm{d}w \int_e^w \mathrm{d}v \int_e^v \mathrm{d}u\,(v-u)\,(n\cdot r_a(u))\,[V_b(v)\cdot V_c(w)]\, e^{i[p_a(o-u)+p_b(o-v)+p_c(o-w)]\cdot n}.$$

Every term of the direct cubic measurement then has at most three independent times.

Independent measurement check. In a spatially homogeneous anisotropic off-shell metric, canonical photon momentum is conserved and $e^{-I(n)} = [n^T R^T e^{H_o/2} e^{-H_e} e^{H_o/2} R n]^{-1/2}$ at the physical roots. Expanding this exactly reproduces (107)–(108) and, with prescribed nonzero H_2, H_3, ν_2, the assembly $J^{(3)}_{\mathrm{meas}} = J_{111} - 3S_{\mathrm{meas},1}[\nu_2, H_2] + J_{\mathrm{lin}}[H_3]$ using the implemented first source. This tests the measurement algebra, not a radiation solution.

C.5 Exact angular closure and evaluated Green blocks

Entire phase moments. Let $F_r(z) = {}_0F_1(;r+\frac{3}{2};-\frac{z}{4})/(2r+1)!!$, so $F_r(x^2) = j_r(x)/x^r$ but with no division. For a monomial of degree $d = \sum m_i$,

$$\langle n_1^{m_1} n_2^{m_2} n_3^{m_3} e^{iK\cdot n}\rangle_\Omega = i^d \sum_{0\le r_i\le m_i/2} (-1)^r F_{d-r}(K\cdot K) \prod_{i=1}^{3} \frac{m_i!\, K_i^{m_i-2r_i}}{2^{r_i} r_i!\,(m_i-2r_i)!}, \qquad r = \sum r_i, \quad (109)$$

valid for all phase vectors, including $K = 0$. Substituting the Volterra forms of the first ray and adjoints, the *entire measurement part* of the quadrupole first source becomes finite combinations of $F_0, \dots, F_7$ at the mixed-time phase $K_{tu} = (p+q)(o-t) + p(t-u)$, with no sky integral and no independently solved ray. The normalisation contribution contains, besides the combined-phase term, the two separated products $\frac{1}{4}\mathcal{U}[h'_{nn}; K_p]\mathcal{A}_C[N_n; K_q] + \frac{1}{4}\mathcal{A}_C[h'_{nn}; K_p]\mathcal{U}[N_n; K_q]$ ($\mathcal{U}$ the plain and $\mathcal{A}_C$ the w-weighted phase average); both are generically nonzero. Mixed-time phases mean this is *not* a closure on the nine equal-time moments of §5.3.1.

The frame covector is a commutator. Since rotations preserve area, $\mathcal{K}_a = [S_a, \mathsf{C}]$ with $S_a = \frac{15}{2}\langle \Theta_a N_n\rangle_\Omega$ the linear quadrupole tensor; for a scalar mode $S_a = -\frac{15}{2} T_2(|p|) Q_{\hat{p}}$. The same tensor gives $J_a = \mathsf{C} : S_a$ and $A_a = -\mathsf{C} : \partial_o S_a$.

The forced homogeneous tensor drops out after isotropic contraction. The exact pair residuals at $(\alpha, -\alpha)$ are $f_H(t, |\alpha|)Q_\alpha$ and $f_N(t, |\alpha|)$, so on the regular branch $H_{bc} = g_H Q_\alpha$. It enters the cubic response linearly, through a functional independent of $\hat{\alpha}$; since $\langle Q_\alpha\rangle_{\hat{\alpha}} = 0$, it cancels from the *entire* isotropically contracted cubic response, including measurement and boundary terms. The homogeneous lapse and its initial row $c_{(q,-q)} = 5q^2/9$ do not cancel. The uncontracted map still contains the tensor.

Evaluated Green kernels. Let $\mathsf{T}_j(a,d) = \sum_{m\ge 0} (-a)^m d^{2m+j}/(2m+j)!$ (entire; $\mathsf{T}_0, \mathsf{T}_1$ are the entire cosine and sine-over-frequency). On STF matrices $\mathfrak{B}_q^3 = \frac{2z}{3}\mathfrak{B}_q^2 + \frac{z^2}{3}\mathfrak{B}_q$ ($z = q\cdot q$), and the retarded spatial inverse of $\partial_t^2 + \frac{2}{t}\partial_t + \mathfrak{B}_q$ is the quadratic operator polynomial

$$\mathfrak{G}_q(t,s) = \frac{s}{t}\Big[d\,\mathbf{1} - \frac{\mathsf{T}_3(z,d) + 3\mathsf{T}_3(-\frac{z}{3},d)}{4}\mathfrak{B}_q + \frac{3\mathsf{T}_5(z,d) + \mathsf{T}_5(-\frac{z}{3},d)}{4}\mathfrak{B}_q^2\Big], \qquad d = t-s, \quad (110)$$

which is the closed form of the series in Prop. 8. With $a = z/3$, $G_N^+ = \frac{1}{3}[\mathsf{T}_1(a,d) + (d\mathsf{T}_2 - \mathsf{T}_3)(a,d)/(ts)]$, and the nested lapse-to-spatial convolution is evaluated:

$$\int_s^t \frac{u}{t}\mathsf{T}_1(-a, t-u) G_N^+(u,s)\mathrm{d}u = \frac{s\mathcal{A} + \mathcal{B} + \mathcal{D}/s}{3t},$$

with $\mathcal{A} = \frac{1}{2}[\mathsf{T}_3(a) + \mathsf{T}_3(-a)]$, $\mathcal{B} = \frac{1}{2}[d\mathsf{T}_3(a) + \mathsf{T}_4(-a) - \mathsf{T}_4(a)]$, $\mathcal{D} = \frac{1}{2}[d\mathsf{T}_4(a) - 2\mathsf{T}_5(a) + \mathsf{T}_5(-a)]$ (all at d). The complete proper-clock pairing is evaluated too. For a physical first input, set $a = p^2/3$ in the

clock formulas: $D_p = \frac{p^2}{9o} W_c$, with $W_c = P_c(o) - P_c(\iota) + \iota^2 v_p(\iota)$ and $P_c(T) = 3[T\mathsf{T}_3(a,T) - 2\mathsf{T}_4(a,T)]$; at zero momentum $W_c = (o^4 + 3\iota^4)/4$.

For the physical first scalar mode, the direct primordial spatial forcing $-2Q\mathsf{T}_3(-a,t)/t$ and the initial-row column $c_1 H_c$ ($c_1 = -a/3$) combine to $H_1 = -2Q[\mathsf{T}_2(a,t) - 2\mathsf{T}_3(a,t)/t]$: all hyperbolic ($-a$) terms cancel between the source and the initial matching. §10.2 shows why, and extends this to every physical pair.

Time arity. Counting independent integration variables after distributing products (a product of three one-time means counts as three): background adjoints need one time, first adjoints at most two, the single–pair pairings and the direct cubic measurement at most three, initial rows at most two. Hence the complete stationary cubic *quadrupole* has at most three independent time integrations per distributed term, for arbitrary triples. This bounds the representation; it does not evaluate the surviving kernels.

C.6 The constraint-row transpose

The transpose, with its boundary terms. Let a weak spatial functional be $\mathcal{W}[j] = \int_0^o (\Omega_0 : j + \Omega_1 : j')\mathrm{d}t + \Omega_e : j'(e) + \Omega_o : j'(o)$, plus lapse terms. On compatible sources at $K \neq 0$, its non-TT part equals

$$\int_0^o Z(s) \cdot \big[4K(\nu' - \nu/s) + 2ic_M\big]\mathrm{d}s + \sum_{x=e,o} \mathcal{P}_K^T[\Omega_x] \cdot \big[4K(\nu' - \nu/t) + 2ic_M\big]_{t=x},$$
$$Z(s) = \mathcal{P}_K^T\Big[\Omega_1(s) + \int_s^o \Omega_0\Big], \tag{111}$$

with $\mathcal{P}_K^T[\Omega] = \frac{2}{K^2}\Omega K - \frac{K\cdot\Omega\cdot K}{2K^4}K$. The lapse weights then pass through the acoustic transpose and its origin covector (102). No hyperbolic function occurs. For the linear quadrupole, $\Omega_1 = -\frac{1}{2}C_K \mathbf{1}_{(e,o)}$ and the momentum-row weight is the *local* $Z_0 = -\frac{1}{2}\mathcal{P}_K^T[C_K(t)]$. This has the structure of the momentum form of Prop. 22: an acoustic lapse Wronskian plus a local momentum-source term.

Application to the single–pair term. Each pair source $D_b D_c F$ is a physical second-order source and so is compatible. Since $\langle \lambda_a, D_b D_c F\rangle = -\mathcal{S}_a[y_{bc}]$, the hyperbolic scalar part of the first adjoint at the probe momentum $k_b + k_c \neq 0$ drops out exactly: (111) applied to $\mathcal{S}_a$ replaces it by acoustic kernels, one cumulative integral and the explicit point terms at e and o that carry (106). The *background* adjoint inside $\mathcal{S}_a$ (total momentum) is paired with the mixed cubic residual, which is not compatible by itself. Only the sum of the direct third residual and the three field-variation parts is the full physical cubic source, to which the same replacement applies (Prop. 33). The separate background terms of (101) keep hyperbolic parts that cancel only in that sum. The individual terms could be made hyperbolic-free only by rewriting the whole stationary functional in the momentum rows. That would require their mixed variations, which are not derived here and, by Prop. 33, are not needed.

C.7 Four-gradient first-adjoint coefficients

The cubic quadrupole through four gradients. Scale momenta by κ. The leading first spatial adjoint is evaluated in closed form, with observer trace $p^2(o^4 + 3\iota^4)\mathsf{C}/(72o)$ and emission jump $-p^2 e^3 \mathsf{C}/18$; its integral has no logarithms,

$$\overline{\Lambda}^{[2]}_{H,a} = \frac{L}{3}U + \tau^4 V - \frac{E}{120}\rho_{p,q} + \frac{p^2(3o^4 + 5\iota^4 - 8e^4)}{360}\mathsf{C}, \qquad \Lambda^{[2]}_{c,a} = \frac{3o^4 + 5\iota^4 - 8e^4}{20}\mathsf{C} : Q_p + \frac{E}{5}(p+q)^T \mathsf{C} q,$$

with $\tau^2 = (o^2 - e^2)/6$, $E = o^4 - e^4$, $L = \int_e^o t(o-t)^2 \mathrm{d}t$, and $U, V, \rho_{p,q}$ explicit quadratic tensors in p, q, C (note `quadrupole_adjoint_gradient_expansion.md`). Both initial-data pairings enter at four gradients, below the bulk lapse terms. These coefficients use the averaged objective $J = \langle w\Theta \rangle_\Omega$ and its equivalent full-solid-angle weight $W = w/(4\pi)$.

D A prospective longitudinal-gauge route

D.1 Follow-up: fourth order inside the horizon

This is a proposed route, not an evaluated fourth-order asymptotic result. Its starting point is the exact longitudinal-redshift identity already derived in the report.

Why change variables. The sampled finite-wavelength maps have large cancelling contributions in the temperature frame. No fourth-order decay law is established. The exploratory recombination in App. F suggests where the residual cancellation occurs. It sits between the field terms and the photon displacement, and it has two likely sources: the static relabelling carried by $H_1 \to -\frac{6}{k_0^2}(\partial\partial - \frac{1}{3}\delta\nabla^2)g$, and the $O(1)$ oscillation of the temperature-defined slicing, since the radiation temperature itself oscillates inside the horizon. A purely spatial relabelling addresses the first but not the second, and lowers the cancellation measure only from 10.3 to 3.5 at $x_o = 12$. In longitudinal variables both are removed at first order. The potentials $\Phi = \Psi = -2(\sin x/x - \cos x)/x^2$ decay as x^{-2}, there is no first-order scalar shift, and the $O(1)$ acoustic physics appears explicitly in the endpoint Sachs–Wolfe and Doppler factors (Remark 4). The change is exact: the longitudinal redshift and its identity with the temperature-frame exponent are already proved there.

What has to be built.

1. The second- and third-order gauge generators (time shift and spatial vector) from the temperature frame to longitudinal gauge. At each order these solve linear equations with sources from lower orders, so the triangular scheme of §5 applies. The existing field kernels supply the sources.

2. The longitudinal fields Φ_r, Ψ_r, β_r and M_r through third order, obtained by transforming those kernels. Their late-time decay should be checked order by order: the induced second-order potentials are not guaranteed to decay as fast as the first-order ones.

3. The late-time expansion of the shell-integrated spectrum, with the non-uniform angles $\mu = \pm 1$ (collinear) and $\mu = -\frac{1}{3}$ (tensor sound cone) treated explicitly. Either use matched asymptotics in (μ, x_o), or subtract the resonant behaviour analytically before taking the limit. A pointwise limit followed by angular integration is not justified (see above).

Checks available. The result must agree with the temperature-frame map, which is exact at finite times. It must reproduce the superhorizon coefficients of §9.3 at small x_o, and its linear part must reproduce the first-order subhorizon law of §9.5. The second-order part can be compared with the exploratory numerics, which reach $x_o = 16$, as a consistency check. The natural first milestone is the second-order part $\frac{1}{4}C^{(22)}$ alone. It needs only the second-order generator, and by Prop. 23 it is the entire fourth-order answer at the zeros of T_ℓ.

E Verification ledger

The analytical checks use exact symbolic computations (SymPy 1.14). Supplementary numerical validations are identified separately and are not used in the analytical derivation. "Generic 2-jets" means that every function value and derivative is an independent symbol, so a zero residual is a proof.

The checks are of different kinds, and the ledger says which kind each one is:

- *general proofs*: generic jets, arbitrary fields, closed-form identities;
- *finite exact series tests*: all field equations through a stated order in t;
- *exact closed-form substitutions*: completed for the lapse equation at second order only.

In particular, the assembled third-order kernel at $(k, q, -q)$ is checked by an exact series through $O(t^6)$ and by Ward identities in κ. It is not checked by substitution at arbitrary finite time. The exploratory numerics of App. F add finite-k Ward tests, not used in the analytical derivation. At finite k_0 the exact checks verify the kernels and the integrands of $C_\ell^{(22)}$ and $C_\ell^{(13)}$; the remaining μ and time integrals are not evaluated analytically.

Check	Script and scope
Fluid conservation, exact redshift, null-geodesic map (Lemmas 1, 2); generic field equations (Prop. 5)	`solution/foundations/check_exact_lemmas.py`, `check_fw_conformal.py`, `check_exact_field_equations.py`, `check_field_equations_generic.py` (generic 2-jets)
Gradient-expansion data (5)	`solution/foundations/check_initial_data_gradient.py` (exact in ζ)
Mixed trace-free form (T)	`solution/check_evolution_form.py` (generic 2-jets)
First order, two gauges, all ℓ	`solution/foundations/check_first_order.py`; `solution/first_order_engine_check.py`
Entire-function T_ℓ (Prop. 16)	`solution/transfer_entire.py 1 2 3 4 5 6` (`run_all.sh` runs $\ell = 1, 2, 3$; $\ell = 4$–6 take about a minute more). The proof of the proposition is analytic for every ℓ; the script checks it case by case: the $x_\star$-derivative of (42) equals the integrand exactly, as an identity of exponential polynomials, and the representation vanishes at $x_\star = x_o$. A printed high-precision comparison is supplementary only.
Reconstruction from the superhorizon series (Remark 17)	`solution/reconstruct_transfer.py264`: exact linear algebra over $\mathbb{Q}(\sqrt{3}, i)$, unique solution, the orders k_0^{65}–k_0^{72} predicted exactly; finite-k_0 agreement with the direct integral as a check (about 7 min; cached)

Check	Script and scope
Subhorizon T_ℓ (Prop. 30)	`solution/transfer_subhorizon.pylorder`: exact asymptotic expansion from Prop. 16 with the E_1 series as the only asymptotic input; asserts exact cancellation of all growing, constant and logarithmic terms, the leading Sachs–Wolfe + Doppler form for $\ell = 2, 3, 4$, and the $\ell = 2$ observer term; high-precision spot checks of the expansion
Green functions, $K \to 0$	`solution/check_green.py`
2nd-order sources and operators	`solution/second_order.py`
2nd order, all 10 equations	`solution/second_order_consistency.py`: exact Taylor series through $O(t^8)$, general K and $K = 0$
2nd-order closed forms	`solution/closed_second_order.py`: every antiderivative by exact differentiation; (N) by closed-form substitution (`--check N`); (M) and (T) via the theorem and the series test
3rd-order sources and operators	`solution/third_order.py`
3rd order, all 10 equations	`solution/fast_series.py 8`: the pair kernels, including the exact $K = 0$ pair, then the triple at $(k, q, -q)$ with general momenta. The *full nonlinear* equations are recomputed with the solved kernels and vanish through $O(t^6)$. The arithmetic is exact (sparse polynomials over $\mathbb{Q}(i)$ with formal inverse momenta; denominators are cleared exactly at the zero test). Negative controls (a wrong c_3, a corrupted $K = 0$ pair kernel) fail at the expected orders. The run takes about 70 s. `third_order_consistency.py` is the slower SymPy reference implementation of the same test.
Dilation Ward identities (fields)	`solution/check_ward_dilation.py`, `check_ward_dilation3.py` (closed form; negative controls)
Observable expansion formulas	`solution/check_observable_expansion.py` (arbitrary fields, general jets)
Observable Ward identities	`solution/observable_generator.py W2|W3` and the fast port `solution/observable_fast.pyW2|W34`, through $O(\kappa^4)$. These test the fields and dilation covariance only: mutating the ray, endpoint or normalisation leaves them intact, and dropping $e^{-\zeta}$ in the initial direction makes them fail

Check	Script and scope
Superhorizon observable	`solution/observable_generator.py SH` (orders 1 and 2, general (k, q)). Fast exact port `solution/observable_fast.pySH` (11 s) and `SH3` (third order at $(k, q, -q)$, general momenta, about 12 min), with an $e^{-\zeta}$ negative control. The port keeps the generator's definitions operation for operation and replaces only the arithmetic, using sparse polynomials over $\mathbb{Q}(i)$ with formal inverse momenta and times
Tensor vs. Kohri–Terada	`solution/check_tensor_newtonian.py`: an independent Christoffel/Ricci reconstruction of the shell TT source; the quadratic gauge map (pullback, conformal factor, matrix logarithm); the exact identity $L_K F_T = \mathcal{S}_T$ at arbitrary t and μ; matching regular initial data; the entire-function KT kernel, including the resonance; negative controls. `solution/check_shell_reduction.py` matches the repository's own c_T to $\mathcal{S}_T$. Derivation in `report/tensor_newtonian_crosscheck.md`
Superhorizon C_2	`solution/superhorizon.py`
Report identities	`solution/check_report_identities.py`: background residual Bianchi identities and zero-mode constraint matching; conformal driver and cubic commutator shear; Fermi–Walker rotation; centred normalisation and polarised map kernels; the two-norm spectrum; general-moment and monochromatic leading-gradient resummation. These are exact symbolic identities, not an arbitrary-time cubic-observable verification.
Gradient-mode tests	`solution/check_gradient_modes.py`: exact special-conformal Jacobian and flat seed; zero shear and hence zero map for the completed adiabatic profile; quadratic completion and its conversion to the convention of [1]; nonzero uncompensated-gradient quadrupole; plane-wave soft generator and the leading-gradient check of (36). The general finite-time Ward identity is proved by covariance, not by full-kernel substitution.
Log-map decomposition	`solution/check_logmap_decomposition.py` (about 2 min): (44) and the $3j$ lognormal term (46) for $\ell = 1, \ldots, 4$, in a band-limited Gaussian sky with non-Gaussian L_2, L_3; exact sphere integrals and Gaussian moments
Shell two-amplitude form (Prop. 10)	`solution/check_shell_reduction.py`: $c_{M,z} = c_{T,xz} = c_{T,yz} = 0$ for general momenta; $K \times c_M = 0$ on the shell identically in t and μ; off-shell negative control

Check	Script and scope
Shell locality	`solution/check_shell_locality.py`: the local form of the shell first-order fields and of θ; $\sigma^{(2)} = \frac{1}{2}A_2' + \frac{1}{2}(H_1^2)' + 2gH_1'$ for generic matrices. Item (3) of Prop. 24 is by inspection.
Leading octupole; $\mathcal{M}_\ell$ cancellation	`solution/check_gradient_octupole.py`: the conformal identity $\nabla_{\langle a}R_{bc\rangle} = \frac{f}{2}[\partial^3 f^2]_{\mathrm{STF}}$ for generic ζ; the leading T_ℓ coefficient (68) for $\ell = 2, \dots, 7$; β_3; exact isotropic Wick moments giving (70) and the shell ratios $148\sigma^2$, $27\sigma^2$; $\langle\theta\theta L_2\rangle = 0$ at leading gradient order. The octupole map itself rests on the proof of Thm. 28.
Next superhorizon order	`solution/superhorizon_next.pykernels2|kernels3|spectrum`: map kernels through $O(\kappa^4)$ from the fast generator; $C_2^{(11)}$ at $O(k_0^6)$ checked against the transfer integral; the κ^4 cubic kernel gives $\frac{1}{3}\,\mathrm{Re}\,C^{(13)}/C^{(11)} = 4\sigma^2$ and the total $\frac{29}{3}$ (asserted), and the complete $\epsilon^4 k_0^6$ coefficient. Kernel run about 13 h; assembly about 25 min the first time, 4 s once cached. `solution/check_quadrupole_next_hand.py--compare`: an independent hand-map calculation reproducing $C_2^{(11)}$ and $\frac{1}{4}C_2^{(22)}$ at $O(k_0^6)$ and every component of the projected kernel; endpoint and clock pieces nonzero; negative controls. `solution/check_cubic_curvature_hand.py`: an independent covariant-curvature calculation (no generator, field engine or cache) of all three lines at $O(k_0^6)$, the fourth-gradient field equations, the three-tensor form (73) and the vanishing $(\mathsf{A}^2)_{\mathrm{TF}}$ contractions; negative controls
Separated sky factors (Prop. 20)	`solution/check_bessel_translation.py`: the translation form equals the Funk–Hecke form exactly through degree 10, $\ell \le 4$, symbolic μ; negative control
Projector-free Cartesian solution (Prop. 8)	`solution/check_cartesian_green.py`: eigenvalues of $\mathfrak{B}_K$ $(K^2, 0, -K^2/3)$; the Cartesian series, with coefficients polynomial in K, equals the projector-based kernels exactly through $O(t^7)$ for (k, q), $(k, -q)$, the $K = 0$ pair $(q, -q)$, and third order at $(k, q, -q)$
Orthonormal redshift driver (Prop. 3)	`solution/check_orthonormal_driver.py`: $\frac{1}{2}e^{-H/2}(e^H)'e^{-H/2} = \frac{1}{2}H' + \frac{1}{48}[H,[H,H']] + O(\epsilon^4)$ for generic non-commuting matrices; the shell commutator vanishes. `solution/check_orthonormal_map.py`: the first-order orthonormal direction equation is transverse, and the second-order integrand in v equals the original one (the local products cancel against $e - v$), for generic fields

Check	Script and scope
Adjoint top-order cubic response (Prop. 22)	`solution/check_adjoint_exact.py`: the lapse Green function, its diagonal jump and the contact coefficient, the factor 14, the source compatibility $c'_{\mathrm{Ham}} - 4c_N/t = 2iK \cdot c_M$, and the absence of d_3, D_3 with d_1, d_2, D_1, D_2 retained, all exact. Fubini by the estimate in the proof. Numerical comparison (exploratory, `solution/numerics/explore_adjoint.py`) with the direct solve and the full map
Longitudinal-gauge redshift	`solution/check_longitudinal_redshift.py`: the ADM energy law with a shift on generic jets; exact equality with the temperature-frame exponent including the endpoint conformal and Doppler factors; $\Phi = \Psi$ and zero first-order scalar shift under the full time-and-space map; clock identity; negative controls
Dilation factorisation	`solution/check_dilation_factorisation.py`: $\Theta_{\mathrm{lead}}[\zeta + c] = e^{-2c}\Theta_{\mathrm{lead}}[\zeta]$ exactly in ζ and scaling for a generic cubic seed; mean shift $\equiv$ dilation of the moments; invariant ratio (66); Gaussian average of $e^{-\zeta_o D}$; superhorizon observer-centred kernels of Prop. 19 ($\mathcal{K}_2^c = -\tau^2\,\mathrm{TF}(kq+qk)_{nn}$, $\mathcal{K}_3^c(k,q,-q) = 0$)
Propagation adjoint	`solution/check_propagation_adjoint.py`: mixed cubic driver from matrix exponentials and pulled-back orthonormal directions; general linear ray response; arbitrary plane-wave-jet duality and bulk integration by parts; moving-emission and physical-clock coefficients, centred normalisation, shell emission projection and factorials; boundary and longitudinal-displacement omission controls. The second-order ray is eliminated, not the full mixed-time angular response or its interference with the top-order field.

Check	Script and scope
Contracted cubic source reduction	`solution/check_shell_cubic_sources.py--compare-engine`: exact cubic lapse and longitudinal momentum sources for arbitrary time profiles and pair amplitudes; comparison with the original Christoffel-based engine at finite time; vanishing direct 111 momentum source; homogeneous tensor cancellation; general-momentum lapse normal form; elementary quadratic sources; derivative-free momentum adjoint and boundary coefficients; Green equations and zero momentum; all-order angular-series coefficient identities and finite Laurent integration with at most J_3. The angle-free moment formulas follow by analytic substitution and Fubini, not by a new end-to-end spectrum implementation.
Cubic products; fast curvature map	`solution/check_cubic_product_identity.py`: generic non-commuting driver, ray pullback and moving-boundary/clock composition, with omission controls. `solution/curvature_kernels.py --compare --spectrum`: regenerates the complete map kernels through four gradients without the field/ray engine; exact denominator-cleared comparison to the raw saved kernels at general $(k, q, -q)$; general-moment C_2 through degree six and the separate leading octupole contributions. Does not provide the six-/five-gradient maps needed for the next quadrupole/octupole spectrum coefficients.
Stationary identity, normalisation adjoint	`solution/check_unified_adjoint.py`: polarised identity (82) against recursive substitution in a non-symmetric nonlinear system; background transpose with origin covector (dropping it fails); exact normalisation adjoint; leading-gradient map; a 15-variable root/clock/normalisation fixture with omission controls (not a radiation solution)
First field-adjoint source	`solution/check_first_field_adjoint_source.py`: (104) and all weak transposes against the independent Christoffel engine at general momenta, off shell; homogeneous and constant-shift controls; no-probe-ray Hessian; emission, clock and observer terms; frame generator; Green reconstruction, observer trace, emission jump and origin covector, exactly

Check	Script and scope
Direct cubic measurement	`solution/check_stationary_cubic.py`, `check_direct_cubic_measurement.py`, `check_phase_integral_contraction.py`: second-ray forcing on arbitrary jets; free-algebra temporal identities; polarisation conventions; the common fourth time; conserved-momentum off-shell comparison with noncommuting Fermi–Walker frame, physical roots and clock
Angular closure; Green blocks	`solution/check_entire_sphere.py` (120 moments through rank seven), `check_quadrupole_angular_source.py` (Volterra signs, both normalisation products, frame commutator, homogeneous-pair covariance), `check_regular_field_green.py` and `check_quadrupole_green_source.py` (operator polynomial, evaluated convolutions and clock primitives, first-order hyperbolic cancellation, time arity). Quadrature in these scripts is labelled validation only
Four-gradient cubic quadrupole	`solution/check_quadrupole_adjoint_expansion.py`; `check_stationary_quadrupole_four_gradients.py--symbolic`: the uncontracted cubic quadrupole equals the independent curvature reference as a polynomial in six momentum components, all five components, symbolic times; shell value (83); known $\frac{1}{3}C_2^{(13)}$ through $O(k_0^6)$
Dilation channel	`solution/check_dilation_ward.py`: source and initial-row identities with the engine at symbolic momentum and arbitrary profiles; phase identity at arbitrary momentum and times; algebraic stationary Ward fixture
Constraint rows (Props. 34–12)	`solution/check_constraint_rows.py`: (79)–(80) exactly for a physical pair at unequal momenta (failing off shell); projector and transpose; shell $\mathfrak{m}$, c_r, $\mathcal{I}$ and the local scalar identity; the evaluated kernel $\mathcal{G}$; (27) against the hyperbolic Cartesian solution through $O(t^{16})$; $K \cdot A = V$ for an unequal pair through $O(t^{10})$, with a wrong-initial-row control; at third order, with physical second-order fields, (79)–(80) and $K \cdot A_3 = V_3$ through $O(t^{10})$ for noncoplanar unequal, zero-pair and near-zero-pair triples, with a control
Observer-centred cubic at κ^4 ((85))	`solution/check_observer_centred_kappa4.py`: exact; leading cancellation asserted; uses the independent four-gradient quadratic hand map. `solution/check_observer_dilation_k6.py`: all four terms of (51) at $O(k_0^6)$, summing to the known complete coefficient

Check	Script and scope
Finite-wavelength cubic quadrupole (numerical validation)	`solution/numerics/compare_cubic_quadrupole.py`: independent forward map (fields from (N),(T) only, Fermi–Walker triad, geodesic, roots, clock, normalisation) against (77) for noncoplanar unequal, zero-pair and near-zero-pair triples, agreement $\leq 3 \times 10^{-9}$; J_{111} formula, Hessian, constraint rows and dilation channel separately; frame and clock controls. `observer_centred.py`, `shell_c13.py`: observer-centred split and shell ratios (App. F)
Final manuscript regressions	`solution/check_report_final_review.py`: reads the printed degree-six cross and total formulas and compares them with the hand-map reference, including sign-flip controls; generic STF sky moments fix w versus $W = w/(4\pi)$ and the normalization adjoint; differentiates both fixed-label clock conventions; checks both linear Bianchi identities on arbitrary off-shell fields; checks the report input graph and all labels/references.
Printed formulas, independent code	`solution/check_printed_formulas.py`: coefficients retyped from this report and tested with a separate multilinear algebra sharing no code with the other modules. Exact: seed-curvature identities; the degree-six quadrupole from the three-tensor map; pair and contracted cubic sources against exact-in-ζ t^2, t^4 data; acoustic amplitude; Green polynomial; V_2, V_3. Numerical validation: Kohri–Terada amplitude, T_ℓ table values and node positions
Normalisation; Wick and Fourier measures	`solution/foundations/normalization_expansion.py`; `solution/foundations/check_algebra.py`

F Numerical validation and exploratory observations

The computations in this appendix test implementations and suggest analytical questions; they are not used in the analytical derivation. Reported agreements refer to the sampled configurations and stated grids, not to unrestricted error bounds.

F.1 Implementation comparisons at finite wavelength

The analytical statements above do not rest on numerics. To test the assembled representation away from the gradient expansion, `solution/numerics/compare_cubic_quadrupole.py` compares (77) with an independent forward calculation (`solution/numerics/forward_cubic.py`) at $\bar{t}_\star = 1.2$, $\bar{t}_o = 3$, $\bar{t}_I = 0.5$, $|k| \simeq 1$, so that $|k|\bar{t}_o \approx 3$.

- *Forward side.* The fields are solved through third order from (N) and the Cartesian (T) rows

with the initial row. The constraint rows are not used. The Fermi–Walker triad is transported with $\mathrm{d}E/\mathrm{d}t = -SE$, and the photon geodesic is integrated by multilinear Picard iteration. The emission and clock roots and the exact normalisation follow. Time integrals are spectral and sky averages use Gauss quadrature.

- *Assembled side.* J_{111} is the second-ray-free direct measurement, with its nested integrals evaluated by quadrature. S_{meas} is (105), and J_{lin} is (78).

For three triples the two sides of (77) agree to $\leq 3 \times 10^{-9}$: a noncoplanar triple with unequal frequencies, a triple with an exactly zero pair momentum, and one with $|k_b + k_c| = 0.05$. The pieces themselves are of order one to ten. In addition:

- J_{111} agrees with the forward first-field-only map to $\leq 3 \times 10^{-8}$;
- J_{lin} in constraint rows agrees with the Cartesian third-order field to 4×10^{-14};
- the forward map reproduces the exact channel (84) to 4×10^{-8};
- the no-probe-ray Hessian alone matches the forward mixed response to 10^{-9}.

Two controls are resolved: replacing the Fermi–Walker triad by the symmetric one changes the generic-triple value by 9×10^{-5}, and dropping the clock changes it by 0.24.

What survives the observer dilation. With the exact observer-centred identity of §8.3, $\mathcal{K}_3 = \mathcal{K}_3^{\mathrm{c}} - \sum_{\mathrm{pairs}} \mathfrak{D}\mathcal{K}_2 - \sum_a \mathfrak{D}^2\mathcal{K}_1$, the same code isolates the genuinely cubic observer-centred kernel. For the noncoplanar triple it is 0.9% of $\mathcal{K}_3$: $+0.063$ of -6.759. For the configuration $(k, q, -q)$ that enters $C^{(13)}$ it is 33%: -2.29 of -6.89. In these two samples the dilation of the linear and quadratic maps carries most of the cubic response, though the centred remainder is substantial in the second. Prop. 19 converts that dilation part into $\frac{1}{2}\sigma^2 D^2 C^{(11)}$ and a total derivative. The shell values are in the exploratory spectra below.

Gradient coefficients. The forward calculation reproduces the degree-two and degree-four coefficients of (83) to 1.3×10^{-9} and 1.6×10^{-7}, at $\iota = 0.5$ and 1. Their difference reproduces the observer-local term to 10^{-6} in the script `solution/numerics/check_kappa4_shell.py`. This is supplementary to the exact curvature and stationary calculations. At $x_o = 0.3$, the leading centred coefficient gives -0.194 in units of $\mathcal{K}_1$, against the numerical value -0.195; higher gradients remain in the latter.

Top-order adjoint and linear-asymptotic comparisons. In `solution/numerics/explore_adjoint.py`, the momentum-source form of Prop. 22 agrees with the direct cubic Green response to a few parts in 10^6 at $x_o = 1, 3$. The equivalent contact form, which differentiates the Hamiltonian source numerically, agrees to 10^{-3}–10^{-4}; the full map comparison at $x_o = 1.5$ is limited by its grid to 4×10^{-4}. At $x_o = 1$, $\mu = 0.9$, the contact form has large cancelling homogeneous, lapse and Hamiltonian contributions, approximately 14, -73 and 58 in units of T_ℓ, with an unrounded sum of -0.24. These are implementation diagnostics, not additional analytical cancellations. For the linear subhorizon transfer at $\ell = 2$, $R = 3$, the expansion through $O(x_o^{-3})$ has relative differences 1.8×10^{-2}, 3.5×10^{-4} and 7.6×10^{-8} from the exact transfer at $x_o = 40, 120, 360$, respectively.

F.2 Higher-gradient projection fits

The exact centred-22 proportionality of §8.3 is established through spectral degree six. Numerical fits suggest that it does not extend to degree eight: `solution/numerics/centred22_ratio.py` gives ratios 52.66 at $(e, o) = (1.2, 3)$ and 52.29 at $(0.8, 3.5)$, stable to ≤ 0.015 under a change of fit points. The lower-order ratios 5/3 and 28/3 are reproduced to 3×10^{-6}. With the fixed template $\mathrm{TF}(\hat{k}\hat{q}+\hat{q}\hat{k})$, `solution/numerics/half_dilation_projection.py` gives relative mismatches from the $-\mathfrak{D}\delta\mathcal{K}_1/2$ relation of 1.3×10^{-4} at $k_0 = 0.1$ and 2.4×10^{-3} at 0.2, suggesting a higher-order departure. Neither fit proves the next analytical coefficient or an all-orders failure. A degree-eight spectrum calculation needs both the four-gradient square and the two-/six-gradient interference; that calculation remains open.

F.3 Exploratory finite-wavelength spectra

What is computed. By Thm. 18, every part of the fourth-order spectrum except $\frac{1}{3}\operatorname{Re} C_\ell[\theta, L_3]$ needs only second-order quantities. Those parts are $\frac{1}{4}C_\ell^{(22)}$, $\mathcal{N}_\ell$, $\mathcal{M}_\ell$ and $\frac{1}{4}C_\ell[L_2, L_2]$. `solution/numerics/explore_fourth_order.py` evaluates them for $\ell = 2$ on the shell ($k_0 = 1$, $\sigma = 1$), at fixed $\bar{t}_o/\bar{t}_\star = 3$ and $\bar{t}_I = \bar{t}_\star/2$. The ingredients are:

- the exact second-order sources of `second_order.py`, inserted into the Green formulas of §5 and integrated by quadrature;
- the ray, emission, clock and normalisation formulas of §6;
- Gauss–Legendre quadrature in μ and on the sphere.

Validation.

1. The first-order part reproduces the entire-function T_2 of Prop. 16 to 10^{-6} for $x_o \leq 6$, and to 1.2×10^{-4} at $x_o = 12$ (ray grid $n_t = 160\, x_o$).
2. $\frac{1}{4}C_2[\theta^2, \theta^2]$ equals the first term of $\mathcal{N}_2$ to all printed digits.
3. The dilation Ward identity holds at finite k ($x_o = 1.5$) for the complete second-order map: $[\mathcal{K}_2(k, q) + \mathcal{K}_2(k, -q)]/2 = -k\partial_k\mathcal{K}_1$, with residual 4.5×10^{-3} at $q = 0.04$ and 1.1×10^{-3} at $q = 0.02$, i.e. $O(q^2)$. As negative controls, dropping the ray terms or A_2 gives $O(1)$ failures. The clock and endpoint terms vanish in this limit and are not tested by it.
4. The clock shift agrees with a direct root of the exact clock equation to eight digits.
5. The moving-emission endpoint term agrees, to all printed digits, with the mixed $z_a z_b$ coefficient obtained by solving the emission condition $\ln t_e + \nu(t_e, X(t_e)) = \ln \bar{t}_\star$ directly for real two-mode fields.
6. A_2' is computed analytically from the Green representation: the boundary terms of the cumulative integrals cancel, ν_2'' comes from the lapse equation, and c_{Ham}' is differentiated symbolically. It agrees with a finite-difference derivative to 10^{-4}.
7. Moving the early-time cutoff of the Green integrals from $10^{-3}\bar{t}_o$ to $2 \times 10^{-5}\bar{t}_o$ changes the results in the seventh digit.

8. The ray quadrature error scales as n_t^{-2}. On a coarse grid ($n_t = 320$) it gives a 0.4% error in T_2 at $x_o = 12$. With $n_t = 160\,x_o$, the first-order error is below 10^{-4} for $x_o \le 12$. Grid refinement in μ, the sphere and the ray changes a node value at $x_o \approx 7.9$ only in the sixth digit.

9. As $x_o \to 0$, $\frac{1}{4}C_2^{(22)}/\sigma^2 C_2^{(11)}$ approaches $\frac{17}{3}$.

10. At every node of T_ℓ below, the pipeline gives $C_\ell^{(11)} \sim 10^{-12}$ and $\mathrm{Re}\, C_\ell[\theta, \theta L_2] \sim 10^{-6}$. Both vanish exactly there (Prop. 23).

The convergence tests were made at $x_o \le 3$, at one node near $x_o = 7.9$, and at $x_o = 12$ on both trajectories, where $24 \to 40$ μ nodes and a 1.5× finer ray grid change $\frac{1}{4}C_2^{(22)}$ only in the sixth digit. At $x_o = 16$ the μ quadrature needs 32 nodes, and the $x_o = 16$ rows use them (see the note below the tables).

Results. Absolute values with $\sigma = 1$ and $\ell = 2$, along the fixed-ratio trajectory $x_\star = x_o/3$, $x_I = x_o/6$. The first-order error is at most 1.2×10^{-4}.

x_o	$C_2^{(11)}$	$\frac{1}{4}C_2^{(22)}$	$\frac{1}{4}C_2[L_2, L_2]$	$\mathcal{M}_2$	$\mathcal{N}_2$
0.5	0.00243	0.00867	0.00867	1.9×10^{-6}	-6.0×10^{-7}
1	0.0258	0.0172	0.0177	-2.9×10^{-4}	-6.7×10^{-5}
1.5	0.0596	0.388	0.396	−0.0119	-3.1×10^{-4}
2	0.0423	2.61	2.64	−0.0453	9.5×10^{-5}
3	0.116	6.26	6.32	−0.320	−0.0012
4	0.958	8.32	8.65	−2.10	−0.048
6	0.0458	5.23	4.96	0.697	0.126
8	0.00059	6.80	4.65	1.26	0.746
12	0.00036	2.43	2.44	−0.057	0.020
16†	0.0533	1.29	1.33	0.267	0.035

Fixed-emission trajectory, $x_\star = 1$ and $x_I = \frac{1}{2}$:

x_o	$C_2^{(11)}$	$\frac{1}{4}C_2^{(22)}$	$\frac{1}{4}C_2[L_2, L_2]$	$\mathcal{M}_2$	$\mathcal{N}_2$
2	0.0340	2.12	2.14	−0.032	-6.7×10^{-5}
4	0.946	8.31	8.53	−1.98	−0.028
8	0.0014	4.21	2.57	0.907	0.563
12	0.0302	1.76	1.63	−0.304	0.041
16†	0.00091	0.787	0.719	−0.061	0.047

†At $x_o = 16$, 32 μ nodes are needed. With 24 nodes, which suffice at $x_o \le 12$, $\frac{1}{4}C_2^{(22)}$ comes out 3% low (fixed ratio: 1.2514 against 1.28688) and 15% low (fixed emission: 0.66617 against 0.787269). Increasing to 48 nodes leaves both values unchanged to six digits, and a 1.5× finer ray grid has no effect. The relative-momentum integrand therefore acquires structure between $x_o = 12$ and 16 that 24 nodes do not resolve. The $x_o = 16$ rows above use 32 nodes.

Fourth-order spectra at linear-transfer nodes (Prop. 23; exploratory values, not analytical evaluations). At these points $C_\ell = \epsilon^4 \sigma^4\, c_\ell$ exactly through fourth order, with:

	trajectory	nodes x_o of T_ℓ	$c_\ell = \frac{1}{4}C_\ell^{(22)}/\sigma^4$
$\ell = 2$	$x_\star = x_o/3$	2.49867, 6.38035, 7.90609	5.686, 3.982, 7.285
$\ell = 2$	$x_\star = 1$	2.51338, 6.33088, 7.50682	5.538, 2.732, 6.222
$\ell = 3$	$x_\star = x_o/3$	3.05037, 7.00073, 8.69752	3.886, 4.334, 2.353
$\ell = 3$	$x_\star = 1$	3.05208, 6.92329, 8.48546	3.920, 2.582, 1.156

The node positions are roots of the entire-function T_ℓ. Further nodes for $x_o \le 14$ are at 9.46677 and 11.96208 ($\ell = 2$, fixed ratio), at 8.52178, 10.89081 and 13.06476 ($\ell = 2$, $x_\star = 1$), at 10.60041 and 13.04624 ($\ell = 3$, fixed ratio), and at 9.38590, 11.66731 and 13.83506 ($\ell = 3$, $x_\star = 1$).

Observations. The following are leads for analysis, not conclusions.

1. *Linear dominance, not perturbative breakdown.* Inside the horizon, $\frac{1}{4}C_2^{(22)}$ stays $O(1)\sigma^4$ while $C_2^{(11)}$ is smaller and passes through nodes of T_2. With $r_{22} = C_2^{(22)}/(4\sigma^2 C_2^{(11)})$, the quadratic contribution is smaller than the linear one only if $\epsilon^2\sigma^2\, r_{22} \ll 1$, and $r_{22} \gg 1$ for $x_o \gtrsim 2$. At a node, the first nonzero prediction is the fourth-order one (Prop. 23). This is a statement about which order dominates a given multipole. It does not establish breakdown of the expansion, which would need the next order. Away from nodes, the 13 term must also be included; its evaluation below changes the fourth-order correction materially.

2. *Measurement nonlinearity.* $\mathcal{N}_2 + \mathcal{M}_2$ is below about 5% of $\frac{1}{4}C_2[L_2, L_2]$ for $x_o \le 3$. On the fixed-ratio trajectory it is 25%, 17% and 43% at $x_o = 4, 6, 8$; at fixed emission it is 23% and 57% at $x_o = 4, 8$.

3. *Large cancelling pieces.* At small x_o almost all of $\frac{1}{4}C_2[L_2, L_2]$ comes from the A_2 part. At larger x_o the local part, the A_2 part and their interference are each large and strongly anticorrelated: at $x_o = 12$ (fixed ratio) they are 12.5, 17.4 and -27.5, summing to 2.4. This suggests recombining the amplitudes *before* squaring. For example, $\int A'_{2,nn} e^{iK\cdot\hat{n}(\bar{t}_o - t)}\mathrm{d}t = \left[A_{2,nn} e^{iK\cdot\hat{n}(\bar{t}_o - t)}\right]_{\bar{t}_\star}^{\bar{t}_o} + iK\cdot\hat{n}\int A_{2,nn} e^{iK\cdot\hat{n}(\bar{t}_o - t)}\mathrm{d}t$ turns the field term into endpoint pieces. Those could combine with the observer-frame, ray and emission terms, before the separately squared norms are formed.

4. *Resonances.* The μ-integrand is smooth at these finite times, as the entire-function representation guarantees. It peaks at $K \approx 0.8$–$1.0\,k_0$ in the tested configurations. Neither fact settles the asymptotics. A resonant layer of width $\propto 1/t$ is regular at every finite time yet can dominate at late times, so a matched asymptotic analysis in angle and time is needed before any late-time claim. In these runs, $\frac{1}{4}C^{(22)}$ divided by $T_2(K)^2$ is not universal in x_o.

5. *Two different limits.* The main table follows $x_\star = x_o/3$ and $x_I = x_o/6$: increasing k_0 at fixed physical labels, which is the k_0-dependence the problem asks for. The late-time limit of §9 instead keeps $x_\star$ and x_I fixed. The second table follows that trajectory with $x_\star = 1$, $x_I = \frac{1}{2}$. The two must not be conflated.

The cubic term and the complete fourth-order spectrum. `solution/numerics/explore_third_order.py` evaluates the remaining piece, $\frac{1}{3}\,\mathrm{Re}\,C_\ell[\theta, L_3]$, at $(k, q, -q)$. Its ingredients are:

- the exact third-order sources of `third_order.py`, together with the third-order Hamiltonian source, extracted for this purpose;

- the numerical second-order pair fields, with the exact $K = 0$ kernel for the pair $(q, -q)$;
- the Green formulas of §5 for A_3.

The map is not taken from the hand-expanded formulas. It is solved from its definitions by a numerical multilinear engine: the ray equation, the fields at the displaced ray, the emission root, the clock equation, the composition with the random observation time, and the normalisation. The checks are:

1. The engine's second-order map reproduces the hand-formula evaluator above to 1.2×10^{-6}, and its first-order map to 3×10^{-9}. These are two independent implementations.
2. The third-order dilation Ward identity $\mathcal{K}_3(k, q, -q) \to (k\partial_k)^2\mathcal{K}_1$ holds at finite k ($x_o = 1.5$). The residual is 3.3×10^{-2} at $q = 0.08$ and 8.2×10^{-3} at $q = 0.04$, i.e. $O(q^2)$. Dropping A_3 gives a 46% failure.
3. Keeping only the scalar part of A_3 gives the same $\frac{1}{3}\operatorname{Re} C_2[\theta, L_3]$ as keeping its vector and TT parts, to every printed digit. This confirms that the contracted cubic response is purely scalar.
4. As $x_o \to 0$, $\frac{1}{3}\operatorname{Re} C_2[\theta, L_3]/\sigma^2 C_2^{(11)}$ is 3.720 at $x_o = 0.15$ and 2.895 at 0.3; removing the $O(x_o^2)$ term extrapolates to 3.995, against the expected 4 (§9.1).
5. At the node $x_o = 2.49867$ of T_2 the cubic term is -2×10^{-5}, consistent with its exact vanishing (Prop. 23).

With $C_2^{(4)} \equiv \frac{1}{4}C_2^{(22)} + \frac{1}{3}\operatorname{Re} C_2^{(13)}$, assembled from Thm. 18 as $\frac{1}{4}C_2^{(22)} + \frac{1}{3}\operatorname{Re} C_2[\theta, L_3] + \operatorname{Re} C_2[\theta, \theta L_2] - C_2^{(11)2}/2\pi$, the fixed-ratio trajectory gives ($\sigma = 1$):

x_o	$C_2^{(11)}$	$\frac{1}{4}C_2^{(22)}$	$\frac{1}{3}\operatorname{Re} C_2[\theta, L_3]$	$C_2^{(4)}$	$C_2^{(4)}/C_2^{(11)}$
0.15	2.209×10^{-5}	1.204×10^{-4}	8.22×10^{-5}	2.03×10^{-4}	9.17
0.3	3.414×10^{-4}	1.653×10^{-3}	9.88×10^{-4}	2.64×10^{-3}	7.74
0.6	0.00475	0.0135	−0.00084	0.0127	2.67
1	0.0258	0.0172	−0.159	−0.141	−5.5
1.5	0.0596	0.387	−0.804	−0.420	−7.0
2	0.0423	2.611	−0.581	2.010	47.6
3	0.116	6.264	−7.57	−1.57	−13.5
4	0.958	8.324	−38.7	−32.2	−33.6

The small-x_o ratios approach $\frac{29}{3} = 9.67$. Inside the horizon the 22 and 13 contributions are of the same size and partly cancel, and the net fourth-order correction to the quadrupole changes sign several times. Away from nodes the cubic term is therefore essential: the second-order part alone would give the wrong sign at $x_o = 1$–1.5 and at 3–4. Convergence of the cubic term: at $x_o = 3$ it is stable to five digits under $16 \to 24$ μ nodes and under a finer sphere ($20 \to 28$ polar nodes) and ray ($300 \to 450$) grid. At $x_o = 4$ it is stable to five digits under $16 \to 24$ μ nodes and to four digits under a finer sphere ($24 \to 32$) and ray ($400 \to 600$) grid.

Independent recomputation of the cubic term. The validated forward map of §F.1, in `solution/numerics/shell_c13.py`, recomputes the shell ratio

$$\frac{\frac{1}{3}\operatorname{Re} C_2^{(13)}}{\sigma^2 C_2^{(11)}} = \frac{\langle \mathcal{K}_3(k,q,-q)\rangle_{\hat{q}}}{\mathcal{K}_1(k)}$$

on the fixed-ratio trajectory. Unlike the calculation above, it keeps all three sectors of A_3, the Fermi–Walker frame, and fields solved without the Hamiltonian-algebraic scalar. It gives:

x_o	μ nodes	cubic ratio	rounded-table ratio	centred part
0.3	8	2.8955	2.89	−0.195
1	12	−6.156	−6.13	−2.96
3	10	−67.29	−67.5	—

The rounded-table column is $(C_2^{(4)} - \frac{1}{4}C_2^{(22)})/C_2^{(11)}$ computed from the printed entries above. Allowing half a last printed digit in each input gives ratio intervals $[2.8745, 2.9076]$, $[-6.1651, -6.0986]$ and $[-67.8745, -67.1974]$, respectively. The recomputed values lie within these rounding intervals; this is compatibility with the rounded table, not agreement to every displayed digit or a numerical error bound. The recomputed $x_o = 3$ value has not been separately refined in its μ quadrature. Its linear quadrupole reproduces $C_2^{(11)} = \frac{4\pi}{25}\mathcal{K}_1^2$ (0.0258 and 0.116). The last column is $\langle \mathcal{K}_3^c\rangle/\mathcal{K}_1$ from Prop. 19; the rest is the dilation part. It grows from 7% of the cubic term at $x_o = 0.3$ to about half at $x_o = 1$. This is consistent with $\mathcal{K}_3^c(k,q,-q) = 0$ at leading gradient order and quantifies its breakdown inside the horizon. As before, these are numbers, not analytical evaluations.

F.4 Amplitude-level recombination and cancellation diagnostics

To locate the cancellations, split the second-order log-map kernel into pieces P_i and measure $\kappa = \sum_i \|P_i\|^2 / \|\sum_i P_i\|^2$. The norm is the $\ell = 2$ projection integrated over μ. $\kappa \approx 1$ means the pieces barely cancel; $\kappa \gg 1$ means they cancel heavily. The pieces are:

- the local shear products, which split into the $e^{2\zeta}$ term $2(H_a' + H_b')$ and the $H_a H_b$ term;
- the A_2' term;
- the ray displacement $X \cdot \partial\sigma$ and the direction terms $\sigma(n, e)$;
- the emission endpoint and the clock.

The results:

grouping	$x_o = 1$	3	6	12
local | A_2 | ray | endpoint | clock	11.8	1.01	12.6	21.8
$\tilde{\gamma}$ (= local $+A_2$) | ray | endpoint | clock	6.3	0.91	4.9	9.9
$\tilde{\gamma}$ integrated by parts (observer, emission, bulk) | ray | $\cdots$	6.6	6.0	43.7	–
$A_2 + HH + e^{2\zeta}$+direction | displacement | endpoint | clock	1.78	1.24	3.8	10.3
Eulerian: above field group $-\Psi \cdot \partial\sigma$ | $(X + \Psi) \cdot \partial\sigma$ | $\cdots$	1.92	1.21	1.91	3.49
orthonormal: A_2 | $-2\sigma(n, v)$ | displacement | endpoint | clock	1.73	0.99	6.2	13.9
orthonormal: A_2+displacement | $-2\sigma(n, v)$ | endpoint | clock	1.23	0.86	3.45	3.99

Four observations follow.

1. The full second-order metric $\tilde{\gamma} = A_2 + \frac{1}{2}(H_a H_b + H_b H_a) + 2(H_a + H_b)$ (so that $\sigma^{(2)} = \frac{1}{2}\partial_t \tilde{\gamma}$) roughly halves the cancellation at every x_o. Part of it is an artefact of the exponential parametrisation $\gamma = e^{2\zeta} e^H$.

2. Grouping the $e^{2\zeta}$ shear term with the direction terms removes almost all cancellation near horizon crossing ($\kappa = 1.8$ at $x_o = 1$). In orthonormal directions $e^{\zeta} e$, the static conformal factor drops out of $\sigma(e, e)$. Prop. 3 completes this analytically: in the fully orthonormal direction $e^{\zeta} e^{H/2} e$ the driver is $\frac{1}{2}H'$ through third order on the shell. In the map the local products, the $e^{2\zeta}$ term and the $\sigma(n, e)$ terms combine exactly into the single piece $-H_b'(n, v_a) - H_a'(n, v_b)$; the numerics reproduce this identity to quadrature accuracy. That piece is small: its share of the norm is 0.13, 0.04, 0.76 and 1.3 at $x_o = 1, 3, 6, 12$, against about 5 for each of its constituents. Its three constituents were cancelling among themselves, not against the rest of the kernel. The orthonormal rows of the table show what is left. Near horizon crossing almost nothing cancels, and grouping A_2 with the displacement gives $\kappa = 1.23$ at $x_o = 1$, the lowest of any grouping. The late-time cancellation between A_2 and the displacement is untouched: orthonormal directions do not address it (observation 4).

3. Integrating the field term by parts makes things worse inside the horizon: the observer endpoint and the bulk term cancel.

4. At late times the residual cancellation is between the field group and the photon displacement $X \cdot \partial\sigma$. The first-order kernel tends to a static profile, $Q \to 6/k_0^2$ up to oscillations, so $H_1 \to -\frac{6}{k_0^2}(\partial_i \partial_j - \frac{1}{3}\delta_{ij}\nabla^2) g$. This is a pure static relabelling of the fluid labels, $X \to X - \frac{3}{k_0^2}\nabla g$. Its products appear both in $\tilde{\gamma}$ and in the ray displacement, and they must cancel in the observable. This suggests removing the static part of H_1 by an explicit relabelling before expanding. That would be the analogue of the cancellation of bulk displacements in large-scale-structure perturbation theory. A first test supports this. Write $\Psi_a = -\frac{i}{2} Q_a(t)\, k_a e^{i k_a \cdot X}$ for the first-order fluid displacement that removes the scalar part of H_1; it tends to $-\frac{3}{k_0^2}\nabla g$ at late times. Split the photon displacement term as $X \cdot \partial\sigma = (X + \Psi) \cdot \partial\sigma - \Psi \cdot \partial\sigma$, and group $-\Psi \cdot \partial\sigma$ with the field and direction terms. This lowers κ from 3.8 to 1.9 at $x_o = 6$ and from 10.3 to 3.5 at $x_o = 12$, and leaves it unchanged near horizon crossing (last row of the table). The test keeps only the first-order scalar displacement. Including the full first-order relabelling does not change this. That relabelling is $e = (1 - \partial\Psi)(e^Y - v)$ with $v = \partial_t \Psi$, and it adds the Jacobian rotation of σ and the fluid-velocity term. Kept as separate groups, the new Jacobian and direction pieces cancel each other (7.1 against 6.5 at $x_o = 12$). Combined, they return the same $\kappa = 3.5$. The residual is between the Eulerian field group and the photon's displacement relative to the fluid. A likely reason is that only space has been relabelled. In the Eulerian spatial coordinates the conformal factor becomes $\zeta_Y = \zeta(1 - Qk^2/6)$, which still oscillates with $O(1)$ amplitude ($\to -g \cos x$) at late times. The temperature-defined time slicing is also displaced, because the radiation temperature oscillates with $O(1)$ amplitude inside the horizon. The natural next step is therefore a full change to longitudinal gauge, including the time shift, in which the potentials decay as x^{-2} inside the horizon.

The observer-centred decomposition $g = g(0) + [g - g(0)]$ was also tested. Exact dilation covariance, $\Theta[\zeta + c]$ with momenta rescaled by e^{-c}, implies two identities. The purely dilation parts of $\frac{1}{4}C^{(22)}$ and $\frac{1}{3}C^{(13)}$ combine exactly into $\frac{1}{2}\sigma^2 D^2 C^{(11)}$, with $D = k_0\, \mathrm{d}/\mathrm{d}k_0$ at fixed times. The dilation–centred cross terms combine into a total derivative $-2D\,\mathrm{Re}\, X$, with $X = \iint PP\, A_2^c A_1^*$. This generalises the superhorizon $8 = 4 + 4$. At finite k_0, however, $\frac{1}{2}\sigma^2 D^2 C^{(11)}$ is not the dominant part of $C^{(4)}$: the remainder is $-8.3\, C^{(11)}$ at $x_o = 1$ and $-67\, C^{(11)}$ at $x_o = 3$. The cross term

of Prop. 19 was then evaluated (`cross_term` in `solution/numerics/explore_fourth_order.py`), which determines the centred part $C_2^{c,4} = C_2^{(4)} - \frac{1}{2}\sigma^2 D^2 C_2^{(11)} + 2D_{k_0} X_2$. In units of $C_2^{(11)}$, fixed ratio:

x_o	$C_2^{(4)}$	dilation	cross	centred
0.15	9.17	7.89	−0.25	1.53
0.3	7.74	7.54	−0.95	1.15
0.6	2.67	6.16	−3.30	−0.19
1	−5.49	2.79	−5.95	−2.33
1.5	−7.05	−4.03	−0.18	−2.83
2	47.6	−10.4	28.0	30.0
3	−13.5	53.9	−168.5	101.1

The small-x_o entries follow the proposition: the centred part tends to $\frac{5}{3}$, and the cross term vanishes as x_o^2. Inside the horizon the three pieces are each large and cancel, as at $x_o = 3$, so the observer-centred split relocates the cancellation rather than removing it. Its value is the exact superhorizon structure, not a simplification at finite k_0.

The cubic term above uses the q-contracted scalar response of Prop. 21; the second-order vectors and tensors inside its sources and in L_3 are kept.

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